

國立陽明交通大學
生化暨分子生物研究所
碩士論文

Institute of Biochemistry and Molecular Biology
National Yang Ming Chiao Tung University
Master Thesis

藉由演化分析與自監督學習探討固有無序蛋白質之序列特徵
Characterizing the molecular features of intrinsically disorder
proteins by evolutionary analysis and self-supervised learning

研究生：何玟霖（Wen-Lin Ho）
指導教授：黃介嶸 博士（Jie-rong Huang, Ph.D.）

中華民國 一一一年七月
July 2022

藉由演化分析與自監督學習探討固有無序蛋白質之序列特徵

Characterizing the molecular features of intrinsically disorder
proteins by evolutionary analysis and self-supervised learning

研 究 生： 何玟霖

Student： Wen-Lin Ho

指導教授： 黃介嶸 博士

Advisor： Jie-rong Huang, Ph.D.

國立陽明交通大學
生化暨分子生物研究所
碩士論文
NYCU

A Thesis
Submitted to Institute of Biochemistry and Molecular Biology
College of Life Science
National Yang Ming Chiao Tung University
in partial Fulfillment of the Requirements
for the Degree of
Master of Science
in
Biochemistry and Molecular Biology

July 2022
Taiwan, Republic of China
中華民國 一一一年七月

國立陽明交通大學

博碩士學位論文電子檔著作權授權書

(請裝訂於紙本論文書名頁之次頁For authors to bind after the title page)

本授權書所授權之學位論文，為本人於 國立陽明交通大學 生化暨分子生物研究所 _ _ _ _ _ 組，110 學年度第 二 學期取得 碩士 學位之論文。

論文題目：藉由演化分析與自監督學習探討固有無序蛋白質之序列特徵

指導教授：黃介嶸

本人授權將本著作以非專屬、無償授權國立陽明交通大學、台灣聯合大學系統圖書館及國家圖書館。

論文上載網路公開之範圍及時間：	
中英文摘要	■ 中華民國 116 年 7 月 18 日公開
論文全文於本校及台灣聯合大學系統區域網路	■ 中華民國 116 年 7 月 18 日公開
論文全文於國家圖書館內部網路	■ 中華民國 116 年 7 月 18 日公開
論文全文於校外網際網路	■ 中華民國 116 年 7 月 18 日公開

說明：基於推動「資源共享、互惠合作」之理念與回饋社會及學術研究之目的，得不限地域、時間與次數，以紙本、光碟或數位化等各種方式收錄、重製與利用；於著作權法合理使用範圍內，讀者得進行線上檢索、閱覽、下載或列印。

授權人： 何致霖 (親筆簽名)

中華民國 111 年 07 月 19 日

國立陽明交通大學 / 國家圖書館

博碩士學位論文【延後公開】申請書

Application for Embargo of Thesis/Dissertation

(請裝訂於電子檔授權書之次頁 For authors to bind after authorization to copyright of ETD)

申請日期：中華民國 111 年 07 月 19 日

Application Date: 2022 / 07 / 19 (YYYY/MM/DD)

申請人姓名 Applicant Name	何玟霖	學位類別 Graduate Degree	碩士	畢業年月 Graduation Date (YYYY/MM)	民國 111 年 07 月 2022 / 07 (YYYY/MM)
學號 Student ID	309303004	系所名稱 Department	生化暨分子生物研究所		
論文名稱 Thesis/Dissertation Title	藉由演化分析與自監督學習探討固有無序蛋白質之序列特徵				
延後公開原因 Reason for embargo	☐ 準備申請專利，請說明 Filing for patent registration. Please specify : ☒ 涉及國家機密，請說明 Contains information pertaining to the national secret. Please specify. : 涉及機密 ☐ 依法不得提供，請說明 Withheld according to the law. Please specify. :		公開日期 Delayed Until	民國114年7月18日 2025/7/18 (YYYY/MM/DD)	

申請人簽名：何玟霖
Applicant
Signature: _____

系所審議章戳：
Seal of the
Department: 

指導教授簽名：黃介牛
Advisor Signature: _____

【說明】

1. 以上所有欄位請據實填寫並檢附證明文件，缺項或簽章不全，恕不受理。
2. 延後公開須訂定合理期限，請依實際需求設定延後公開日期，自申請日期起算至多3年，若超過3年或未填寫延後公開日期者，一律以3年期限辦理。

【Notes】

1. Please fill in all blanks. The application form will not be accepted for processing until all information, signatures, and stamps are included.
2. The delay should be a reasonable period of no more than 3 years. If the applicant fills in a date that creates a period longer than 3 years, the delayed period for public viewing period will be fixed at 3 years.

國立陽明交通大學 碩士 學位論文審定同意書

生化暨分子生物

研究所 何玟霖 君

所提之論文

題目：(中文) 藉由演化分析與自監督學習探討固有無序蛋白質之序列特徵

(英文) Characterizing the molecular features of intrinsically disorder proteins by evolutionary analysis and self-supervised learning

經學位考試委員會審查通過，特此證明。

學位考試委員會（簽名）

口試委員：林達顯（召集人）

黃介升

許世宜

論文已完成修改

指導教授

黃介升

（簽名）

所 長

鄭子豪

（簽名）

中 華 民 國 111 年 7 月 14 日

致謝

碩士生活過得很快也很充實，兩年前的我很幸運也很有眼光，二話不說選擇了黃介嶸老師實驗室與陽明交大生化所，並在令人放心的環境中做著自己感興趣的題目，我很慶幸我對人生做出了很棒、也很合適的選擇。

感謝黃介嶸老師，給了我很有趣、也很有挑戰的研究題目，並讓我使用一直以來的興趣作為方法，雖然過程中很長的一段時間，我什麼都沒做出來，但您還是有耐心與寬容的讓我繼續試，並在討論時都能翻譯出我想表達，但講不清楚的事情，謝謝您，希望將來能與您發現並解決更多有趣的東西。而每次進度報告與口試的林達顯老師、許世宜老師、黃宣誠老師，您的建議總能為我研究的方向做出指引，替我講解我說不清楚的地方，並釐清問題的盲點，謝謝您們與每次實用的意見。

謝謝實驗室的每位成員，儘管在實驗室我常常就是戴耳機、打電腦，活在自己的世界，實驗內容也鮮少與你們互動，但實驗室氣氛是讓我能自在做研究的關鍵，實驗室大小事都可以問，一起玩過許多遊戲的永宸、畢業後常回來反而才變比較熟的明軒、前期帶我實驗與喝紅茶的逸之、帶我們去鶯歌玩，畢業後很快就回來分享有趣職場生活的詩惠，還附有蛋糕，學長姐們沒有高姿態的架式，營造了和諧的實驗室氣氛，謝謝你們。與我同屆的靜柔、佳儀，謝謝你們當我的同學，一起準備可怕的考試，吃遍了免錢便當，你們也帶給實驗室熱鬧的氣氛，碩二時，於班級事務我比較狀況外，平時忙於實驗也較少互動，但有什麼重要事，你們總會提醒我，也會問我中午要不要吃東西，謝謝你們，與你們相處很輕鬆，期許我們的未來都能順利。兩位學弟妹，裕先、昱蓁，謝謝你們能陪我玩遊戲，陪我聊天，祝福你們能順利畢業，將來一定能與你們有更多有趣的互動。

我的父母與哥哥，謝謝你們從小到大無條件地贊成我做的任何事，你們的辛勞，讓我可無後顧之憂地完成學業，父母常主動打來問有沒有吃飽，有沒有吃水果，不要太累，但碩士生活繁忙，回家次數變少，對你們感到愧疚，誠心感激你們的付出與支持，謝謝你們的栽培。

中文摘要

固有無序區段 (Intrinsically disordered regions, IDRs) 與結構蛋白相比，雖然沒有穩定的結構且高度動態，但仍對生物體有著重要的角色。IDRs 藉由疏水與 π - π 等作用力，透過液-液相分離 (liquid-liquid phase separation, LLPS) 形成小滴 (droplets)，以調控生物學功能。一些序列模式與模體 (motif) 會促進上述作用力的形成。而已有很多研究發現 IDRs 中的 RNA 結合蛋白 (RNA-binding proteins, RBPs) 會經由特別的序列模式促進液-液相分離。為了系統性地分析 IDRs 中有重要功能的序列模式，我們透過演化分析與自監督機器學習 (self-supervised machine learning) 方法分析 IDRs 序列。透過演化分析，我們發現了與人類蛋白質組相比，RBPs 中的 IDRs 有一些特別的特徵，包含: RBPs 有較多的 IDRs、一些特別的短序列模式在 RBPs 較多、芳香族胺基酸在 RBPs 較多。然而，芳香族胺基酸通常被認為是結構蛋白的穩定劑 (stabilizers)，為了瞭解為何芳香族胺基酸在 RBPs 中比較多，我們透過演化分析發現這些芳香族胺基酸具有高度保守性。結合先前研究，我們認為儘管 IDRs 總體來說較不保守，若一些位點具高度保守性，其可能為功能重要位點。另一方面，我們使用自監督機器學習方法，經由對比學習 (contrastive learning) 由蛋白質的同源物 (homologs) 訓練出能找出潛在重要位點的模型，其經訓練後已可辨認出 IDRs 中重要的位點，目前我們正透過更多的下游任務來改善模型的效能。總結來說，我們發現了 RBPs 中 IDRs 的一些獨特特徵，且做出了可以找出重要位點的深度學習模型。

關鍵字: 固有無序區段, RNA 結合蛋白, 液-液相分離, 演化分析, 自監督機器學習。

Abstract

Intrinsically disordered regions (IDRs) do not have stable structures and are highly dynamic compared to structural proteins, but they still carry important roles for the organism. IDRs form droplets by liquid-liquid phase separation (LLPS) to regulate biological functions through interactions like hydrophobic and π - π interactions. Special sequence patterns and motifs are the driving force to promote molecular interactions. Several studies have found that IDRs in RNA-binding proteins (RBPs) contribute to LLPS by many kinds of special patterns. To systemically uncover special IDRs patterns which are important sites, we analyzed the IDRs sequences by evolutionary and self-supervised machine learning approach: In evolutionary analysis, we first found that IDRs in RBPs have some unique properties compared to the human proteome, including that RBPs have more IDRs, special short sequences patterns are rich, and aromatic residues are more. However, aromatic residues usually act as stabilizers in structural proteins and are less in IDRs. To figure out why aromatic residues are more in RBPs, we conducted sequences conservation analysis, and we found that aromatic residues are highly conserved in many RBPs. Consistent with former studies, we conclude that even though IDRs are overall less conserved, some residues express high conservation, and these sites may be functionally important. In the self-supervised machine learning approach, we built a model trained with the protein homologs by contrastive learning to discover potentially important sites. After training, the model can determine important sites, and we are now trying to improve the confidence of the model's output by applying more downstream tasks to evaluate performance. To sum up, we discovered some unique properties in RBPs disorder regions, and we built a machine learning model to discover important sites.

keywords: intrinsically disordered regions, RNA-binding proteins, liquid-liquid phase separation, evolutionary analysis, self-supervised machine learning.

Contents

致謝	i
中文摘要	ii
Abstract	iii
Contents	iv
List of Figures	vi
1. Introduction.....	1
1.1 Intrinsically Disordered Regions (IDRs)	1
1.2 Molecular features in IDPs	2
1.3 RNA binding proteins (RBPs)	3
1.4 Machine learning in protein sequence analysis	3
1.5 Aim	6
2. Methods	7
2.1 Retrieve protein sequences from UniProt and homologs from OMA.....	8
2.2 Disorder regions identification	9
2.3 Percentage of IDRs and relative amino acid ratio	10
2.4 Polyalanine propensity score and α -helix propensity	11
2.5 The analysis of k -mer polypeptide compositions	13
2.6 Sequence conservation analysis.....	14
2.7 Training dataset preprocessing.....	16
2.8 Dataset loading for training.....	18
2.9 Model architecture and training	20
3. Results.....	23
3.1 RBPs have more IDRs compared to the human proteome	23
3.2 There are some special polypeptide patterns in RBPs disorder regions.....	25
3.3 RBPs have more polyalanine and α -helix propensity.....	26
3.4 Aromatic residues are more in disorder regions of RBPs	27
3.5 Aromatic residues are early conserved among Chordata	29
3.6 Self-supervised machine learning: reproducing previously published results and the limits in their approach.....	31
3.7 Discover the distal features by the self-attention based improved model	32
3.8 Model performance evaluation by downstream tasks.....	36

3.9 Clustering of the LLPS dataset	40
4. Discussion	41
4.1 IDRs properties in RBPs	41
4.2 Evolutionary analysis and conserved aromatic residue in RBPs	42
4.3 Potentially important sites in IDRs by self-supervised model	46
4.4 Future works	48
4.4.1 Applying more downstream tasks	48
4.4.2 Verifying the levels of importance for residues in a sequence	48
4.4.3 Training our models with other homologs	48
4.4.4 Future aims: to build a platform for the public	49
4.5 Summary	49
5. Reference	50
6. Appendix	58
6.1 Publications	58

陽明交大
NYCU

List of Figures

Figure 1.1 Overview of contrastive learning	5
Figure 2.1 Data format for disorder regions predicted by PONDR.....	9
Figure 2.2 Examples of Polyalanine propensity score and α -helix propensity.....	11
Figure 2.3 Example for analysis of 3-mer polypeptide compositions	13
Figure 2.4 Momentum update mechanism in MoCo v3	22
Figure 3.1 Percentage of consecutive IDRs between RBPs and the human proteome.	23
Figure 3.2 Percentage of consecutive IDRs in subsets of RBPs	24
Figure 3.3 The k -mers analysis of IDRs between RBPs and human proteome IDRs.....	25
Figure 3.4 The abundance of polyalanine and α -helix between RBPs and the human proteome	26
Figure 3.5 Amino acid relative content ratios between RBPs and human proteome IDRs	28
Figure 3.6 Conservation score of Proteins.....	30
Figure 3.7 Sequence logos of features extracted by the contrastive learning model	31
Figure 3.8 Attention map visualization for human and zebrafish galectin-3 IDR.....	33
Figure 3.9 Attention map visualization for human hnRNPA1 (176~372).....	34
Figure 3.10 Attention map visualization for human FUS (1~286)	35
Figure 3.11 Model performance evaluation with different <i>embed_dim</i>	38
Figure 3.12 Model performance evaluation with different <i>depth</i>	39
Figure 3.13 PCA of the LLPS dataset	40
Figure 4.1 Conservation score of proteins with aromatic residues has not been found to be functionally important	44
Figure 4.2 Examples show that sequences are misalignment	45

1. Introduction

1.1 Intrinsically Disordered Regions (IDRs)

It is the ordinary sense that proteins perform biological functions through their 3D structures and dysfunction because of misfolding [1, 2]. Recently, intrinsically disordered proteins (IDPs) have been discovered, they do not have stable tertiary and quaternary structures and are highly dynamic compared to structured proteins [3, 4]. In the human proteome, there are about 50% of proteins are IDPs or IDRs (intrinsically disordered regions, structure proteins consisting a part of disorder regions) [5, 6]. Eukaryotes have more IDPs than prokaryotes, implying that IDPs can increase the complexity of biological functions by the flexibility of IDRs, and enable more precise regulation of biological reactions [7, 8].

Although IDRs are dynamic and do not have stable structures, they still carry important roles: Short IDRs act as a linker between structure regions of proteins, increasing flexibility to promote allosteric regulation of proteins [9]; and several studies have found that IDRs phase separate to form droplets by intermolecular or intramolecular interactions [10–12]. The droplets are membrane-less organelles (MLO) in the cells, and this process is also called liquid-liquid phase separation (LLPS) [13, 14]. The droplets formed by LLPS are assembled by molecular forces like hydrogen bonding, π - π , cation-anion, cation- π , dipole-dipole, hydrophobic, and electrostatic interaction [15]. These interactions promote biochemical reactions by increasing the concentration of proteins and other biomolecules, and the formation of droplets can be regulated by temperature, pH, charge, and post-translational modification (PTM) [16–18].

1.2 Molecular features in IDPs

As we said, there are some molecular forces between IDRs. These forces are mainly formed by special sequence patterns: The π - π interaction is due to aromatic stacking, which is caused by three aromatic residues (phenylalanine, tyrosine, and tryptophan); cation-anion interaction is mainly because of interactions between positive charge (arginine and lysine) and negative charge (glutamic acid and aspartic acid); hydrophobic interaction provided by glutamine/asparagine-rich regions promote formation hydrogels [15, 19]. Some special patterns among sequences also contribute to the LLPS ability of IDRs: aromatic residues content regulates proteins like TDP-43, galectin-3, hnRNPA1, and FUS [11, 12, 20–22]; positive and negative charge content controls molecular interaction for G3BP1, nuclear speckle, and TAZ1-binding TADs [10, 23, 24]. The short motifs in IDRs also have been discovered and may contribute to function, like RGG/RG motifs in many IDRs of RNA-binding proteins (RBPs) [25].

Although IDRs are less conserved compared to fold regions by alignment analysis, these special features are highly conserved among IDRs, because these sequences are functionally important during evolution [26, 27]. However, sometimes it is hard to align sequences because the concept of conservation between fold regions and IDRs is slightly different: when it comes to disorder regions, conserved sequences are not only high similarity and have perfect alignment [28]. In some situations, disorder regions are also seen as conserved with the same repeats of special features, the similar amino acid composition, and arrangement of charge/hydrophobic residues [28, 29]. This is because these patterns in IDRs carry similar functions to organisms, even though their sequences are sometimes not so identical through sequence alignment.

1.3 RNA binding proteins (RBPs)

Several studies found that IDRs in RNA-binding proteins (RBPs) contribute to LLPS [30]. RBPs regulate the whole RNA's life, including transcription, splicing, translation, localization, and degradation [31]. RBPs have more IDRs than the human proteome, and many RBPs have been proved that their IDRs contribute to LLPS to form MLOs like stress granules (SGs) [32]. Many RBPs have been found in SGs like G3BP1, TDP-43, Musashi-family proteins, and FUS. These proteins help biomolecules to condensate, promoting biological reactions. For example, G3BP1 is a key factor to assemble SGs [10]; Musashi-1 enhances drug resistance in cancer cells [33]; TDP-43 responds to oxidative stress to regulate SGs [34]. These RBPs, however, are often the reasons for many diseases in humans, mainly neurodegenerative diseases [35]. Some mutations of proteins decrease IDRs flexibility and increase abnormal molecular forces, leading to protein aggregation, which is harmful to cells [15]. An example is that misfolding of TDP-43 causes many neurodegenerative diseases like amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTLD), and Alzheimer's disease (AD) [36, 37].

1.4 Machine learning in protein sequence analysis

Machine learning has achieved many successes in many fields like language translation, text-to-speech, facial recognition, object detection, and text generation [38]. In recent years, machine learning also has been used to solve biological problems. Many studies try to build models and applications to solve problems like specific protein function determination (classification) [39], enzyme binding site prediction [40], protein engineering [41], disorder region prediction [42], protein structure prediction [43], and medical image recognition [44].

I attempt to use the machine learning approaches to identify functional sites in disorder regions. Our data does not have any label with many disorder sequences (e.g., a database told

us some residues in disorder regions are important), so we considered self-supervised learning as our model architecture by using the pretext task to learn the representations of our data [45]. Specifically, I used contrastive learning for representation learning and try to find functional sites in disorder regions [46].

Contrastive learning becomes a popular method in Computer Vision (CV), briefly speaking, in the CV field, they try to get important elements in a picture by defining an objective function, which encourages the same picture with the data augmentations are more similar, and different pictures are more dissimilar [47–50]. On the other hand, Moss and coworkers took advantage of this concept in protein sequence machine learning [51, 52]. They consider homologs as phylogenetic augmentations because proteins between species in homologs have the same function, even though their sequences differ slightly. As in **Figure 1.1**, the model can determine important sites by considering sequences when the same homolog are similar, while sequences from different homologs are dissimilar. Part of my work is to improve their model's disadvantages. In addition, for any input sequences, we can use attention map visualization to see what residues that model pays attention to, helping us to find potentially important sites. Details are in the results.

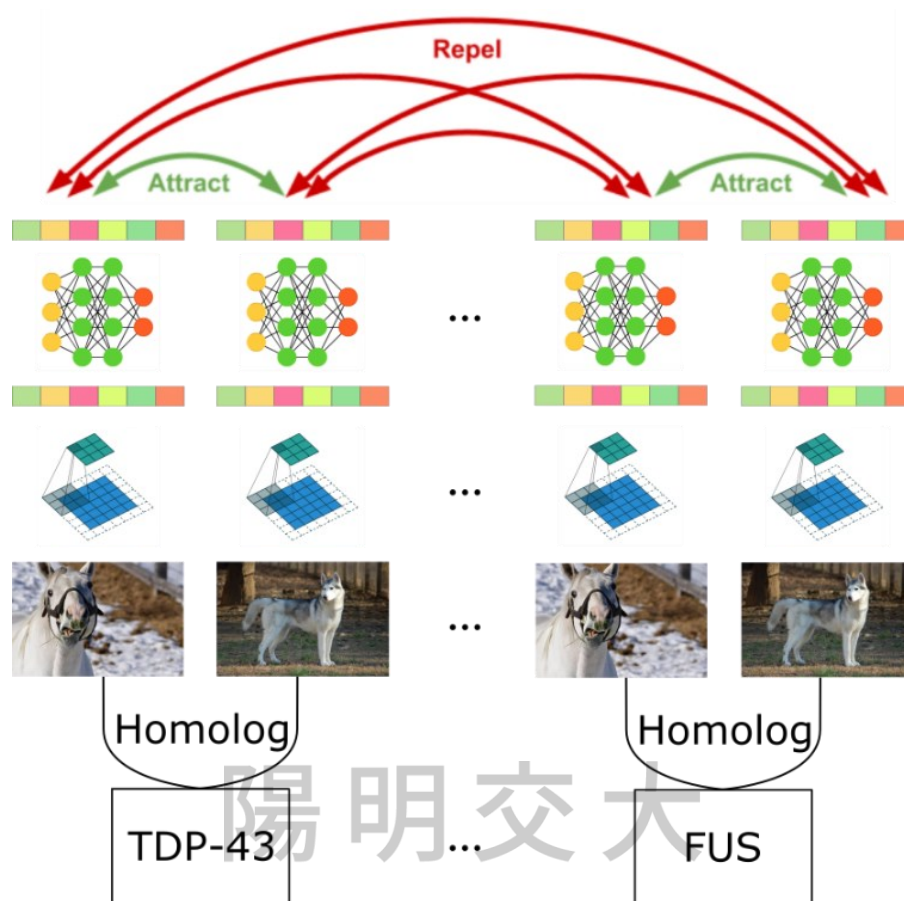


Figure 1.1 Overview of contrastive learning

The model architecture is adapted from Moss and coworkers [52]. Sequences are sampled from different homologs as the model's input during each round of training. By transforming encoded sequences to representation vectors through the model, the training object is that the model can output representation vectors for sequences that are similar to the same homologs, and dissimilar to different homologs. After training, the model should output a well-represented vector of each input sequence and identify important sites. The model composes of two parts: the base encoder network by Convolutional Neural Network (CNN) for transforming encoded sequences into representation vectors, and the projection head is simple multilayer perceptron (MLP), which maps the representation vector to another vector to calculate the similarity of two input sequences.

1.5 Aim

In this thesis, I aim to systemically find potentially important sites in IDRs by the bioinformatics approaches, because many studies found that proteins experiencing LLPS are regulated by special IDRs patterns. Our goal is to build a tool that inputs protein disorder regions from any IDRs sequence, and the tool will output some sites that are important to IDRs, and these sites may have key roles to regulate proteins' function. This thesis mainly has two parts:

First, I attempt to determine the properties of IDRs in RBPs (amino acid composition and content of IDRs), because RBPs have more IDRs than the human proteome. Many studies have proved that RBPs experience LLPS, bringing important functions. Therefore, we infer that these IDRs may have special patterns and properties that could be discovered and characterized.

On the other hand, I use evolutionary analysis and self-supervised learning approaches to find potentially important sites in IDRs. We want to build models or methods that can determine these sites: In evolutionary analysis, we discover that some residues in IDRs are early conserved compared to other IDRs sites, and these sites may be potentially important; In the self-supervised learning approach, I built the model that can identify important sites, and I applied some downstream task and make literary research to evaluate that our results are accurate.

The purpose of this thesis is to achieve:

1. Discovering the properties for IDRs in RBPs.
2. Using evolutionary analysis to find potentially important sites in IDRs.
3. Identifying potentially important sites in IDRs by building a self-supervised model.

2. Methods

Unless otherwise specified, all analysis and codes were implemented by Python 3.9.6 [53] and its relative packages are listed below.

Packages	Version	Description
biopython	1.79	biological computation tool [54]
comet-ml	3.24.2	machine learning results logging tool [55]
jellyfish	0.9.0	calculate Levenshtein distance [56]
jupyter notebook	6.4.6	python interactive interface by web application
matplotlib	3.5.0	plotting tool [57]
numpy	1.21.4	multi-dimensional arrays computing [58]
pandas	1.3.4	numerical tables manipulation [59]
plotly	5.4.0	plotting tool [60]
pytorch	1.10.0	deep learning framework [61]
requests	2.26.0	used to get API responses for OMA browser [62]
scikit-learn	1.0.1	machine learning and data preprocessing tool [63]
selenium	4.1.0	web crawler tool [64]
tqdm	4.62.3	progress bar in python [65]
visualizer	0.0.1	extracting attention map from model easily [66]

Table 2.1 Used python packages.

2.1 Retrieve protein sequences from UniProt and homologs from OMA

All human protein sequences were retrieved from UniProt (UniProtKB_2021_01, 20396 sequences). Gene names of RBPs (1542 sequences) and RBPs targeting subsets (689 for mRNA, 151 for tRNA, 89 for snoRNA, 122 for rRNA, and 167 for ribosome RNA) are from Gerstberger, et al. [67]. Gene names were then converted to UniProt Entry ID. Nine genes from Gerstberger, et al. are not in our analysis mainly because of duplicate gene names, so the total number of RBPs is 1533. Details of absence genes are listed in **Table 2.2**. Homologs of human proteins are retrieved from the OMA database (19909 OMA groups) [68].

No	Gene names from Gerstberger, et al [67]	Reason
1	PABPC1L2B	duplicates with PABPC1L2A
2	AARSD1	duplicates with PTGES3L-AARSD1
3	FRG1B	does not have UniProt Entry ID
4	SMN2	duplicates with SMN1
5	RPS17L	duplicates with RPS17
6	RBM11J	duplicates with RBMY1F
7	RPL41	the sequence is too short (P62945)
8	hCG_2045905	unreview from UniProt (D3DX98)
9	VARSL	unreview from UniProt (A2ABL6)

Table 2.2 Nine absence genes are not in our analysis.

2.2 Disorder regions identification

Non-standard codes of amino acids (B, Z, and X) in the sequences were removed [69]. Disorder regions were predicted by PONDR VL3, VLXT, and VSL2 algorithms [70]. All sequences were sequentially sent to the PONDR web server and parsed results automatically by the python web crawler package, Selenium [64]. **Figure 2.1** is an example output for predicted data: *uniprot_id* is UniProt Entry ID, *order_sequence* and *disorder_sequence* are protein sequences while residues predicted to be fold and disorder regions are masked by ‘*’, *od_ident* is the identification of fold or disorder regions, 1 means disorder and 0 means fold. **Code 2.1** shows the overall workflow.

	uniprot_id text	order_sequence text	disorder_sequence text	od_ident text
1	A0A024RBG1	*****FKKRAACLCFRSEQE*****	MMKFKPNQTRTYDREG*****DEVLLV...	11111111111111110000000...
2	A0A075B6H7	MEAPAQLLFLLLLWLPDTRREIVMTQ*****	*****SPPTLSLSPGERVTL...	0000000000000000000000...
3	A0A075B6H8	MDMRVPAQLLGLLLWLPGVRFDIQMTQSPSFLSA...	*****	0000000000000000000000...
4	A0A075B6H9	MAWTPLLFLTLLHCTGSLSQLVLTQSPS*****	*****ASASLGASVKL***	0000000000000000000000...
5	A0A075B6I0	MSVPTMAWMMLLLGLLAYGSGVDSQTVVTQE****	*****PSFSVSPGGTV...	0000000000000000000000...
6	A0A075B6I1	MAWTPLLLLFPLLHC*****	*****TGSLSQPVLTQSSASASLGSS...	0000000000000000111111...
7	A0A075B6I3	*****CTLSSDLS...	MALTPLLLLLSHCTGSLSRPVLTPPPSLSASP...	1111111111111111111111...
8	A0A075B6I4	MPWALLLLTLTHSAVSVVQAGLTQPPSVSKGLRQT...	*****	0000000000000000000000...
9	A0A075B6I6	MAWSSLLLTLLAHCTGSWAQSVLTQPPSVSGAPG...	*****	0000000000000000000000...

Figure 2.1 Data format for disorder regions predicted by PONDR

Data format for outputs of PONDR VL3, VLXT, and VSL2 algorithms. Proteins from UniProt with *uniprot_id* and protein sequences are sent to the PONDR server. Predicted disorder information is parsed to the following columns for further use: *order_sequence*, *disorder_sequence*, and *od_ident*.

```

1 # code2.1 workflow for disorder regions identification by PONDR
2
3 # input: sequence, a sequence from UniProt
4
5 # step 1. remove non-standard codes (B, Z, and X)
6 sequence = seq_aa_check(sequence)
7
8 # step 2. send sequence to PONDR, and get outputs
9 cutpondr = CutPONDR(show_progress_window=False) # initialize PONDR function
10 algorithm = "VSL2" # PONDR algorithm ('VLXT', 'XL1_XT', 'CAN_XT', 'VL3-BA', 'VSL2')
11
12 cutpondr.cut(sequence=a_sequence, algorithm=algorithm) # perform PONDR
13 order_sequence = cutpondr.get_order_sequence() # order_sequence, disorder regions are masked by "*"
14 disorder_sequence = cutpondr.get_disorder_sequence() # disorder_sequence, order regions are masked by "*"
15 od_ident = cutpondr.get_od_ident() # 1 means disorder and 0 means fold

```

Code 2.1 Workflow for disorder regions identification by PONDR

2.3 Percentage of IDRs and relative amino acid ratio

Proteins that consecutive disorder regions longer than designated conditions (20, 30, 40, 50, and 60) were considered. The percentage of IDRs (%) is defined as

$$\text{Percentage of IDRs (\%)} = \frac{\text{disorder region longer than conditions in protein}}{\text{numbers of protein in subset}} \times 100\%,$$

and the relative amino acid ratio of disorder regions is defined as

$$\text{Amino acid ratio} = \ln \left(\frac{\frac{a.a \text{ in subset} \in \text{IDRs}}{a.a \text{ in subset}}}{\frac{a.a \text{ in human proteome} \in \text{IDRs}}{a.a \text{ in human proteome}}} \right),$$

where $a.a$ is one of the 20 amino acids, subset includes the human proteome, RBPs, and mRBPs [67].

2.4 Polyalanine propensity score and α -helix propensity

For polyalanine, each amino acid was given a polyalanine propensity score according to former experimental studies [71] and our lab's observations (**Figure 2.2 a top left**). Continuous multiplication was performed in the disorder regions of a protein, and the protein was considered to have polyalanine propensity when the score exceeded the specified threshold (≥ 1.5 , 2.0, and 2.5). **Code 2.2** shows the procedure for calculating the propensity. For α -helix propensity, we used the standalone version of RaptorX [72] (**Figure 2.2 b**), and outputs were processed by python script.

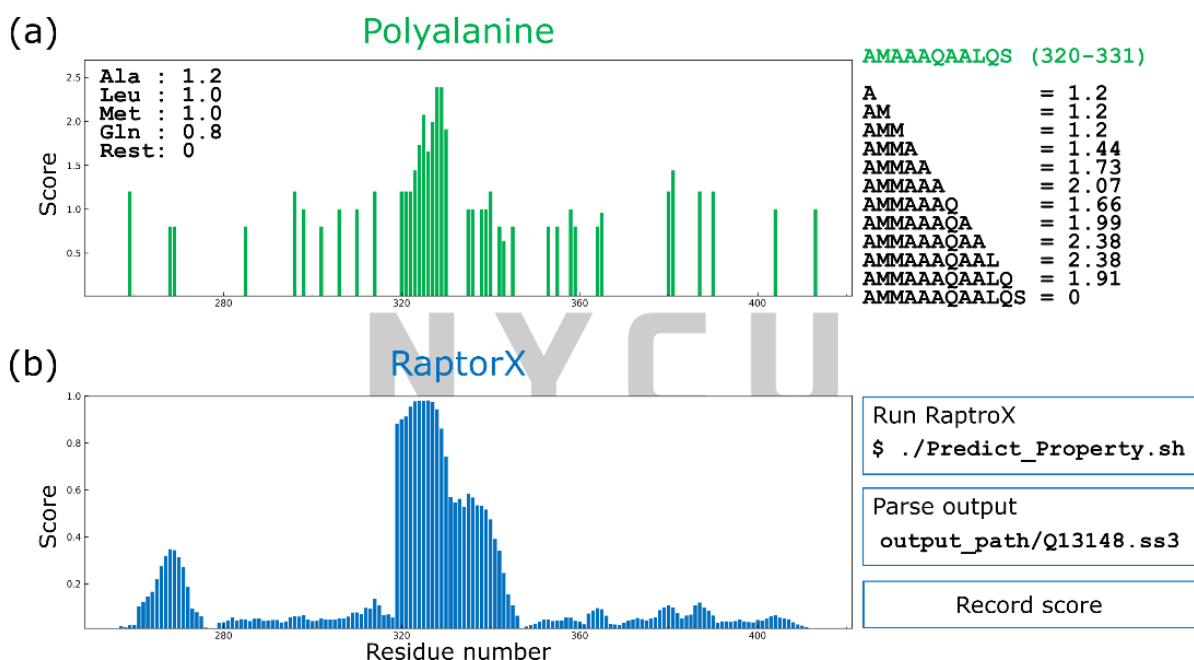


Figure 2.2 Examples of Polyalanine propensity score and α -helix propensity

Using TDP-43 IDR (257~414) as an example, for polyalanine propensity **(a)**, continuous multiplication was performed with a propensity score for each residue (**top left**). In this case, the protein has a higher propensity score around position 320, if the score exceeds the designated threshold (≥ 1.5 , 2.0, and 2.5), then this protein is considered as having polyalanine propensity. **(b)** α -helix propensity calculated by RaptorX through sequences (**left panel**). Proteins are sent to RaptorX and output is recorded for further use (**right panel**).

```

1 # code2.2 examples of Polyalanine propensity score and  $\alpha$ -helix propensity
2
3 # input
4 # sequence: a sequence from UniProt
5 # od_ident: identification of fold or disorder regions from Section 2.2
6
7 # step 1. only consider fold/disorder consecutive regions longer than 10/40
8 od_ident = seqfilter.length_filter_by_od_ident(od_ident,
9                                                disorder_filter_length=40,
10                                               order_filter_length=10)
11
12 # step 2. get result, prop=1 means having polyalanine propensity, and prop=0 is not
13 # polyalanine propensity score
14 propensity_dict = {
15     'A': 1.2,
16     'L': 1,
17     'M': 1,
18     'Q': 0.8,
19     'D': 0, 'E': 0, 'F': 0, 'G': 0, 'H': 0,
20     'I': 0, 'K': 0, 'C': 0, 'Y': 0, 'N': 0,
21     'P': 0, 'T': 0, 'R': 0, 'S': 0, 'V': 0, 'W': 0,
22 }
23 # initialize Polya function by loading polyalanine propensity score
24 polya = Polya(propensity_dict=propensity_dict)
25 prop = polya.get_seq_prop(sequence=sequence,
26                           od_ident=od_ident,
27
28                           # score larger than designated score is seen as having polyalanine propensity
29                           prop_score_threshold=1.5,
30
31                           # 1 means only one score larger than polya_thresholds is seen as having polyalanine propensity
32                           prop_score_filter_length=1
33                           )

```

Code 2.2 Workflow for calculating polyalanine propensity score for a sequence

2.5 The analysis of k -mer polypeptide compositions

For k -mer ($k = 2$ to 7), all possible consecutive k residues in disorder regions of the human proteome and RBPs (mRBPs) were considered. The number of occurrences of all k -mer in each subset was counted. Taking 3-mer as an example (**Figure 2.3**), for a single sequence, 3-mer was scanned starting at sequence position 1, 2, and 3. Each type of 3-mer and its number of occurrences were recorded. For all sequences in each subset, the occurrences of all possible 3-mers were added up. After that, the proportion of subset and relative ratio to human proteome could be calculated, and the relative ratio to the human proteome is defined as

$$\ln(\text{relative to the human proteome}) = \ln \frac{\text{Proportion of } k\text{-mer in RBPs (mRBPs)}}{\text{Proportion of } k\text{-mer in the human proteome}}.$$

3-mer
Position 1: GML ASQ QNQ SGP SGN NQN QGN MQ
Position 2: GMLA SQQ NQS GPS GNN QNQ GNM Q
Position 3: GMLAS QQN QSG PSG NNQ NQG NMQ

Figure 2.3 Example for analysis of 3-mer polypeptide compositions

In position one (red), 3-mer was starting to scan at the first residue, and each three residues was considered as a mer, scanning to the whole sequence, and possible mers were recorded. In position two (green) and position three (purple), 3-mer was scanning from the second and the third residue respectively. After considering all positions, mers were added up as all possible 3-mers for this protein.

2.6 Sequence conservation analysis

Homologs from the OMA database were grouped by taxonomy (Chordata, Vertebrata, Tetrapoda, and Mammalia). Each group was aligned using Clustal Omega (v.1.2.4) [73]. The sequences were discarded when they had more than 20 gaps unless more than 3 sequences had the same gaps simultaneously. **Code 2.3** shows the overall workflow.

The level of conservation was calculated using Jensen-Shannon divergence (JSD), which is defined as

$$D^{JS} = \sum_{\alpha \in \text{a.a}} \left(0.5 \cdot p_c(\alpha) \cdot \log_2 \frac{p_c(\alpha)}{r(\alpha)} + 0.5 \cdot q(\alpha) \cdot \log_2 \frac{q(\alpha)}{r(\alpha)} \right),$$

where a.a is one of the 20 amino acids, $p_c(\alpha)$ is amino acids occurrence frequency along the column of alignment, $q(\alpha)$ is the background distribution in the BLOSUM62 alignment, and $r(\alpha) = 0.5p_c(\alpha) + 0.5q(\alpha)$. JSD was normalized by Z-Score normalization, and the mean (μ) and standard deviation (σ) were derived from Chordata homologs.

```

1 # code2.3 workflow for retrieving and processing homologs
2
3 # some parameters
4 uniprot_id = 'Q13148' # UniProt entry
5 tax_ids = [7711, 7742, 32523, 40674] # (chordata, vertebrata, tetrapoda, mammalia) taxonomy id for grouping
6 oma_fasta_path './oma_fasta/Q13148.fasta' # path for saving downloaded fasta
7
8 # step 1. retrieve homologs
9 fetchomaseq.get_oma_seq(uniprot_id, oma_fasta_path) # retrieve homologs by uniprot_id
10
11 # step 2. group by taxonomy id
12 for tax_id in tax_ids:
13
14     # make grouped fasta saveing paths
15     # e.g. chordata group homolog is in "./b_grouped/7711/Q13148.fasta"
16     grouped_fasta_path = create_save_path(tax_id)
17
18     # group by taxonomy id
19     taxfilter = TaxSeqFilter(tax_id) # initialize TaxSeqFilter given taxonomy id
20     taxfilter.taxfilter(oma_fasta_path, grouped_fasta_path) # Group and save output
21
22 # step 3. sequence alignment
23 for tax_id in tax_ids:
24
25     # make alied fasta saveing paths,
26     # e.g. chordata group homolog is in "./c_alied/7711/Q13148.fasta"
27     alied_fasta_path = create_save_path(tax_id)
28
29     # perform sequence alignemnt
30     alignment.alignment_single(grouped_fasta_path, alied_fasta_path)
31
32 # step 4. discard special sequences
33 for tax_id in tax_ids:
34
35     # make processed fasta saveing paths,
36     # e.g. chordata group homolog is in "./d_extre_filtered/7711/Q13148.fasta"
37     extre_filtered_tax_fasta_path = create_save_path(tax_id)
38
39     # perform discarding special sequences
40     extre_index = fastaextrefilter.fasta_extre_filter(alied_fasta_path,
41                                                       extre_filtered_tax_fasta_path,
42
43                                                       # more than 20 gaps
44                                                       gap_filter_num=20,
45
46                                                       # more than 3 sequences have the same gaps simultaneously
47                                                       max_filter_seq=3,
48                                                       )

```

Code 2.3 Workflow for retrieving and processing homologs

2.7 Training dataset preprocessing

Homologs belonging to Chordata were from the OMA database and aligned using Clustal Omega (v.1.2.4) [73]. 22 human proteins were not used because there is only one homolog. Human protein sequence in each homolog with disorder regions information identified by PONDR VSL2 was used to define the boundary of other aligned sequences disorder regions. The disorder regions with the same boundary were a new sub-homolog, and then the gap created by the alignment was removed. If sequences were at the start of protein, methionine was removed because it might reduce the contrastive task effect [52]. Similarity for homologs and human protein sequences in sub-homolog were calculated by Levenshtein distance [74] for probability sampling during training. Overall, 28,955 sub-homologs were collected from 19,887 homologs and 2,402,346 sequences. **Code 2.4** shows the overall workflow.

```
1 # code2.4 workflow for training dataset preprocessing
2
3 # input:
4 # fasta_path: processed fasta's path from Code 2.3
5 # od_ident: disorder regions information from human sequence
6
7 # step 1. get processed fasta from Code 2.3
8 seq_list = fasta_to_seqlist(fasta_path)
9
10 # step 2. remove non-standard codes (B, Z, and X)
11 seq_list = fasta_seq_aa_check(seq_list)
12
13 # step 3. disorder regions information from human sequence as the boundary of other aligned sequences
14 od_ident = seqfilter.length_filter_by_od_ident(od_ident,
15                                              disorder_filter_length=40,
16                                              order_filter_length=10)
17 # add gap on od_ident to fit other alied sequences
18 od_ident = seqfilter.od_add_alignment(fasta_path,od_ident)
19 # get disorder region infomation
20 od_index = seqfilter.get_od_index(od_ident)['disorder_region']
21
22 # step 4. create sub-homologs
23 # create homology_id for each sub-homolog
24 homo_frag_seq_id = []
25 for i in od_index:
26     homo_frag_seq_id.append(get_rand_id())
27
28 # step 5. extract sequence and make sequence data
29 seq_frag_list = []
30 for seq in seq_list:
31     for index,element in enumerate(od_index):
32
```

```

32
33     # get disorder sequence by the boundary from human sequence
34     start, end = element['start'], element['end']
35     frag_seq = str(seq.seq)[start:end]
36
37     # use homology_id from step 4
38     homology_id = homo_frag_seq_id[index]
39
40     # remove alignemnt gap
41     frag_seq = frag_seq.replace("-", "")
42     if len(frag_seq) < 40:
43         continue
44
45     # remove methionine
46     if frag_seq[0] == "M" and start == 0:
47         frag_seq = frag_seq[1:]
48
49     # mark human sequences for further use
50     oma_protein_id = seq.id # e.g. id HUMAN47411 for TDP-43 in OMA database
51     human_seq_match = (oma_protein_id is '^HUMAN*')
52     if human_seq_match:
53         is_human_seq = 1
54     else:
55         is_human_seq = 0
56
57     # save sequeunce data
58     seq_frag_list.append({ "homology_id":homology_id,
59                           "frag_seq":frag_seq,
60                           "is_human_seq":is_human_seq,
61                           })
62
63 # step 6. calculate Levenshtein distance after get all sub-homologs
64 # df: save all sequences produced by step 1 ~ step 3
65
66 # get table for only human sequences
67 human_df = df[df["oma_protein_id"]=="^HUMAN*"]
68
69 def get_sim_to_human_seq(x):
70     '''
71     create a function that input a row of data
72     including homology_id, frag_seq, and is_human_seq
73     and return probablity of this sequence
74     '''
75     # get human sequence by the same homology_id
76     human_seq = human_df[human_df['homology_id'] == x['homology_id']]['frag_seq'].tolist()[0]
77
78     # calculate Levenshtein distance
79     sim_score = calculate levenshtein_distance(human_seq, x['frag_seq'])
80
81     return sim_score
82
83 # apply get_sim_to_human_seq function to all sequences
84 df['q_prob_score'] = df.apply(get_sim_to_human_seq, axis=1)
85
86 # convert Levenshtein distance score to 0~1
87 df['q_prob_pure'] = df.groupby('homology_id')['q_prob_score'].transform(lambda x : x/sum(x))

```

Code 2.4 Workflow for training dataset preprocessing

2.8 Dataset loading for training

During each training batch, sequences were adjusted to the same length: repeated padding to 512 if the length is shorter than 512, and truncated to 512 if the length is longer than 512. CLS token (character O) was added at the start of the sequence. The CLS token stands for classification, used to represent the entire sequence and visualized by the attention map [75]. All sequences with the same length 513 were tokenized by Label encoding [76]. Each dataset consists of tokenized query sequence which is the human sequence; tokenized key sequence which is the other species' sequences sampled by Levenshtein distance probability. The probability is higher when the sequence is more dissimilar to the human sequence. In all experiments, batch=50 was used, meaning 50 datasets were put into the model simultaneously.

Code 2.5 shows the overall dataset implemented by PyTorch dataset.

```
1  # Code 2.5: pytorch dataset for training
2
3  # input:
4  # df: processed training dataset from Code 2.4
5
6  class SeqDataset(Dataset):
7      def __init__(self):
8
9          self.df = df
10         self.seq_length = 512 # padding length to 512
11         self.homology_ids = df['homology_id'].unique() # get all homology_id
12
13         # initialize Label encoder
14         le = LabelEncoder()
15         le.fit(list('OACDEFGHIKLMNPQRSTVWY'))
16         self.encoder = le
17
18     def __len__(self):
19         return len(self.homology_ids)
20
21     def __getitem__(self, index):
22         homology_id = self.homology_ids[index]
23
24         #get human q
25         q = self.df[(self.df['homology_id'] == homology_id) & (self.df["is_human_seq"]==1)]
26         q_seq = q['frag_seq'].tolist()
27         q_seq_tokened = self.seq_process(q_seq)
28
29         #get k as postive sample
30         q_as_k = self.df[self.df['homology_id'] == homology_id]
31         q_as_k_seq = q_as_k.sample(1,replace=True, weights='q_prob_pure')['frag_seq'].tolist()
32         q_as_k_seq = q_as_k.sample(1,replace=True)['frag_seq'].tolist()
33         q_as_k_seq_tokened = self.seq_process(q_as_k_seq) # max_length + encode
```

```

34
35     return {"q" : {'token' : q_seq_tokened,
36                   'seq'   : q_seq,
37                   },
38            "k" : {'token' : q_as_k_seq_tokened,
39                   'seq'   : q_as_k_seq,
40                   }
41           }
42
43 def seq_process(self, seq_list):
44     encoded_seq_list = []
45     for seq in seq_list:
46         #repeat_string
47         seq = self.repeat_string(seq, target_length=self.seq_length)
48         #add cls token '0'
49         seq = '0' + seq
50         #encode
51         seq = self.encoder.transform(list(seq))
52         encoded_seq_list.append(seq)
53     encoded_seq = torch.tensor(np.stack(encoded_seq_list))
54
55     return encoded_seq
56
57 def repeat_string(self, a_string, target_length=256):
58     #https://www.kite.com/python/answers/how-to-repeat-a-string-in-python
59     number_of_repeats = target_length // len(a_string) + 1
60     a_string_repeated = a_string * number_of_repeats
61     a_string_repeated_to_target = a_string_repeated[:target_length]
62     return a_string_repeated_to_target
63
64 def collate_fn(self, data): # collate_fn for dataloader
65     #https://li.allmwh.org/tLibhH
66     q_seq_tokened, q_seq, k_seq_tokened, k_seq, = \
67     zip(*[(s['q']['token'], s['q']['seq'],
68           s['k']['token'], s['k']['seq'],
69           ) for s in data])
70     q_seq_tokened = torch.stack(q_seq_tokened).squeeze(dim=1)
71     k_seq_tokened = torch.stack(k_seq_tokened).squeeze(dim=1)
72
73     return {"q" : {'token' : q_seq_tokened,
74                   'seq'   : q_seq,
75                   },
76            "k" : {'token' : k_seq_tokened,
77                   'seq'   : k_seq,
78                   }
79           }

```

Code 2.5 PyTorch dataset for training.

2.9 Model architecture and training

MoCo v3 [50] was used as our model architecture but replaced with vision transformer and patch embedding with classical self-attention to meet our tasks. The MoCo v3 uses the momentum encoder, which only performs backpropagation on the query encoder, while the key encoder is updated by momentum from the query encoder (**Figure 2.4**). The momentum update can save memory usage with the same performance as end-to-end training [49]. In each experiment, 400 epochs were trained using AdamW [77] optimizer with Cosine Annealing and Warmup [78, 79], and backpropagation was computed by InfoNCE with temperature parameter [80]. The model was all built by PyTorch [61], results were logged by Comet.ml [55], and trained models were saved and managed by Data Version Control [81]. InfoNCE is defined as:

$$Loss = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}$$

where q is a human sequence, k^+ is other species sequences with the same homolog as q , k^- is other species sequences that are different from homologs of q , and τ is temperature as a model hyperparameter. All hyperparameters used in training are listed in **Table 2.3**.

Hyperparameter	Value	Description
batch_size	50	50 datasets are put into the model altogether
seq_length	512	all sequences are repeated padding to length 512
depth	6, 3, 2, 1	attention layers used in experiments
num_heads	8	8 attention heads in each attention layer
embed_dim	128, 64, 32, 16	each residue in the sequence is embedded to the following dimensions for experiments
atten_mlp_ratio	4	from MoCo v3, dimension of projection head $\text{embed_dim} \times 4 = 512$
m	0.999	from MoCo v3, momentum update ratio for momentum encoder
nce_temp	0.2	temperature used for InfoNCE
epochs	400	training epochs
max_lr	1e-3	max learning rate during cosine scheduling
min_lr	1e-5	min learning rate during cosine scheduling
warm_up_step	10	epochs of learning rate warm-up to max_lr from starting training
cycle_steps	30	epochs of learning rate from max_lr to min_lr and vice versa

Table 2.3 Hyperparameter used in training

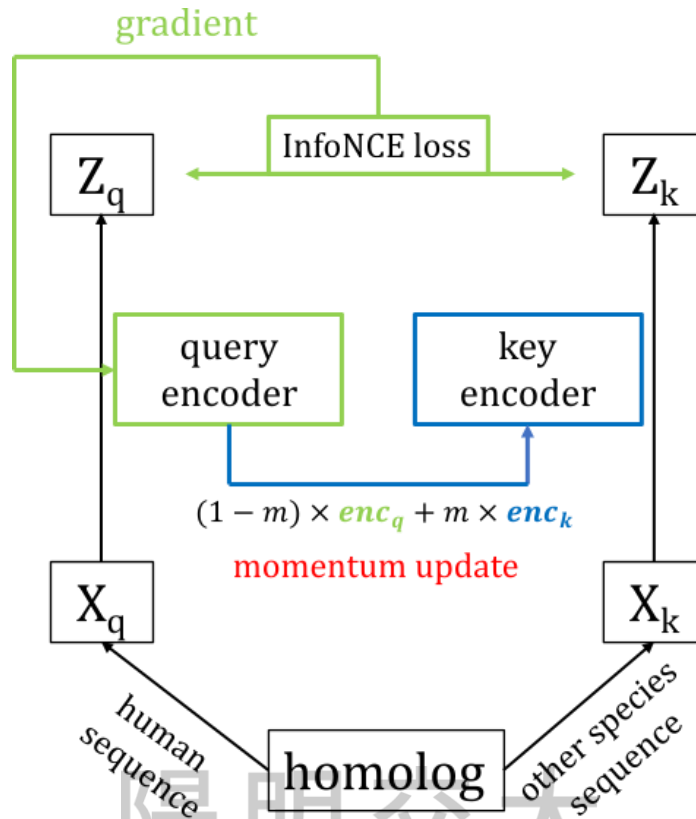


Figure 2.4 Momentum update mechanism in MoCo v3

X_q is human sequence form a sub-homolog, and X_k is other species' sequence sampled from the same sub-homolog. Two sequences are put into the two encoders (query and key) respectively and output representation vectors (Z_q and Z_k), which are used to calculate similarity (InfoNCE loss). InfoNCE loss is then used to update the query encoder (backpropagation), and the key encoder is updated by a part of the query encoder.

3. Results

3.1 RBPs have more IDRs compared to the human proteome

We first compared the percentage of consecutive IDRs between RBPs and the human proteome. As in **Figure 3.1**, three PONDR VL3, VLXT, and VSL2 algorithms show similar trends. Overall, RBPs have more consecutive disordered regions than the human proteome, with more significant in RBPs targeting mRNA (mRBPs), and RBPs targeting other subsets are in **Figure 3.2**. Except for tRNA and ribosome RNA, all subsets (snRNA, snoRNA, ncRNA, and rRNA) have more consecutive disordered regions.

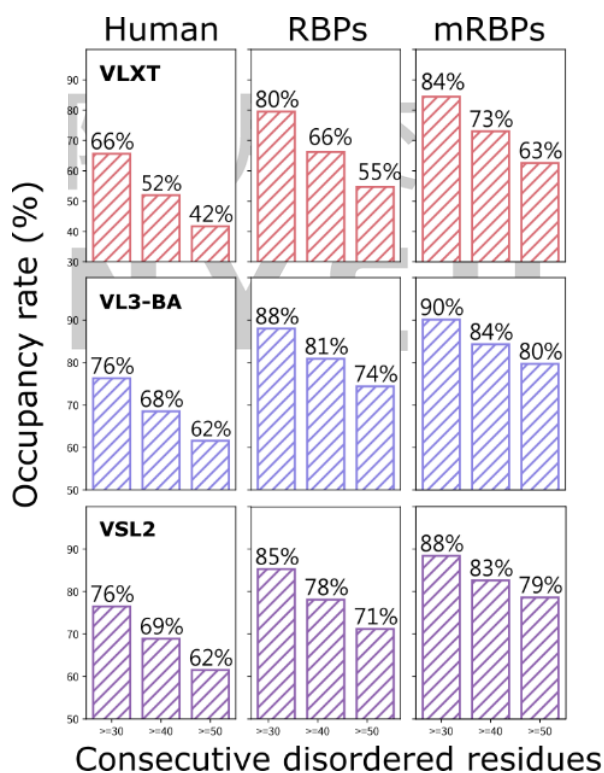


Figure 3.1 Percentage of consecutive IDRs between RBPs and the human proteome.

The percentage of IDRs between RBPs and the human proteome. The number of proteins is 20396 for the human proteome, 1533 for RBPs, and 688 for RBPs targeting mRNA (mRBPs). RBPs and mRBPs have more consecutive disordered regions than the human proteome within three PONDR algorithms (VL3, VLXT, and VSL2). Consecutive disordered regions are proteins with disorder regions longer than 30, 40, and 50 residues.

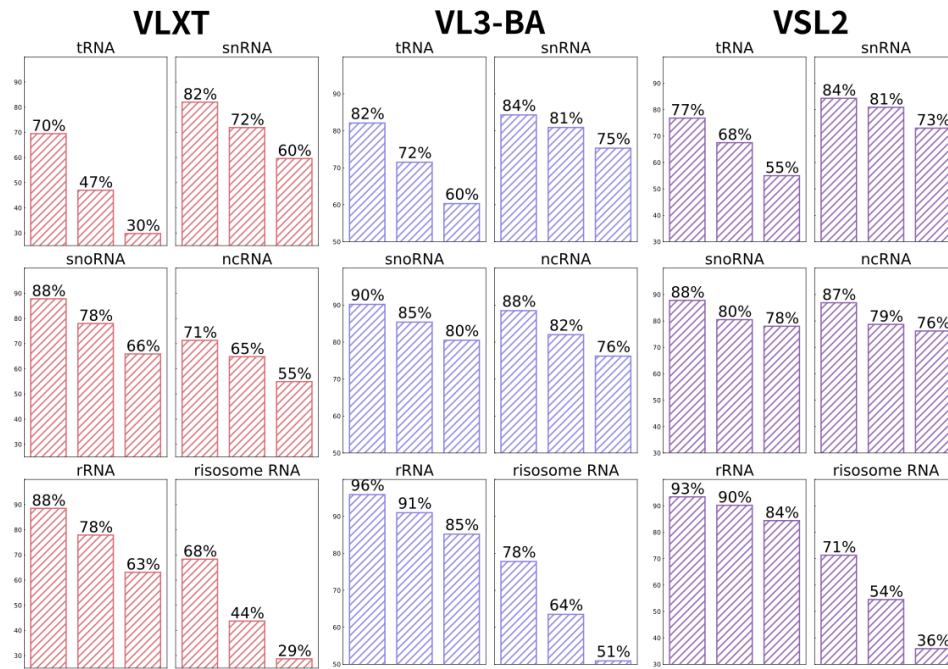


Figure 3.2 Percentage of consecutive IDRs in subsets of RBPs

The percentage of IDRs in RBPs targeting subsets (N = 151 for tRNA, 89 for snRNA, 41 for snoRNA, 122 for ncRNA, 122 for rRNA, and 167 for ribosome RNA).

3.2 There are some special polypeptide patterns in RBPs disorder regions.

We tried to explore overall common patterns in RBPs disorder regions, and we first conducted *k*-mer analysis compared to the human proteome as shown in **Figure 3.3**. In 3-mers, there are some special patterns are more in RBPs and mRBPs, mainly RXR (RSR, RGR, RER, RDR), XRX (KRK, SRS, ERE, GRG), and 3 letters repeat (KKK, RRR, GGG, RRR). In 5-mers and 7-mers, special patterns are less obvious because it is hard to appear when polypeptide are long and become complicated. Overall, we could find special patterns of polypeptides in RBPs, but only for the short polypeptide.

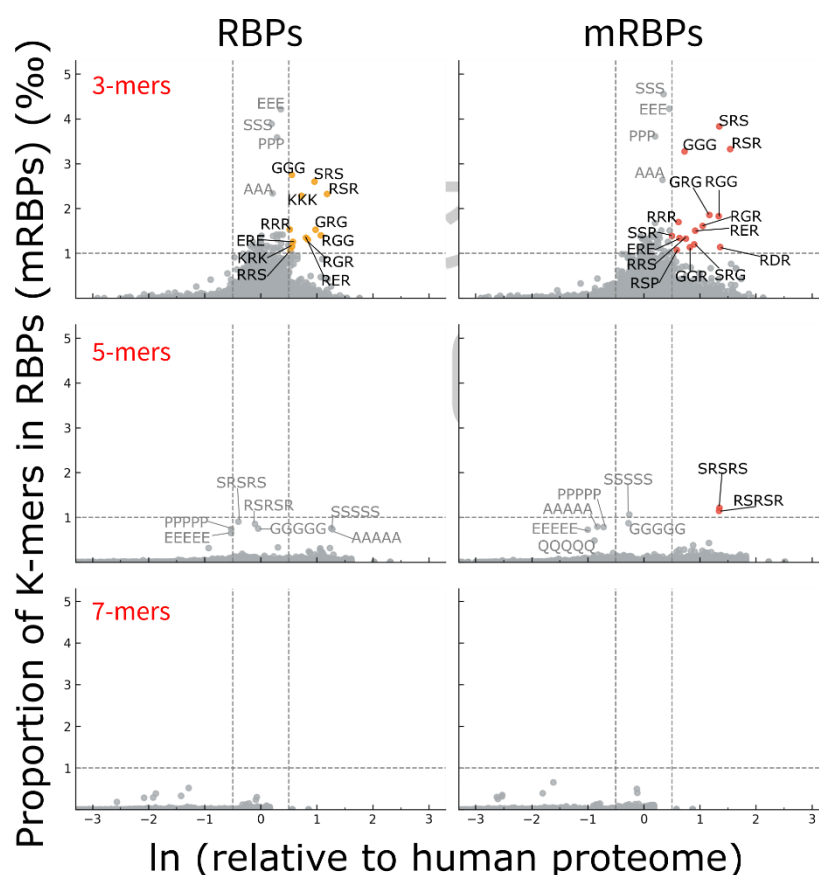


Figure 3.3 The k -mers analysis of IDRs between RBPs and human proteome IDRs

Disorder regions predicted by PONDR VSL2. Log ratio of k-mers proportion relative between RBPs(left)/mRBPs(right) and the human proteome. The dotted line of y-axis is proportion at 1% in RBPs/mRBPs, and the lines of x-axis are relative ratios at 0.5 and -0.5. Points with color are significant in RBPs/mRBPs (proportion more than 1% and relative ratio more than 0.5).

3.3 RBPs have more polyalanine and α -helix propensity

We also considered polyalanine propensity as overall common patterns in RBPs disorder regions, and therefore we designed a polyalanine score (see Methods) to evaluate the abundance of polyalanine. As in **Figure 3.4 a**, the higher percentage, the more protein sequences with polyalanine propensity scores above 3 thresholds respectively (≥ 1.5 , 2.0 , and 2.5). Results reveal that RBPs contain more polyalanine propensity compared to the human proteome. To further confirm our results, we also used RaptorX [72], a typical structure property prediction tool, to calculate the α -helix propensity (**Figure 3.4 b**). Although α -helix propensity is not only caused by polyalanine, results from RaptorX show similar trends to our polyalanine scores, with the percentage of sequences population is higher than our results ($\sim 60\%$ for RaptorX and $\sim 10\%$ for polyalanine).

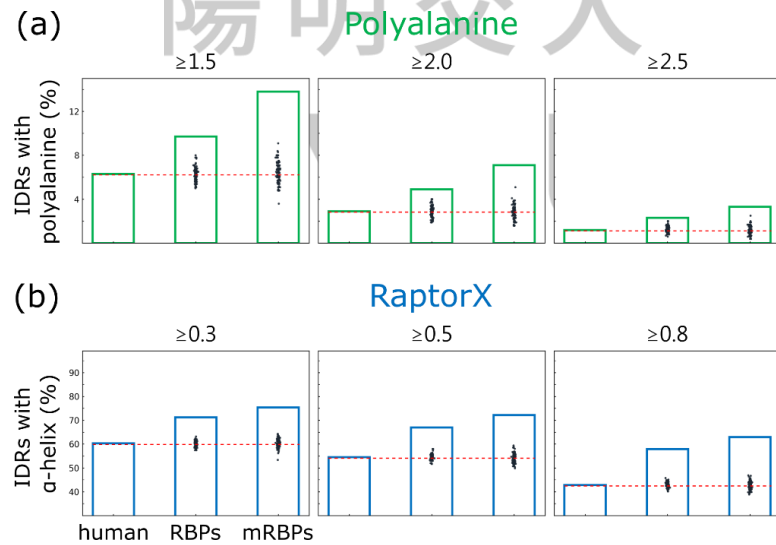


Figure 3.4 The abundance of polyalanine and α -helix between RBPs and the human proteome

Black dots are the significant test sampling the same number of proteins (1533 for RBPs and 688 for mRBPs) in the human proteome. Red dotted line is the population proportion for the human proteome. **(a)** The population proportion of proteins with polyalanine scores is greater than 1.5, 2.0, and 2.5 in disorder regions. Details of the score are in methods. **(b)** The population proportion of proteins with more than 5 consecutive residues and predicted α -helix propensity score is more than 0.3, 0.5, and 0.8 by RaptorX.

3.4 Aromatic residues are more in disorder regions of RBPs

Furthermore, we attempted to find which types of amino acids are more between RBPs and the human proteome by the relative ratio of amino acid content (Figure 3.5). Except for cystine, three different PONDR algorithms have similar tendencies. The high relative ratio means that RBPs have more amino acids compared to the human proteome, and almost all relative ratios are high. This is because our approach for the relative ratio is affected by the disorder region contents (the more disorder regions, the higher relative ratio, see Method). To reduce the influence of the content of disorder regions, we conducted significant tests by sampling the same number of protein subsets (1533 for RBPs and 688 for mRBPs) in the human proteome, and the black dots are relative ratio sampling 100 times. With the significant tests for all types of amino acids, the results indicate that RBPs disorder regions are rich in aromatic residues (tryptophan, phenylalanine, and tyrosine) among different three PONDR algorithms used compared to the human proteome, and more significant in mRBPs. That is, aromatic residues are more prevalent in RBPs disorder regions. Noticeably, statistics of overall proteins in the DisProt database report that disorder regions have fewer aromatic residues than folded regions [82, 83], but our results show that there are many aromatic residues in disorder regions of RBPs.

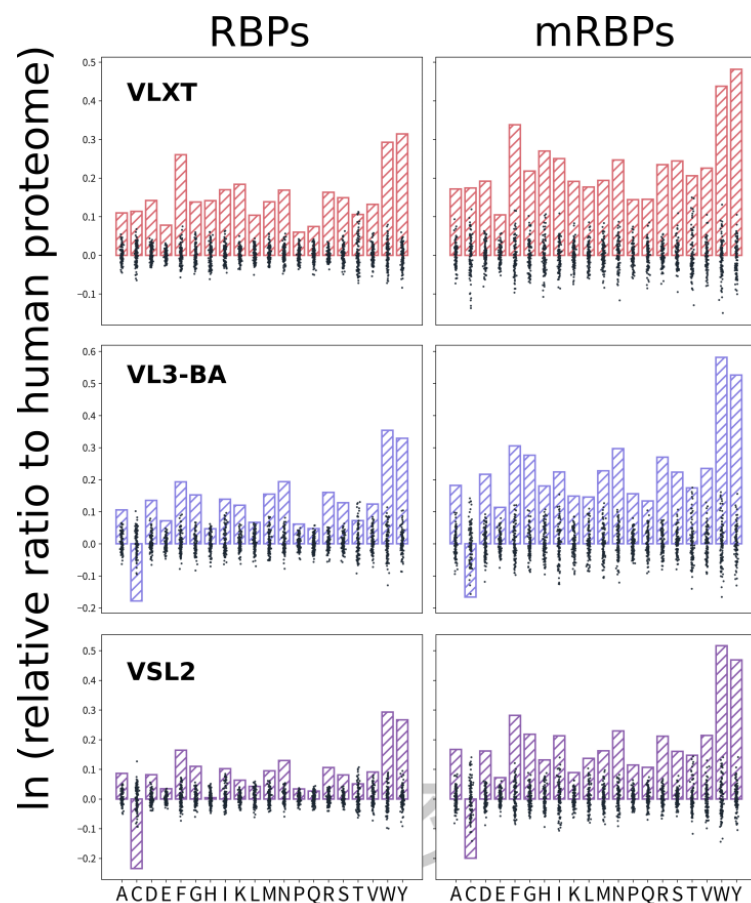


Figure 3.5 Amino acid relative content ratios between RBPs and human proteome IDRs

The higher relative ratio, the more amino acid content in RBPs (mRBPs). Black dots are significant tests sampling the same number of proteins (1533 for RBPs and 688 for mRBPs) in the human proteome. For three different PONDR algorithms (red for VLXT, blue for VL3-BA, and purple for VSL2), aromatic residues (tryptophan, phenylalanine, and tyrosine) are more in disorder regions of RBPs than the human proteome.

3.5 Aromatic residues are early conserved among Chordata

To understand why aromatic residues are more in RBPs disorder regions, we conducted sequence conservation analysis because functionally important sites are conserved during evolution. We used Jensen-Shannon divergence with Z-Score normalization as metrics [26]. **Figure 3.6 a** shows the level of conservation for TDP-43 among four taxonomy (Chordata, Vertebrata, Tetrapoda, Mammalia). The red bar at the bottom indicates the disorder region, and the level of conservation in Chordata (dark blue) is more divergent in the disorder region, which means that the disorder region is less conserved than the fold region. However, we point out the Chordata conservation level of each aromatic residue, and we notice that many aromatic residues have higher conservation levels than other residues, meaning that these aromatic residues are early conserved compared to other residues. On the top panel is the overall score for 20 amino acids among four different ranks of taxonomy (**Figure 3.6 a**), the scoreline is nearly horizontal indicating that aromatic residues are early conserved, while scorelines of other amino acids are steep, which means these amino acids are diverse during evolution.

Not only TDP-43 but many proteins have similar properties (**Figure 3.6 b, c, and d**), like tyrosine in FUS's N-terminal, phenylalanine in NUP153, and tryptophan in TNRC6A. These aromatic residues also have been shown to control LLPS in other research and will be discussed later. In sum, through sequence conservation analysis, we could discover early conserved sites that are potential functionally important in disorder regions.

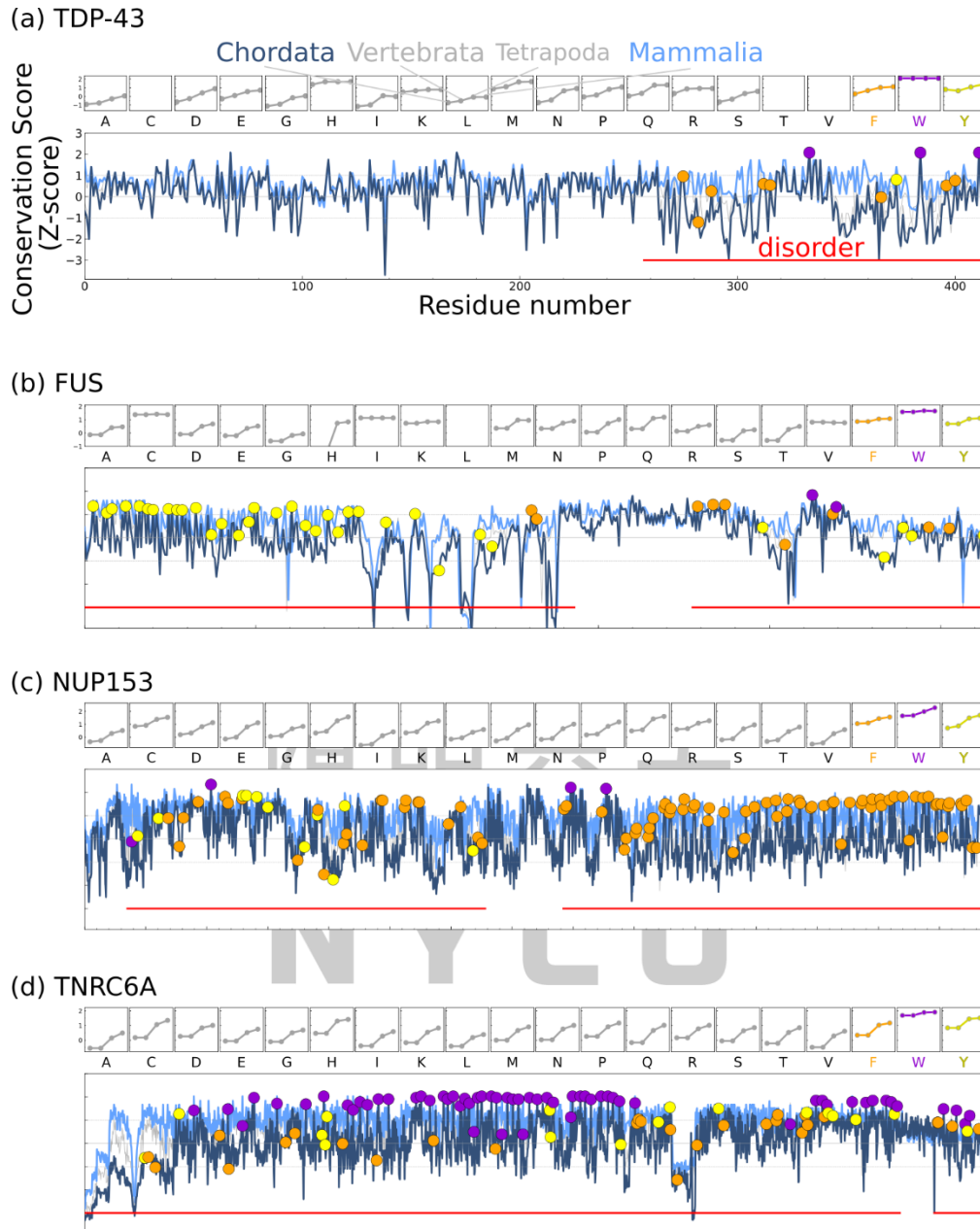


Figure 3.6 Conservation score of Proteins.

Conservation score of (a) TDP-43, (b) FUS, (c) NUP153 and (d) TNRC6A. Levels of conservation by Jensen-Shannon divergence with Z-Score normalization within Chordata (dark blue), Vertebrata (gray), Tetrapoda (gray), and Mammal (light blue). Mean and Standard Deviation for normalization are derived from the score of Chordata. **Bottom Panel:** conservation score along the whole sequence; red line is the disorder region predicted by PONDR VSL2 with consecutive disorder regions longer than 40. Aromatic residues points (orange for phenylalanine, purple for tryptophan, and yellow for tyrosine) are Chordata conservation scores. **Top Panel:** mean conservation scores along 20 amino acids.

3.6 Self-supervised machine learning: reproducing previously published results and the limits in their approach

In self-supervised machine learning approach, we first reproduced contrastive learning results provided by Moss and coworkers [52] and trained by RBPs homologs. After training, each CNN neuron may respond to sequences that have special patterns. As in **Figure 3.7**, we extracted evidence features from the final CNN layer. Each sequence logo represents a single neuron with higher output for a specific pattern and we see that neurons can detect various patterns, like positive (KR), negative (DE) charge, repeat SR repeat, Q rich region, and P rich region, suggesting that these patterns are more among RBPs so they can easily be detected by the model.

Although their approach can detect special patterns in sequences, the challenging work is that the base encoder (CNN) used in their model only recognizes regional features. Therefore, in our attempts, we replaced CNN with the self-attention mechanism for the base encoder [84], which can overall discover distal potential important sites.

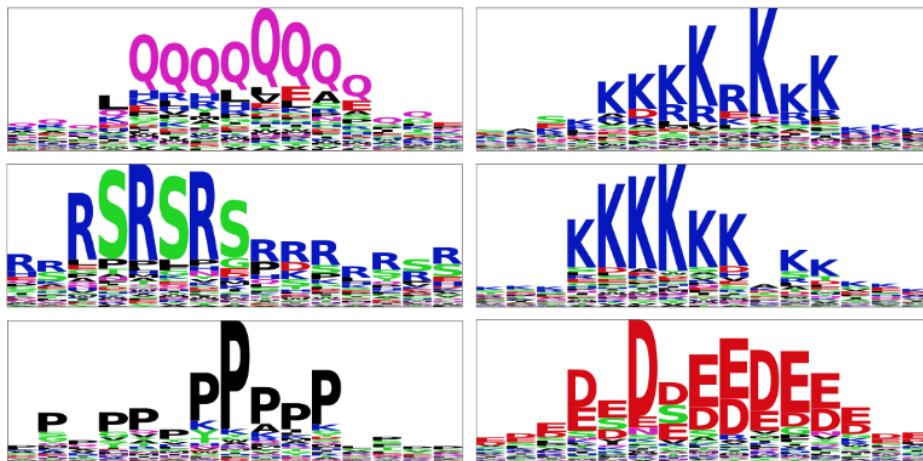


Figure 3.7 Sequence logos of features extracted by the contrastive learning model

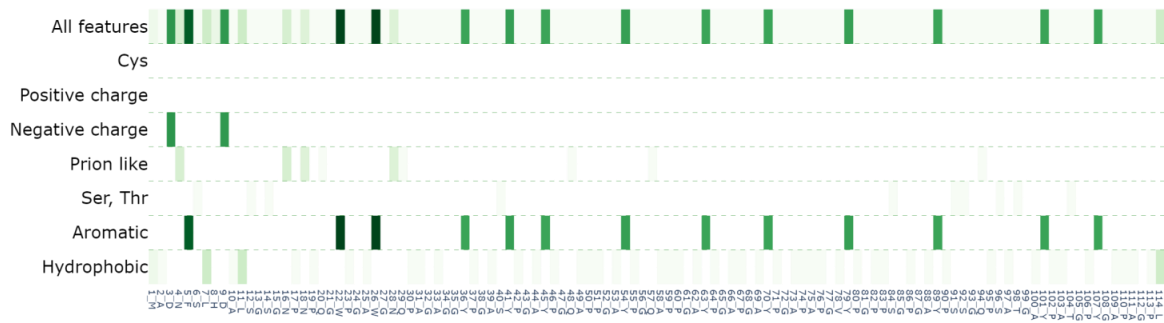
Each sequence logo represents a sequence of features from the final CNN layer (256 neurons in total in Moss and coworker's model architecture). Most obvious patterns are picked as examples by collecting the sequences with the top thirty percent high neuron outputs and using these sequences to create sequence logos.

3.7 Discover the distal features by the self-attention based improved model

To improve that the CNN base encoder only captures regional features, we replaced the CNN with self-attention layers as the base encoder. After training with all human homologs, we can input any sequence and visualize the output through the final attention layer, which tells us what residues the model pays attention to, and these residues may be potentially important sites. **Figure 3.8** is attention maps of galectin-3 IDR of two species, human and zebrafish. We have known that aromatic residues (tyrosine) in human galectin-3 IDR are functionally important for LLPS [12], our model reveals that aromatic residues have higher signals in **Figure 3.8 a**. Other features like negative charge also have higher outputs, which means these residues may be potential functional sites. On the other hand, aromatic residues in human galectin-3 IDR are tyrosine, while in zebrafish are tryptophan, which may as an example that sequences having the similar properties but cannot be identified by sequence alignment (**Figure 3.8 b**), the model also recognizes these aromatic residues may be important. Although the model shows importance through shades of color and other features are less important, we still need to conduct more verification and will discuss this issue later.

Furthermore, whether sequences have been used for training, our model can theoretically input any protein sequence and output an attention map for identifying potential function important sites. We show two other proteins as examples. In hnRNPA1 (**Figure 3.9**), the features are vague because many positions have higher signals, but we can still see some features like positive charge and aromatic residues. In FUS (**Figure 3.10**), aromatic residue and charge are obvious features. In conclusion, our model can discover potential functional important sites through attention map visualization, but the accuracy still needs to be verified by the downstream task (next part) and conduct more literature search to support our results.

(a) human Galectin-3 NTD (0~114)



(b) zebrafish Galectin-3 NTD (0~91)

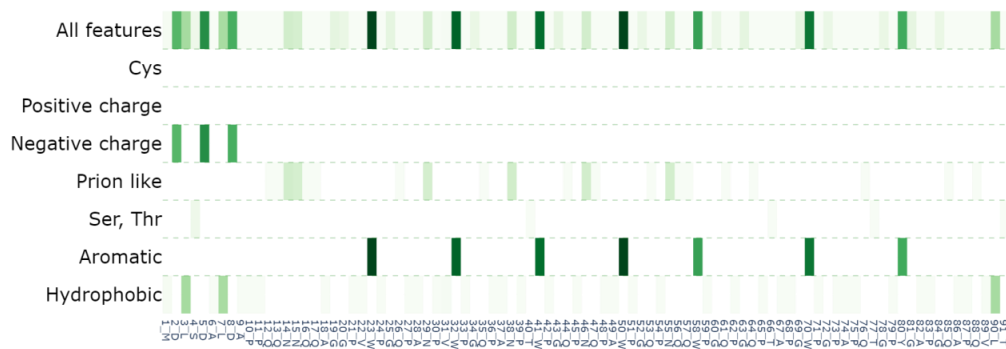


Figure 3.8 Attention map visualization for human and zebrafish galectin-3 IDR

We extracted outputs from the final attention layer as *All features*, deep green means this residue may be important. We separated these features by kinds of protein interaction which often contribute to functions (LLPS). (a) Attention map visualization for human galectin-3 IDR, negative charge, and tyrosine are the most obvious features in predicted outputs. (b) Features in zebrafish galectin-3 IDR are negative charge and tryptophan.

human HNRNPA1 (176~372)

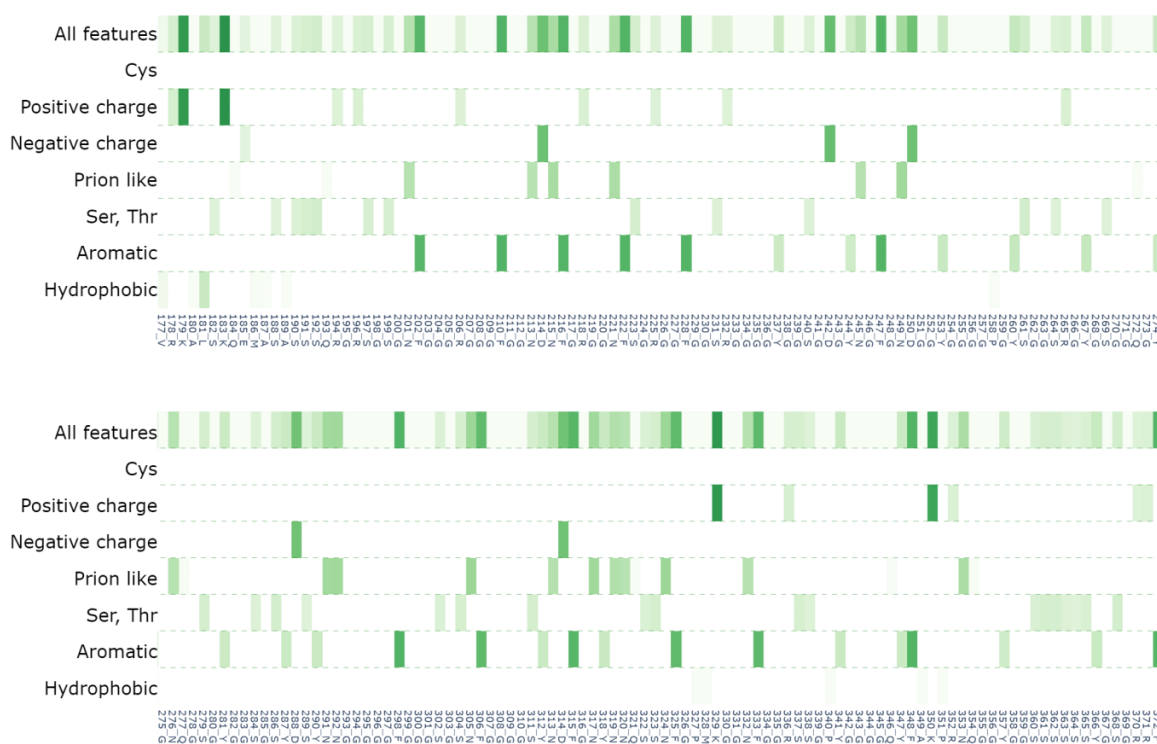


Figure 3.9 Attention map visualization for human hnRNPA1 (176~372)

The features for hnRNPA1 (176~372) are vague because many residues have higher attention outputs. Still, obvious features in hnRNPA1 are aromatic and positive charge residues.

human FUS (1~286)

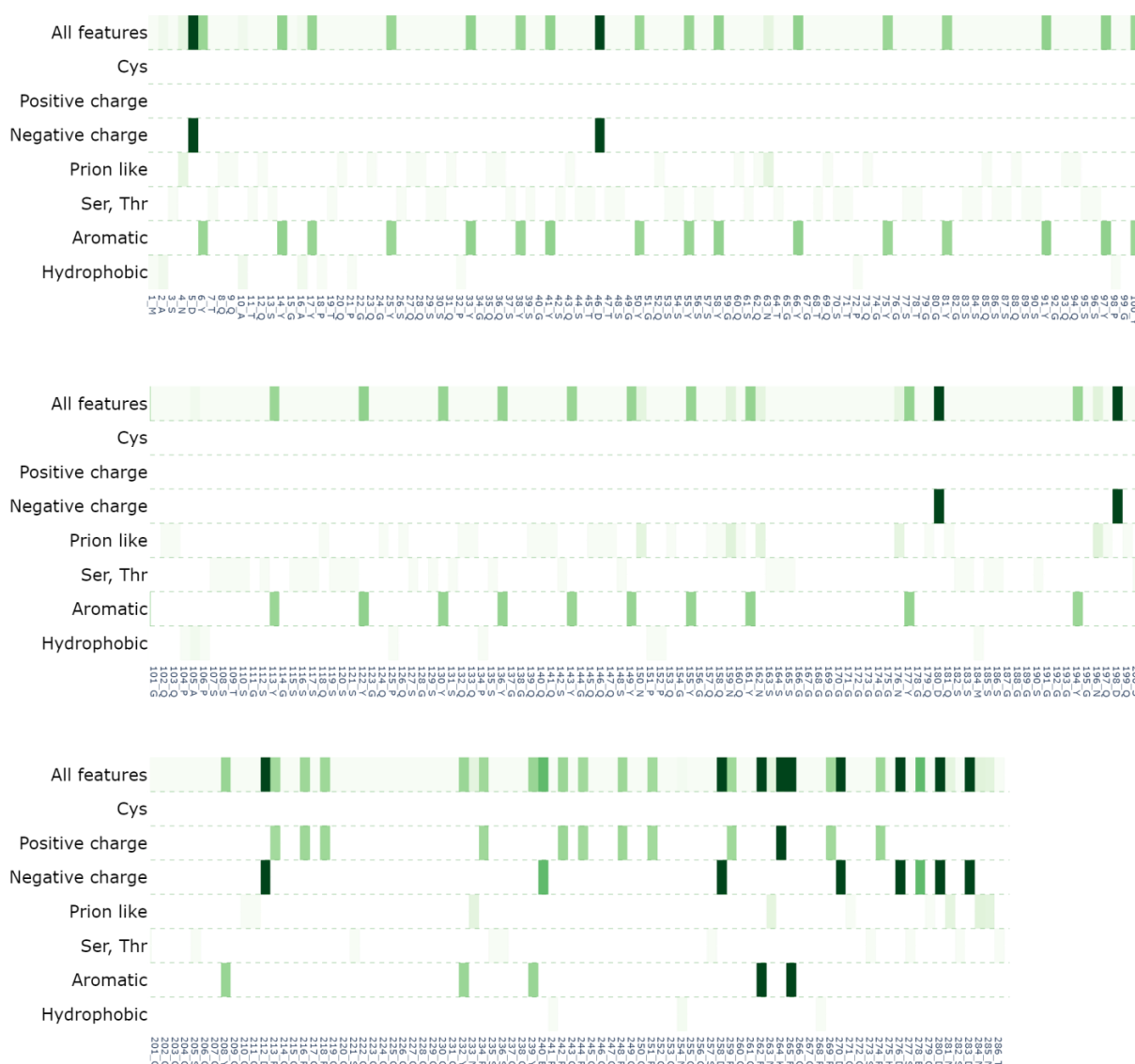


Figure 3.10 Attention map visualization for human FUS (1~286)

In FUS (1~286), the obvious features are negative charge and aromatic residues. The attention outputs of these two features among residues are high with the deep green. For now, however, we can not infer that these features are more important than others. We still need to perform more evaluation tasks to confirm.

3.8 Model performance evaluation by downstream tasks

To better know our model performance, we introduce the LLPS classification downstream task, which has two classes: the label “1” for sequences experiencing LLPS and the label “0” for sequences that are not [85]. By using our self-supervised model to identify sequences belonging to classes, we evaluated model performance by classification performance.

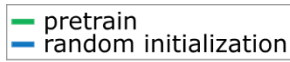
After training the self-supervised model, we only kept the query encoder and used it to create the representation vector for the input sequences from the LLPS classification dataset. The representation vectors were then used as the input to the downstream model and output predict labels. The training step of the downstream model is also called “fine-tune” [46]. **Figure 3.11** and **Figure 3.12** show model performance with different hyperparameters: *embed_dim* (each residue in sequences is embedded to specific dimensions) and *depth* (the number of attention layers).

First, we compared the pretrained model to the baseline model (the self-supervised model without training) by common metrics for classification tasks, including area under the receiver operating characteristic (AUROC), accuracy, and the harmonic mean of precision and recall (F1-score). In **Figure 3.11 b**, three metrics of classification performance per training epochs indicate that the pretrained model (green) has better performance than the model without pretraining (blue). The pretrained model with better metrics means that it learns something during the model pretraining, so it can handle the downstream task better than parameter random initialization.

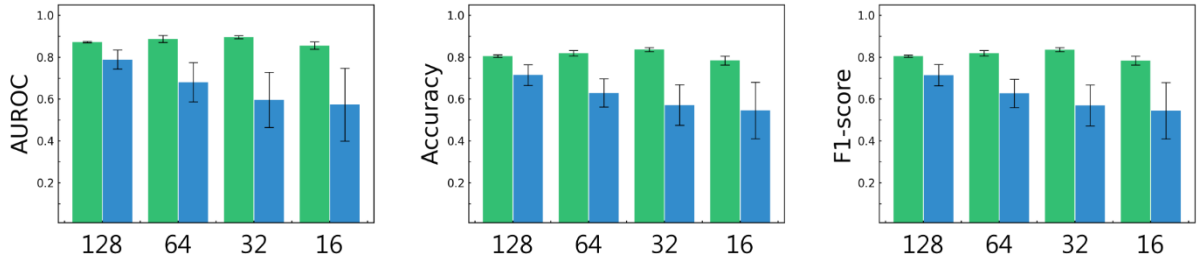
To find the capacity that fits our tasks, we reduced self-supervised model capacity with different hyperparameters including *embed_dim* (**Figure 3.11**) and *depth* (**Figure 3.12**): In hyperparameter *embed_dim*, each residue in sequences is represented by the following

dimensions: 128, 64, 32, and 16. **Figure 3.11 a** is the performance of four conditions, they had similar performance in the pretrain model, with a slight but not significant drop in dimension 16; In hyperparameter *depth*, we reduced the number of attention layers: 6, 3, 2, and 1 (**Figure 3.12 a**). The results were the same as hyperparameter *embed_dim*, even though we reduced *depth*=2, the performance was still similar to *depth*=6. However, in *depth*=1, three metrics dropped significantly, and thus we further examined the performance per training epochs and noticed that three metrics got stuck, suggesting that the model could not learn anything (could not update its parameters) when *depth*=1. In summary, although our model recognizes the potential important sites by attention maps and the results are also consistent with former studies (last part), we still do not know what capacity the model needs. Even though we reduced the model capacity, no significant difference between the tested hyperparameters, and I will discuss the possible reasons and future works to overcome this limit.

陽明交大
NYCU



(a) different *embed_dim*



(b) fine-tuning when *embed_dim*=128

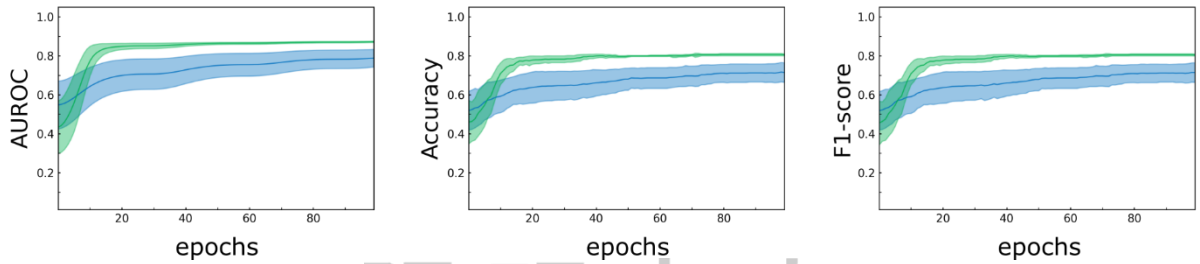
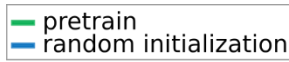


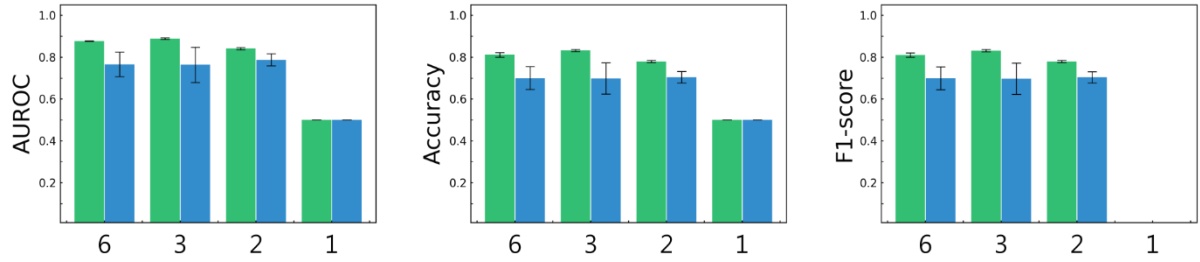
Figure 3.11 Model performance evaluation with different *embed_dim*

(a) We reduced the model capacity and the tested hyperparameters were *embed_dim*. The *embed_dim* is that each residue in sequences is embedded to (represented by) the following dimensions: 128, 64, 32, and 16. In the LLPS downstream tasks, the pretrain model (green) and the model without pretraining (blue) were used to evaluate the pretrain model performances. During the downstream task, both models were fine-tuned by the LLPS classification task and were repeated three times. The decision threshold for the positive and negative labels is derived from the ROC curve where the true positive rate is high and the false positive rate is low. The three metrics indicate that the pretrain model has better performance and low variance than the model without pretraining. But among four different *embed_dim*, their performances are similar.

(b) Model performance per training epochs when *embed_dim*=128, revealing that the pretrain model has better performance and low variance than the model without pretraining.



(a) different *depth*



(b) fine-tuning when *depth*=1

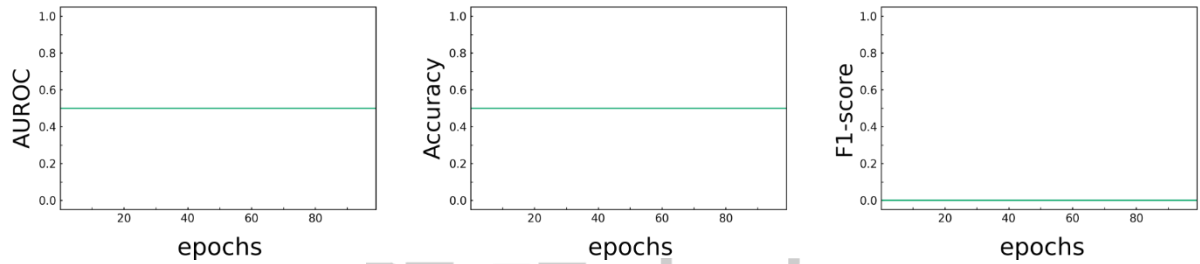


Figure 3.12 Model performance evaluation with different *depth*

(a) Another tested hyperparameter is *depth* (the number of attention layers: 6, 3, 2, and 1). The conditions for the model fine-tuning were the same as different *embed_dim* (fine-tuned by the LLPS classification task and were repeated three times). The results show that except for *depth* =1, the pretrain model (green) has better performance and low variance than the model without pretraining (blue). In the pretrain model, their performances are similar when *depth*=6, 3, and 2. But when *depth*=1, three metrics per training epochs (b) show that both the pretrain model and the model without pretraining got stuck when fine-tuning, meaning that they could not update parameters.

3.9 Clustering of the LLPS dataset

Our model achieves higher accuracy on the LLPS classification downstream task, we further perform representation vector clustering on all sequences in the LLPS dataset. **Figure 3.13** demonstrates principal component analysis (PCA) of representation vectors from outputs of the self-supervised model. Even without any fine-tuning by downstream task, our self-supervised model can roughly demarcate the boundary of the two groups (red dot for sequence experiencing LLPS; and blue dot are non-LLPS). In conclusion, our model makes good representations for input sequences and helps us to find potential important sites through attention maps. However, I currently only perform one downstream task to evaluate model performance and this may not be enough. Also, I still need to test more hyperparameters and do more studies to find out why our model works. I will discuss this later.

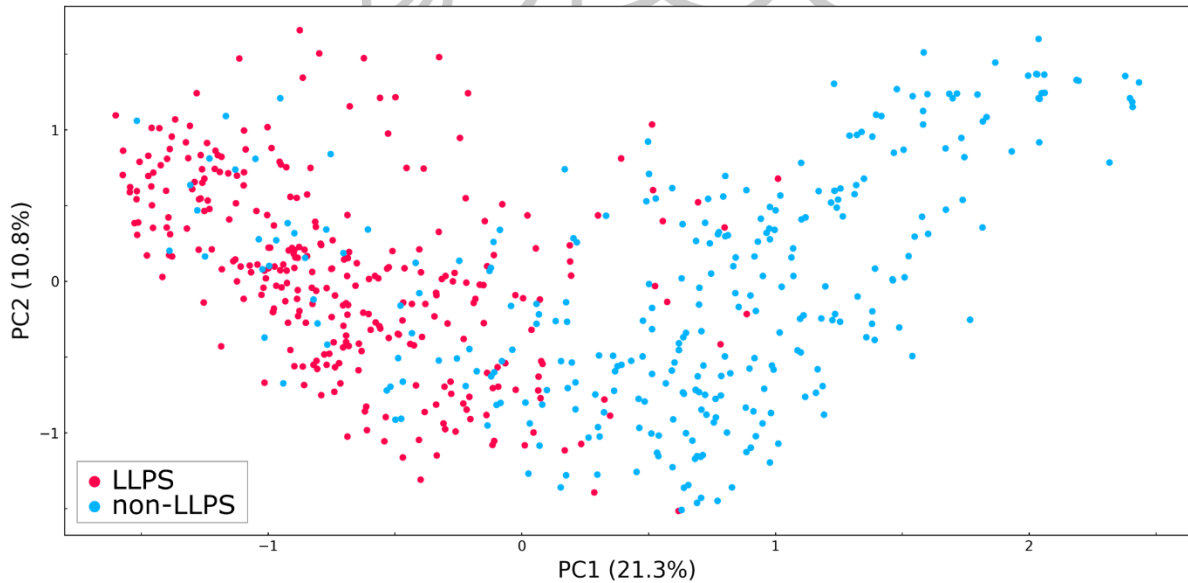


Figure 3.13 PCA of the LLPS dataset

297 sequences are experiencing LLPS (red dot) and 298 sequences are non-LLPS (blue dot) in the LLPS dataset [85]. All sequences are sequentially sent to the based encoder of the self-supervised model to obtain their representation vectors. All representation vectors then conducted dimension reduction by principal component analysis (PCA) to 2 dimensions.

4. Discussion

4.1 IDRs properties in RBPs

With the properties discovering analysis of IDRs (content of IDRs, k -mer patterns, polyaniline propensity, and amino acid composition), some special features were observed: RBPs have many IDRs with more significant in mRBPs compared to the human proteome among three different PONDR algorithms (**Figure 3.1**), and the results are consistent with the former analysis [86].

Some special polypeptides were rich in RBPs (**Figure 3.3**). In 3-mer, most cases had arginine (RXR, XRX, and RRR), studies report that the short RG/RGG motifs contribute to functions like RNA binding [25, 87]. Other cases also relate to functions: glycine-rich domain (GGG) regulates subcellular localization [88]; lysine (KKK) drives LLPS and is enriched in disordered regions [89]. In 5-mer, arginine-serine repeats (SRSRS and RSRSR) are still obvious and relate to regulating mRNA splicing [90]. But in 5-mer and 7-mer, it is hard to have common polypeptides among sequences with longer residues, so the number of special patterns is less. The k -mer analysis could find special polypeptide patterns that are significant compared to the human proteome. However, it is not always relevant between the functional importance and the number of polypeptides. Therefore, by utilizing mer analysis, we find potentially important patterns due to having more numbers of polypeptides in RBPs, but not always correct.

The reason why we investigated polyaniline propensity in RBPs is that polyaniline residues form α -helix, promoting LLPS ability in proteins like TDP-43 [91] and Musashi-1 [92]. We wonder if polyaniline is a common property in RBPs. Our results show that RBPs have a higher polyaniline propensity compared to the human proteome (**Figure 3.4**), meaning that RBPs are more likely to form α -helix. And by using RaptorX, the helix propensity of RBPs also

have a similar trend to polyalanine propensity. Therefore we think that, like other driving forces, polyaniline may be a key factor to promote LLPS, forming biomolecular condensates.

In amino acid composition, aromatic residues are more in RBPs' IDRs among three different PONDR algorithms (**Figure 3.5**). But in the proteome statistics, IDRs always have fewer aromatic residues compared to fold regions [82, 83], mainly because aromatic residues are seen as structure protein stabilizers through the hydrophobic effect [93, 94]. But in RBPs, we observed the opposite results: for some reason, RBPs have more aromatic residues compared to the human proteome in IDRs. Thus, we deduce that these aromatic may play important roles in biological functions, so they reappear in RBPs disorder regions. Still, the number of aromatic residues does not always correlate with bringing important functions. We then make more studies by evolutionary analysis to confirm our thoughts.

4.2 Evolutionary analysis and conserved aromatic residue in RBPs

We analyzed all RBPs' sequence conservation by evolutionary analysis. Although IDRs were less conserved compared to fold regions, some aromatic residues experienced higher conservation levels than other IDRs sites in some proteins (**Figure 3.6**). We then conclude that these aromatic residues are functionally important sites which are highly conserved among IDRs [26, 27]. Our results also consist with the former studies which perform biophysics and point mutation analysis to discover that aromatic residues are the driving force to regulate LLPS: the ability of LLPS in TDP-43 is regulated by the number of tryptophan [11]; the number of tyrosine in FUS's N-terminal collaborate with other residues (glycine, glutamine, and serine) in C-terminal domain to adjust the concentration of biomolecular condensate [22]; nuclear pore complexes (NPCs) form a barrier through liquid droplets to isolate nuclear and cytoplasm. The NPCs are composed of nucleoporins such as NUP153 with phenylalanine and glycine repeats

[95, 96]; and in RNAi, the TNRC6-Ago2 complex is a key factor to induce miRNA gene silencing, and the driving force of complex assembly is LLPS [97]. We also found other proteins with a high aromatic residue conservation score and still have not yet been studied that these sites are functionally important (**Figure 4.1**). These sites in proteins may be potential targets for discovering their properties. Therefore, by using evolutionary analysis, we find potential functional important sites through the high conservation score.

Furthermore, together with the results that aromatic residues are more in RBPs disorder regions, we suggest that aromatic residues in RBPs play important roles, especially LLPS. Even though the aromatic residues in IDRs within the human proteome are fewer, they provide molecular forces like π - π , cation- π , and hydrophobic effects to form biomolecular condensates. This is common property among RBPs, and the aromatic residues reappear in these proteins' IDRs. Therefore, we conclude that aromatic residues have two roles: act as stabilizers in structure proteins and promote LLPS in RBPs.

However, during sequences conservation analysis, we noticed that some IDRs in RBPs are poorly aligned, but their sequences still have similar properties (different repeats of short motifs or different aromatic residues) to IDRs (**Figure 4.2**). This situation causes conservation score to be underestimated for some proteins. To better discover potential important sites and try our best to rule out the influences of misalignment with similar properties, next we try to use machine learning approach that can systematically help us to find potentially important sites in IDRs.

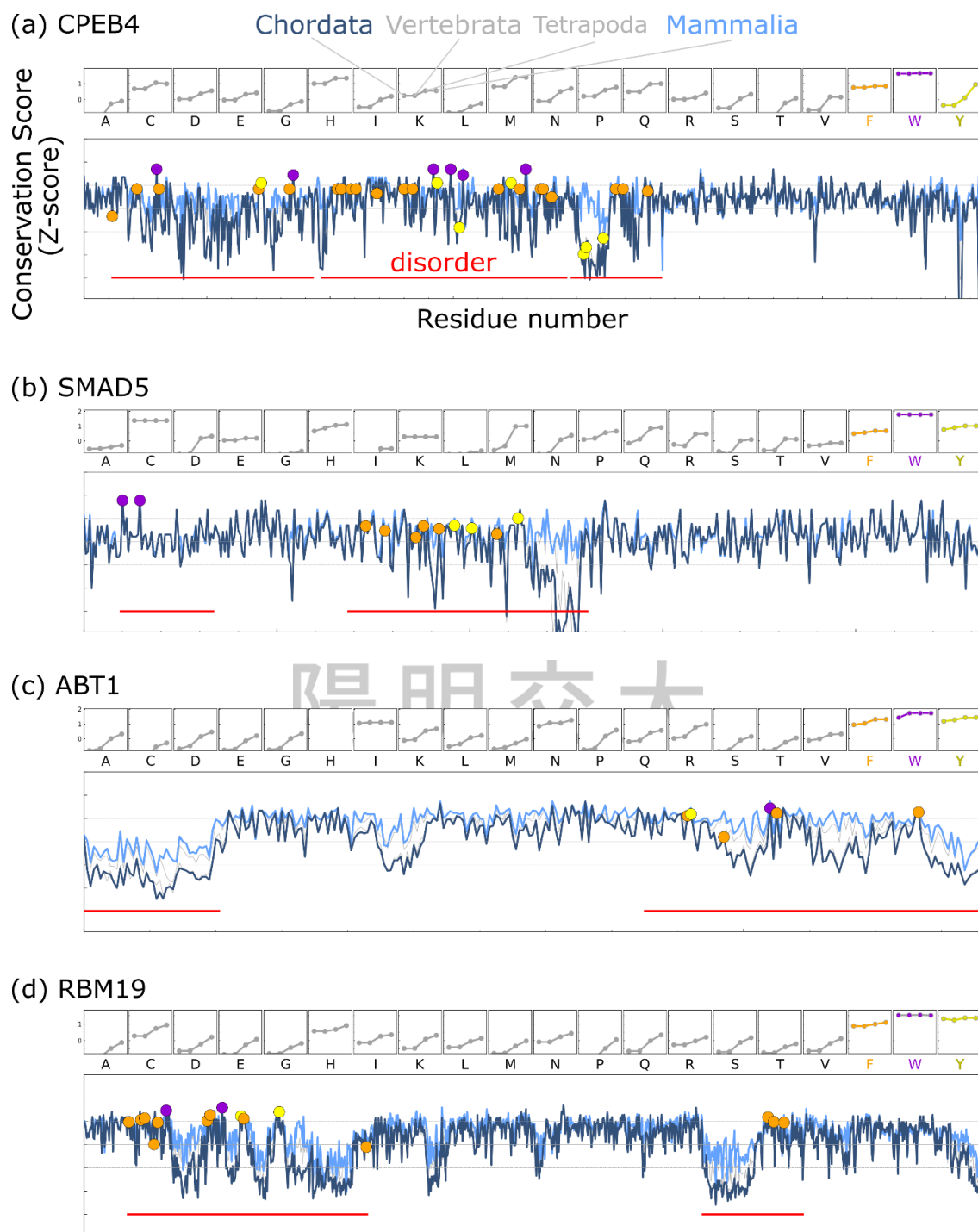
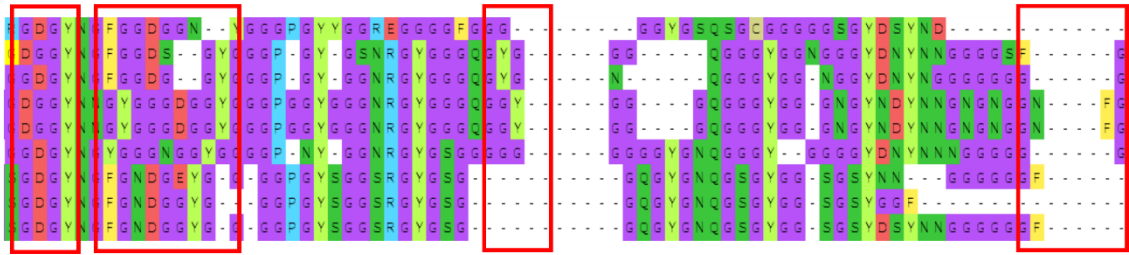


Figure 4.1 Conservation score of proteins with aromatic residues has not been found to be functionally important

Examples show that aromatic residues are highly conserved, while they have not yet been found to be functionally important. UniProt Entry ID of proteins: CPEB4 (Q17RY0), SMAD5 (Q99717), ABT1 (Q9ULW3), and RBM19(Q9Y4C8).

(a) hnRNPA1



(b) FUS

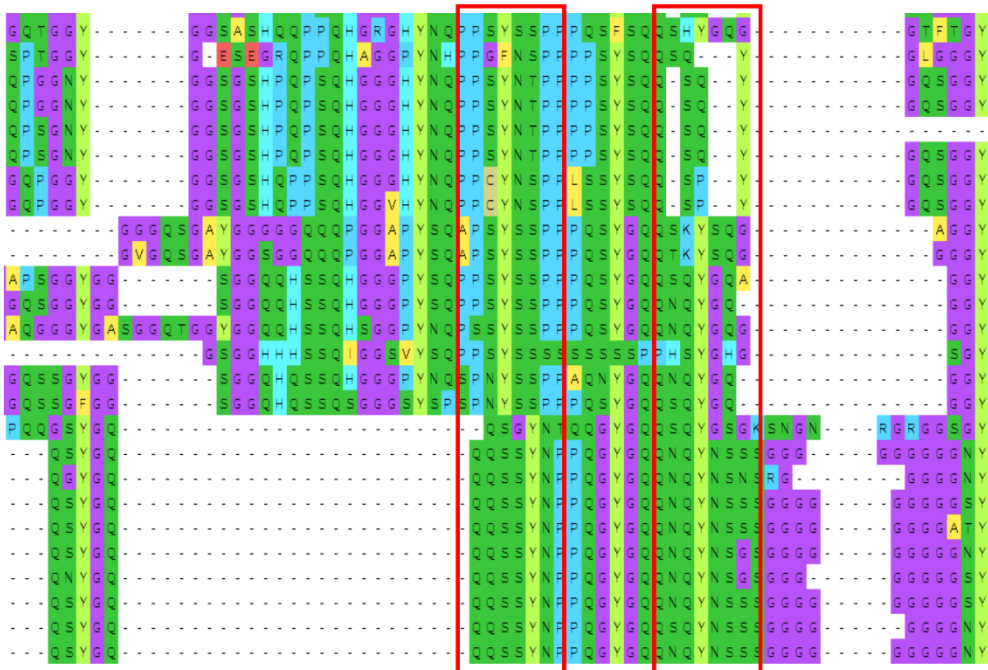


Figure 4.2 Examples show that sequences are misalignment

Some proteins' IDRs are poorly aligned but with similar properties: **(a)** for hnRNPA1, aspartic acid and tyrosine have serious misalignment; **(b)** tyrosine in FUS also shows the same situation. Sequence misalignment causes the conservation score to be underestimated for some proteins.

4.3 Potentially important sites in IDRs by self-supervised model

Our aim in this thesis is to systemically identify potentially important sites. For now, we have nearly achieved this goal, our self-supervised model trained by phylogenetic augmentations could determine potentially important sites by visualizing attention maps.

Through attention map, in human galectin-3 (**Figure 3.8 a**), our model shows that aromatic residues and negative charge residues are potentially important sites. Former research from our lab has found that LLPS ability in galectin-3 is correlated with the number of aromatic residues [12]. For negative charge residues, my classmate in our lab has preliminary data revealing that they also regulate LLPS ability verified by biophysics experiments. In zebrafish galectin-3 (**Figure 3.8 b**), the main difference between zebrafish and other species is that aromatic residues in zebrafish are tryptophan, while in other species are tyrosine [12]. Our model suggests that these tryptophans may be potentially important sites. Still, there is no direct evidence that these sites are functionally important. And these sites may be potential targets to discover their importance.

Our model can discover potentially important sites, and for now, the results are consistent with the former studies. For example, hnRNPA1 (**Figure 3.9**) has positively charged and aromatic residues, in agreement with previous studies that hnRNPA1 undergoes LLPS through these two features by molecular forces such as electrostatic and hydrophobic interactions [98]; FUS (**Figure 3.10**) has positively charged and aromatic residues, and former studies found that N-terminal aromatic residues interact with C-terminal arginines to form cation- π interactions [21]

However, during model performance evaluation by LLPS classification downstream task, some issues need to be resolved: First, for a self-supervised model, it is necessary to apply

supervised downstream tasks to determine the model's performance such as accuracy. For now, we only performed a downstream task (proteins experience LLPS classification or not [85]), and around only 500 sequences in these tasks. The small dataset may be the reason why the models have the similar performance when we reduced the model capacity by changing hyperparameters (*embed_dim* and *depth*), because this downstream task may be too "easy" to discover the difference between the reduced capacity models, leading to misgauging the model performance; On the other hand, our model's outputs (attention maps) show the importance for each residue through shades of color, because the downstream task we used may not be able to evaluate our model properly, these levels of importance showed by color still need to be verified by conducting more downstream tasks (e.g. bigger dataset or more complex tasks) and having more direct evidence (biophysics experiments) from others research to support our model's results; Furthermore, we still do not know what the capacity that our model needs, that is why we conducted performance evaluation for the reduced capacity model. But our results done by the LLPS classification task for now may not so reliable because even though we reduced the capacity, the accuracy among models with different capacity (different hyperparameters) shows no significant difference. After applying more complex downstream tasks, we will attempt to answer this question.

4.4 Future works

4.4.1 Applying more downstream tasks

To better evaluate our model and see if the model capacity is appropriate, we plan to apply other datasets which have more sequences for LLPS classification. For example, recent research collected many properties (evolutionary, physicochemical, and structural properties) of known proteins that experience LLPS [99]. Besides LLPS classification tasks, we want to conduct other tasks, but even though many studies focus on machine learning studies for IDRs, their goal is mainly disorder regions prediction [42], not potential important sites discovering that we want to achieve, so to find other downstream tasks to meet our requirements is the next challenge to overcome.

4.4.2 Verifying the levels of importance for residues in a sequence

If we have suitable downstream tasks and perform more literature searches to confirm our model's results, we want to quantify the importance of each residue in sequences by attention maps. In this part, our goal is not only to know what sites are important but also the levels of importance (some features may be more important than others).

4.4.3 Training our models with other homologs

For now, we only trained our self-supervised model with 19,909 human homologs from OMA, we plan to introduce homologs from all of the other species (about 1 million homologs with 18 million proteins [68]). The bias of the model could be eliminated with the number of the dataset [100]. We still need to confirm whether the model performance is better.

4.4.4 Future aims: to build a platform for the public

We are now trying our best to improve the model, our final goal for the self-supervised learning project is to build a platform: by entering any protein sequences, our model outputs potential important sites with higher confidence. We expect that our study will help others to find out more properties of protein IDRs.

4.5 Summary

In this thesis, we found that IDRs in RBPs have some unique properties: RBPs have more IDRs, special short motifs are rich in RBPs, and aromatic residues in RBPs' IDRs are more compared to the human proteome. By applying evolutionary analysis, we found that aromatic residues are highly conserved, meaning that they are functionally important sites with others biophysics studies that support our thought. We conclude that aromatic residues have two roles: act as stabilizers in structure proteins and promote LLPS in RBPs.

By using a self-supervised machine learning approach, we built a model that could discover potential important sites for IDRs which may bring important functions, and we are now improving our model by conducting more downstream tasks and investigating more research to support our results.

5. Reference

1. Petsko GA, Ringe D. Protein structure and function. New Science Press; 2004.
2. Creighton TE. Proteins: structures and molecular properties. Macmillan; 1993.
3. Chouard T. Structural biology: Breaking the protein rules. *Nature*. 2011;471:151–3.
4. Uversky VN. Unusual biophysics of intrinsically disordered proteins. *Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics*. 2013;1834:932–51.
5. Deiana A, Forcelloni S, Porrello A, Giansanti A. Intrinsically disordered proteins and structured proteins with intrinsically disordered regions have different functional roles in the cell. :16.
6. Xue B, Dunker AK, Uversky VN. Orderly order in protein intrinsic disorder distribution: disorder in 3500 proteomes from viruses and the three domains of life. *Journal of Biomolecular Structure and Dynamics*. 2012;30:137–49.
7. Basile W, Salvatore M, Bassot C, Elofsson A. Why do eukaryotic proteins contain more intrinsically disordered regions? *PLoS Comput Biol*. 2019;15:e1007186.
8. Wright PE, Dyson HJ. Intrinsically disordered proteins in cellular signalling and regulation. *Nat Rev Mol Cell Biol*. 2015;16:18–29.
9. Huang Q, Li M, Lai L, Liu Z. Allostery of multidomain proteins with disordered linkers. *Current Opinion in Structural Biology*. 2020;62:175–82.
10. Yang P. G3BP1 Is a Tunable Switch that Triggers Phase Separation to Assemble Stress Granules. :50.
11. Li H-R, Chiang W-C, Chou P-C, Wang W-J, Huang J. TAR DNA-binding protein 43 (TDP-43) liquid–liquid phase separation is mediated by just a few aromatic residues. *Journal of Biological Chemistry*. 2018;293:6090–8.
12. Chiu Y-P, Sun Y-C, Qiu D-C, Lin Y-H, Chen Y-Q, Kuo J-C, et al. Liquid-liquid phase separation and extracellular multivalent interactions in the tale of galectin-3. *Nat Commun*. 2020;11:1229.
13. Banani SF, Lee HO, Hyman AA, Rosen MK. Biomolecular condensates: organizers of cellular biochemistry. *Nat Rev Mol Cell Biol*. 2017;18:285–98.
14. Hentze MW, Castello A, Schwarzl T, Preiss T. A brave new world of RNA-binding proteins.

- Nat Rev Mol Cell Biol. 2018;19:327–41.
15. Boeynaems S, Alberti S, Fawzi NL, Mittag T, Polymenidou M, Rousseau F, et al. Protein Phase Separation: A New Phase in Cell Biology. *Trends in Cell Biology*. 2018;28:420–35.
 16. Protter DSW, Parker R. Principles and Properties of Stress Granules. *Trends in Cell Biology*. 2016;26:668–79.
 17. Luo Y-Y, Wu J-J, Li Y-M. Regulation of liquid–liquid phase separation with focus on post-translational modifications. *Chem Commun*. 2021;57:13275–87.
 18. Uversky VN. Intrinsically Disordered Proteins and Their “Mysterious” (Meta)Physics. *Frontiers in Physics*. 2019;7.
 19. Brangwynne CP, Tompa P, Pappu RV. Polymer physics of intracellular phase transitions. *Nature Phys*. 2015;11:899–904.
 20. Martin EW, Holehouse AS, Peran I, Farag M, Incicco JJ, Bremer A, et al. Valence and patterning of aromatic residues determine the phase behavior of prion-like domains. *Science*. 2020;367:694–9.
 21. Qamar S, Wang G, Randle SJ, Ruggeri FS, Varela JA, Lin JQ, et al. FUS Phase Separation Is Modulated by a Molecular Chaperone and Methylation of Arginine Cation- π Interactions. *Cell*. 2018;173:720-734.e15.
 22. Wang J, Choi J-M, Holehouse AS, Lee HO, Zhang X, Jahnel M, et al. A Molecular Grammar Governing the Driving Forces for Phase Separation of Prion-like RNA Binding Proteins. *Cell*. 2018;174:688-699.e16.
 23. Greig JA, Nguyen TA, Lee M, Holehouse AS, Posey AE, Pappu RV, et al. Arginine-Enriched Mixed-Charge Domains Provide Cohesion for Nuclear Speckle Condensation. *Molecular Cell*. 2020;77:1237-1250.e4.
 24. Yang J, Zeng Y, Liu Y, Gao M, Liu S, Su Z, et al. Electrostatic interactions in molecular recognition of intrinsically disordered proteins. *Journal of Biomolecular Structure and Dynamics*. 2020;38:4883–94.
 25. Chong PA, Vernon RM, Forman-Kay JD. RGG/RG Motif Regions in RNA Binding and Phase Separation. *Journal of Molecular Biology*. 2018;430:4650–65.
 26. Capra JA, Singh M. Predicting functionally important residues from sequence conservation. *Bioinformatics*. 2007;23:1875–82.

27. Brown CJ. Evolution and disorder. *Current Opinion in Structural Biology*. 2011;:6.
28. Zarin T, Strome B, Nguyen Ba AN, Alberti S, Forman-Kay JD, Moses AM. Proteome-wide signatures of function in highly diverged intrinsically disordered regions. *eLife*. 2019;8:e46883.
29. Moesa HA, Wakabayashi S, Nakai K, Patil A. Chemical composition is maintained in poorly conserved intrinsically disordered regions and suggests a means for their classification. *Mol BioSyst*. 2012;8:3262.
30. Wiedner HJ, Giudice J. It's not just a phase: function and characteristics of RNA-binding proteins in phase separation. *Nat Struct Mol Biol*. 2021;28:465–73.
31. Gebauer F, Schwarzl T, Valcárcel J, Hentze MW. RNA-binding proteins in human genetic disease. *Nat Rev Genet*. 2021;22:185–98.
32. Zeke A, Schád É, Horváth T, Abukhairan R, Szabó B, Tantos A. Deep structural insights into RNA-binding disordered protein regions. *WIREs RNA*. n/a n/a:e1714.
33. Chen H-Y, Lin L-T, Wang M-L, Tsai K-L, Huang P-I, Yang Y-P, et al. Musashi-1 promotes chemoresistant granule formation by PKR/eIF2 α signalling cascade in refractory glioblastoma. *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease*. 2018;1864:1850–61.
34. Khalfallah Y, Kuta R, Grasmuck C, Prat A, Durham HD, Vande Velde C. TDP-43 regulation of stress granule dynamics in neurodegenerative disease-relevant cell types. *Sci Rep*. 2018;8:7551.
35. Wolozin B, Ivanov P. Stress granules and neurodegeneration. *Nat Rev Neurosci*. 2019;20:649–66.
36. Jo M, Lee S, Jeon Y-M, Kim S, Kwon Y, Kim H-J. The role of TDP-43 propagation in neurodegenerative diseases: integrating insights from clinical and experimental studies. *Exp Mol Med*. 2020;52:1652–62.
37. Prasad A, Bharathi V, Sivalingam V, Girdhar A, Patel BK. Molecular Mechanisms of TDP-43 Misfolding and Pathology in Amyotrophic Lateral Sclerosis. *Front Mol Neurosci*. 2019;12:25.
38. Dhawale CA, Dhawale K, Dubey R. A Review on Deep Learning Applications: In: Thomas JJ, Karagoz P, Ahamed BB, Vasant P, editors. *Advances in Systems Analysis, Software Engineering, and High Performance Computing*. IGI Global; 2020. p. 21–31.

39. Bonetta R, Valentino G. Machine learning techniques for protein function prediction. *Proteins*. 2020;88:397–413.
40. Cui Y, Dong Q, Hong D, Wang X. Predicting protein-ligand binding residues with deep convolutional neural networks. *BMC Bioinformatics*. 2019;20:93.
41. Repecka D, Jauniskis V, Karpus L, Rembeza E, Rokaitis I, Zrimec J, et al. Expanding functional protein sequence spaces using generative adversarial networks. *Nat Mach Intell*. 2021;3:324–33.
42. CAID Predictors, DisProt Curators, Necci M, Piovesan D, Tosatto SCE. Critical assessment of protein intrinsic disorder prediction. *Nat Methods*. 2021;18:472–81.
43. Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021;596:583–9.
44. Tu Y, Lei H, Shen H-B, Yang Y. SIFLoc: a self-supervised pre-training method for enhancing the recognition of protein subcellular localization in immunofluorescence microscopic images. *Briefings in Bioinformatics*. 2022;:bbab605.
45. Jing L, Tian Y. Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey. *arXiv:190206162 [cs]*. 2019.
46. Jaiswal A, Babu AR, Zadeh MZ, Banerjee D, Makedon F. A Survey on Contrastive Self-supervised Learning. *arXiv:201100362 [cs]*. 2021.
47. Chen T, Kornblith S, Norouzi M, Hinton G. A Simple Framework for Contrastive Learning of Visual Representations. *arXiv:200205709 [cs, stat]*. 2020.
48. He K, Fan H, Wu Y, Xie S, Girshick R. Momentum Contrast for Unsupervised Visual Representation Learning. *arXiv:191105722 [cs]*. 2020.
49. Chen X, Fan H, Girshick R, He K. Improved Baselines with Momentum Contrastive Learning. *arXiv:200304297 [cs]*. 2020.
50. Chen X, Xie S, He K. An Empirical Study of Training Self-Supervised Vision Transformers. *arXiv:210402057 [cs]*. 2021.
51. Lu AX, Lu AX, Moses A. Evolution Is All You Need: Phylogenetic Augmentation for Contrastive Learning. *arXiv:201213475 [cs, q-bio]*. 2020.
52. Lu AX, Lu AX, Pritisanac I, Zarin T, Forman-Kay JD, Moses AM. Discovering molecular features of intrinsically disordered regions by using evolution for contrastive learning.

- preprint. Bioinformatics; 2021.
53. Van Rossum G, Drake FL. Python 3 Reference Manual. Scotts Valley, CA: CreateSpace; 2009.
 54. Cock PJ, Antao T, Chang JT, Chapman BA, Cox CJ, Dalke A, et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*. 2009;25:1422–3.
 55. Comet.ML. Comet.ML home page. 2021. <https://www.comet.ml/>.
 56. jellyfish · PyPI. <https://pypi.org/project/jellyfish/0.9.0/>. Accessed 31 May 2022.
 57. Hunter JD. Matplotlib: A 2D graphics environment. *Computing in science & engineering*. 2007;9:90–5.
 58. Harris CR, Millman KJ, van der Walt SJ, Gommers R, Virtanen P, Cournapeau D, et al. Array programming with NumPy. *Nature*. 2020;585:357–62.
 59. McKinney W, others. Data structures for statistical computing in python. In: *Proceedings of the 9th Python in Science Conference*. Austin, TX; 2010. p. 51–6.
 60. Inc PT. Collaborative data science. 2015. <https://plot.ly>.
 61. Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. *arXiv*; 2019.
 62. Chandra RV, Varanasi BS. Python requests essentials. Packt Publishing Ltd; 2015.
 63. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine learning in Python. *Journal of machine learning research*. 2011;12 Oct:2825–30.
 64. selenium. 2021. <https://www.selenium.dev>. Accessed 30 May 2022.
 65. Costa-Luis C da, Larroque SK, Altendorf K, Mary H, richardsheridan, Korobov M, et al. tqdm: A fast, Extensible Progress Bar for Python and CLI. 2022.
 66. lyricpoem. Visualizer. 2022.
 67. Gerstberger S, Hafner M, Tuschl T. A census of human RNA-binding proteins. *Nat Rev Genet*. 2014;15:829–45.
 68. Altenhoff AM, Train C-M, Gilbert KJ, Mediratta I, Mendes de Farias T, Moi D, et al. OMA orthology in 2021: website overhaul, conserved isoforms, ancestral gene order and more. *Nucleic Acids Research*. 2021;49:D373–9.

69. Nomenclature and symbolism for amino acids and peptides (Recommendations 1983). *Pure and Applied Chemistry*. 1984;56:595–624.
70. Obradovic Z, Peng K, Vucetic S, Radivojac P, Brown CJ, Dunker AK. Predicting intrinsic disorder from amino acid sequence. *Proteins*. 2003;53:566–72.
71. Pace CN, Scholtz JM. A helix propensity scale based on experimental studies of peptides and proteins. *Biophys J*. 1998;75:422–7.
72. Wang S, Li W, Liu S, Xu J. RaptorX-Property: a web server for protein structure property prediction. *Nucleic Acids Research*. 2016;44:W430–5.
73. Sievers F, Higgins DG. Clustal Omega for making accurate alignments of many protein sequences. *Protein Science*. 2018;27:135–45.
74. Levenshtein VI. Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Union*; 1966. p. 707–10.
75. Devlin J, Chang M-W, Lee K, Toutanova K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. 2019.
76. Greener JG, Kandathil SM, Moffat L, Jones DT. A guide to machine learning for biologists. *Nat Rev Mol Cell Biol*. 2022;23:40–55.
77. Loshchilov I, Hutter F. Decoupled Weight Decay Regularization. *arXiv*; 2019.
78. Loshchilov I, Hutter F. SGDR: Stochastic Gradient Descent with Warm Restarts. *arXiv*; 2017.
79. Katsura N. Cosine Annealing with Warmup for PyTorch. 2022.
80. Oord A van den, Li Y, Vinyals O. Representation Learning with Contrastive Predictive Coding. *arXiv*; 2019.
81. Kuprieiev R, Pachhai S, Petrov D, Redzyński P, Costa-Luis C da, Rowlands (변기호) P, et al. DVC: Data Version Control - Git for Data & Models. 2022.
82. Radivojac P, Iakoucheva LM, Oldfield CJ, Obradovic Z, Uversky VN, Dunker AK. Intrinsic Disorder and Functional Proteomics. *Biophysical Journal*. 2007;92:1439–56.
83. Radivojac P. Protein flexibility and intrinsic disorder. *Protein Science*. 2004;13:71–80.
84. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention Is All You Need. *arXiv*; 2017.

85. Saar KL, Morgunov AS, Qi R, Arter WE, Krainer G, Lee AA, et al. Learning the molecular grammar of protein condensates from sequence determinants and embeddings. *Proc Natl Acad Sci USA*. 2021;118:e2019053118.
86. Zhao B, Katuwawala A, Oldfield CJ, Hu G, Wu Z, Uversky VN, et al. Intrinsic Disorder in Human RNA-Binding Proteins. *Journal of Molecular Biology*. 2021;433:167229.
87. Bayer TS, Booth LN, Knudsen SM, Ellington AD. Arginine-rich motifs present multiple interfaces for specific binding by RNA. *RNA*. 2005;11:1848–57.
88. The glycine arginine-rich domain of the RNA-binding protein nucleolin regulates its subcellular localization. *The EMBO Journal*. 2021;40:e107158.
89. Ukmar-Godec T, Hutten S, Grieshop MP, Rezaei-Ghaleh N, Cima-Omori M-S, Biernat J, et al. Lysine/RNA-interactions drive and regulate biomolecular condensation. *Nat Commun*. 2019;10:2909.
90. Järvelin AI, Noerenberg M, Davis I, Castello A. The new (dis)order in RNA regulation. *Cell Commun Signal*. 2016;14:9.
91. Conicella AE, Zerze GH, Mittal J, Fawzi NL. ALS Mutations Disrupt Phase Separation Mediated by α -Helical Structure in the TDP-43 Low-Complexity C-Terminal Domain. *Structure*. 2016;24:1537–49.
92. Chiu S-H, Ho W-L, Sun Y-C, Kuo J-C, Huang J. Phase separation driven by interchangeable properties in the intrinsically disordered regions of protein paralogs. *Commun Biol*. 2022;5:1–12.
93. Madhusudan Makwana K, Mahalakshmi R. Implications of aromatic–aromatic interactions: From protein structures to peptide models. *Protein Sci*. 2015;24:1920–33.
94. Hong H, Park S, Flores Jiménez RH, Rinehart D, Tamm LK. Role of Aromatic Side Chains in the Folding and Thermodynamic Stability of Integral Membrane Proteins. *J Am Chem Soc*. 2007;129:8320–7.
95. Dormann D. FG-nucleoporins caught in the act of liquid–liquid phase separation. *Journal of Cell Biology*. 2020;219:e201910211.
96. Bastos R, Lin A, Enarson M, Burke B. Targeting and function in mRNA export of nuclear pore complex protein Nup153. *Journal of Cell Biology*. 1996;134:1141–56.
97. Sheu-Gruttadauria J, MacRae IJ. Phase Transitions in the Assembly and Function of Human miRISC. *Cell*. 2018;173:946-957.e16.

98. Molliex A, Temirov J, Lee J, Coughlin M, Kanagaraj AP, Kim HJ, et al. Phase Separation by Low Complexity Domains Promotes Stress Granule Assembly and Drives Pathological Fibrillization. *Cell*. 2015;163:123–33.
99. Badaczewska-Dawid AE, Uversky VN, Potoyan DA. BIAPSS: A Comprehensive Physicochemical Analyzer of Proteins Undergoing Liquid–Liquid Phase Separation. *IJMS*. 2022;23:6204.
100. Le-Khac PH, Healy G, Smeaton AF. Contrastive Representation Learning: A Framework and Review. *IEEE Access*. 2020;8:193907–34.

陽明交大
NYCU

6. Appendix

6.1 Publications

- **Ho WL** and Huang JR. "The return of the rings: Evolutionary convergence of aromatic residues in the intrinsically disordered regions of RNA-binding proteins for liquid–liquid phase separation." **(2022) Prot Sci 31:e4317**
- Chiu SH, **Ho WL**, Sun YC, Kuo JC, and Huang JR. "Phase separation driven by interchangeable properties in the intrinsically disordered regions of protein paralogs" **(2022) Comm. Biol. 5:400**

陽明交大
NYCU

The return of the rings: Evolutionary convergence of aromatic residues in the intrinsically disordered regions of RNA-binding proteins for liquid–liquid phase separation

Wen-Lin Ho¹ | Jie-rong Huang^{1,2,3} 

¹Institute of Biochemistry and Molecular Biology, National Yang Ming Chiao Tung University, Taipei, Taiwan

²Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, Taipei, Taiwan

³Department of Life Sciences and Institute of Genome Sciences, National Yang Ming Chiao Tung University, Taipei, Taiwan

Correspondence

Jie-rong Huang, Institute of Biochemistry and Molecular Biology, National Yang Ming Chiao Tung University,
No. 155 Section 2, Li-Nong Street, Taipei, Taiwan.

Email: jierongh@nycu.edu.tw

Funding information

The Ministry of Science and Technology of Taiwan, Grant/Award Numbers: 109-2113-M-010-003, 110-2113-M-A49A-504-MY3

Review editor: Nir Ben-Tal

Abstract

Aromatic residues appeared relatively late in the evolution of protein sequences to stabilize the globular proteins' folding core and are less in the intrinsically disordered regions (IDRs). Recent advances in protein liquid–liquid phase separation (LLPS) studies have also shown that aromatic residues in IDRs often act as “stickers” to promote multivalent interactions in forming higher-order oligomers. To study how general these structure-promoting residues are in IDRs, we compared levels of sequence disorder in RNA binding proteins (RBPs), which are often found to undergo LLPS, and the human proteome. We found that aromatic residues appear more frequently than expected in the IDRs of RBPs and, through multiple sequence alignment analysis, those aromatic residues are often conserved among chordates. Using TDP-43, FUS, and some other well-studied LLPS proteins as examples, the conserved aromatic residues are important to their LLPS-related functions. These analyses suggest that aromatic residues may have contributed twice to evolution: stabilizing structured proteins and assembling biomolecular condensates.

KEYWORDS

biomolecular condensates, intrinsically disordered proteins, liquid–liquid phase separation, membraneless organelle, RNA-binding proteins

Amino acids with aromatic rings stabilize protein structure¹ and are accordingly rare in intrinsically disordered regions (IDRs).^{2,3} Interestingly, however, aromatic residues in IDRs have recently been identified as being crucial in mediating liquid–liquid phase separation (LLPS).^{4–7} Here, through protein sequence analyses, we

outline how aromatic residues may have contributed twice to evolution.

How did proteins emerge on Earth? Eck and Dayhoff proposed that early polypeptides were short, with simple compositions, but extended constantly by duplicating their sequences.⁸ In their ferredoxin example, new amino acids appeared by mutation, and the addition of cysteine provided sulfide bonding to ferrous sulfide, a catalyst used as a primitive energy source.⁸ The more lately incorporated amino acids, including hydrophobic and aromatic residues, enhanced the folding stability of these polypeptides.¹ These polypeptides, or proteins, became the main workhorses of early cells. When lifeforms became more complex with the appearance of eukaryotes

Short statements for broader audience: Aromatic residues appear frequently in the core of folded proteins and are relatively rare in the intrinsically disordered regions (IDRs). However, many IDRs that mediate liquid–liquid phase separation (LLPS) have aromatic residues that are important for their functional assembly. Focusing on RNA-binding proteins (many undergo LLPS), we show that aromatic residues have contributed twice to evolution: first, in stabilizing structured protein, and later, in facilitating LLPS in IDRs.

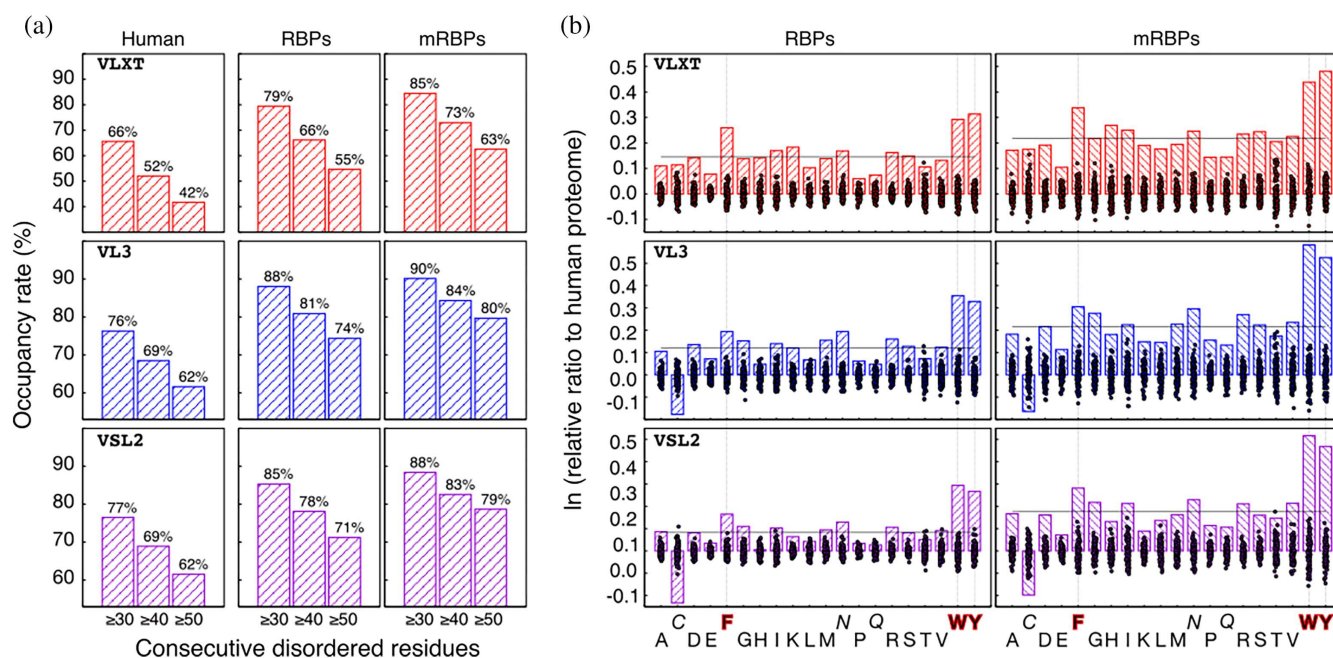


FIGURE 1 Prevalence of disorder and disorder odds ratios relative to the human proteome by amino acid type for RNA-binding proteins. (a) Proportion of proteins with disordered regions longer than 30, 40, or 50 consecutive residues, as predicted using different algorithms, in the human proteome (left column), RNA-binding proteins (middle column), and mRNA binding proteins (right column). (b) Log-odds ratios relative to the human proteome of being in an intrinsically disordered region for amino acids in RNA binding proteins (RBPs) (left) and mRBPs (right). The dashed lines indicate the average value over all amino acid types. The dots represent the values obtained for randomly selected (negative control) subsamples ($N = 1,542$ for RBPs and $N = 689$ for mRBP, the same numbers as considered in the main analysis) of the human proteome

and multicellular organisms, proteins became more efficient and “moonlighted” in different functions.⁹ The acquisition of IDRs substantially enlarged their functional repertoire.¹⁰ Among many functional advantages,¹¹ IDRs’ tunable and reversible assembly, known as LLPS, has recently been recognized as the mechanism that drives the formation of membraneless organelles.¹²

The building blocks of proteins also appeared at different stages of evolution. Eck and Dayhoff found, for example, that the earliest four amino acids to appear were Ala, Asp, Ser, and Gly.⁸ Based on multiple criteria,¹³ the order of appearance of the amino acids in life is now accepted to be: Gly, Ala, Asp, Val, Pro, Ser, Glu, Leu/Thr, Arg, Ile/Gln/Asn, His, Lys, Cys, Phe, Tyr, Met, Trp. Interestingly, as many others have noticed, this suggests that the earliest peptides were disordered because the most primitive amino acids are not structure-promoting.^{14–16} These primitive disordered proteins may have initiated life on Earth and acquired aromatic residues later, thereby increasing their folding stability. The initial set of amino acids was implicated again in the evolution of IDRs for various functions, as these disordered regions were “reinvented.”¹⁷ Although aromatic residues appear relatively late in protein evolution for stabilizing

the folding, they are also present in the molecular recognition features (often phenylalanine) in later evolved IDRs.^{18,19} More importantly, recent studies have shown that they are one of the driving forces of LLPS in the IDRs.^{4–7} Therefore, we would like to understand how general the appearance of “structure-promoting” amino acids is in unstructured regions.

We are particularly interested in RNA-binding proteins (RBPs) because the evidence is accumulating that their subcellular localization with RNA molecules is mediated by LLPS,²⁰ and posttranscriptional gene control is dependent on RNA molecules being in precisely the right place at the right time.^{21,22} Accordingly, we analyzed a set of 1,542 RBPs, a subgroup of 692 mRNA binding proteins (mRBPs), and other types of RBPs from a census study,²³ in comparison with a set of 20,396 human proteins. We collected protein sequences from the UniProt database²⁴ and analyzed their level of disorder using the PONDR server’s VLXT, VL3, and VSL2 algorithms.^{25,26} Although these algorithms predict different results for a single protein, the overall trend in a large-scale analysis is similar (see analysis below and Figure S1). We separated residues into ordered and disordered based on the annotation of these algorithms, and calculated the percentage of proteins with sequences of

consecutive disordered residues longer than 30, 40, and 50 amino acids (Figure 1a; Table S1). Our analysis for the human proteome is similar to a classic study of eukaryotic organisms²⁷ and agrees with the observation that IDRs are more prevalent in RBPs.^{28,29} Our results also show that mRBPs have an even higher proportion of disordered residues (Figure 1a), with ~10 percentage points (pp) more residues than human proteins in consecutive sequences of 20 or more disordered residues, and ~20 pp more residues found in disordered regions longer than 60 amino acids (Table S1), suggesting that the mRBPs tend to have longer IDRs. We also analyzed the subgroups of the other types of RBPs. Our results agree with a recent analysis that mRNA, rRNA, and snRNA binding proteins have a higher portion of disordered regions³⁰ (Figure S2).

Next, we counted the numbers of each amino-acid type in the predicted disordered or folded domains (Tables S2–S4). For each amino acid, we calculated the log-odds ratio relative to the human proteome of occurring in a disordered region in RBPs or mRBPs (Figure 1b):

$$\text{Log-odds ratio of presence in IDRs} = \ln \left(\frac{\frac{\text{RBP}^{\text{disordered}}}{\text{RBP}^{\text{total}}}}{\frac{\text{Human}^{\text{disordered}}}{\text{Human}^{\text{total}}}} \right) \quad (1)$$

The values obtained are positive for all amino acids (except for cysteine with the VL3 and VSL2 algorithms; Figure 1b), and the average values (dashed lines in Figure 1b) confirm the above conclusion that IDRs are more prevalent in RBPs than in the human proteome. We also repeated the analysis 1,000 times for random selections of 1,542 or 692 proteins sequences (the numbers of RBPs and mRBP sequences considered) from the human proteome. The averaged log-odds ratios obtained for the random selections are around zero with standard deviations of mostly less than 0.05 and largest deviations no greater than ± 0.1 (dots in Figure 1b; Tables S5–S7 and Figure S3). This analysis of random selections indicates that the differences for RBPs and mRBPs (bars in Figure 1b) are significant. We also analyzed the other types of RBPs using the same approach, but the deviation of single amino-acid types is not obvious compared to the random distribution (Figure S2).

The results from the different algorithms for each amino acid are consistent, except for cysteine. VL3 and VSL2 predict that the prevalence of cysteines in IDRs is lower in RBPs and mRBPs than in the human proteome, whereas VLXT predicts the opposite, but similar to the range of results of random selection. Although this may reflect the algorithm's use of different scoring functions,

it is also possible that IDRs in human proteins have a higher portion of cysteines than those in RBPs do. Note also that asparagine is slightly more likely to be disordered in RBPs and mRBPs, whereas for glutamine, the difference is similar to those obtained for the randomly selected pools (Figure 1b). Although it has been reported that the IDRs in RBPs are likely to be prion-like, that is, rich in asparagine and glutamine,⁷ these two amino acids are not obviously more prevalent than in the human proteome. Among all amino acids, the most considerable differences obtained with these three algorithms are for phenylalanine, tryptophan, and tyrosine, with values much higher than those obtained for random samples of human proteins, particularly for mRNA targeting proteins (Figure 1b). In other words, *structure-promoting* aromatic amino acids are relatively more abundant in *disordered* regions of RBPs than in the human proteome in general.

Functionally important residues are conserved during evolution.³¹ Therefore, we aligned orthologue sequences for each RBP and calculated the Jensen–Shannon divergence score of each residue³¹ as a measure of sequence conservation (Figure 2; Methods in Supporting Information S1). The level of conservation is lower in the chordate phylum (dark blue lines in Figure 2) than in mammals (light blue lines). The analysis was repeated (gray lines) for vertebrates (subphylum) and tetrapods (superclass). As expected, folded domains are conserved earlier in evolution than disordered regions, as indicated by that the dark blue lines have lower values in the regions underlined with a red bar (sequences of more than 40 consecutive disordered residues). We also calculated the average level of conservation for each amino acid to investigate individual trends. The trends are generally flat for structured domains, indicating early conservation (examples in Figure S4). On the contrary, although increasing trends are observed for most amino acids in the IDRs, the aromatic residues are relatively flat, as observed for residues in structured domains (bottom panels in Figure 2 and Figure S4).

In TDP-43, for example, one of the most extensively studied RBP undergoing LLPS,^{32–35} the aromatic residues in its IDRs are highly conserved (Figure 2a; the orange/purple/yellow circles on the Chordata line respectively indicate Phe/Trp/Tyr). The three tryptophans known to be key residues in driving LLPS⁴ are conserved in all chordates. The phenylalanines are conserved in chordates but to a lesser extent, in keeping with the fact that they contribute less to LLPS.⁴ This interpretation that sequence conservation reflects functional importance is further supported by the fact that the transient α -helical region in TDP-43's IDR (Figure 2a, ~residues 320–340), which is involved in LLPS^{32,36,37} is also conserved.

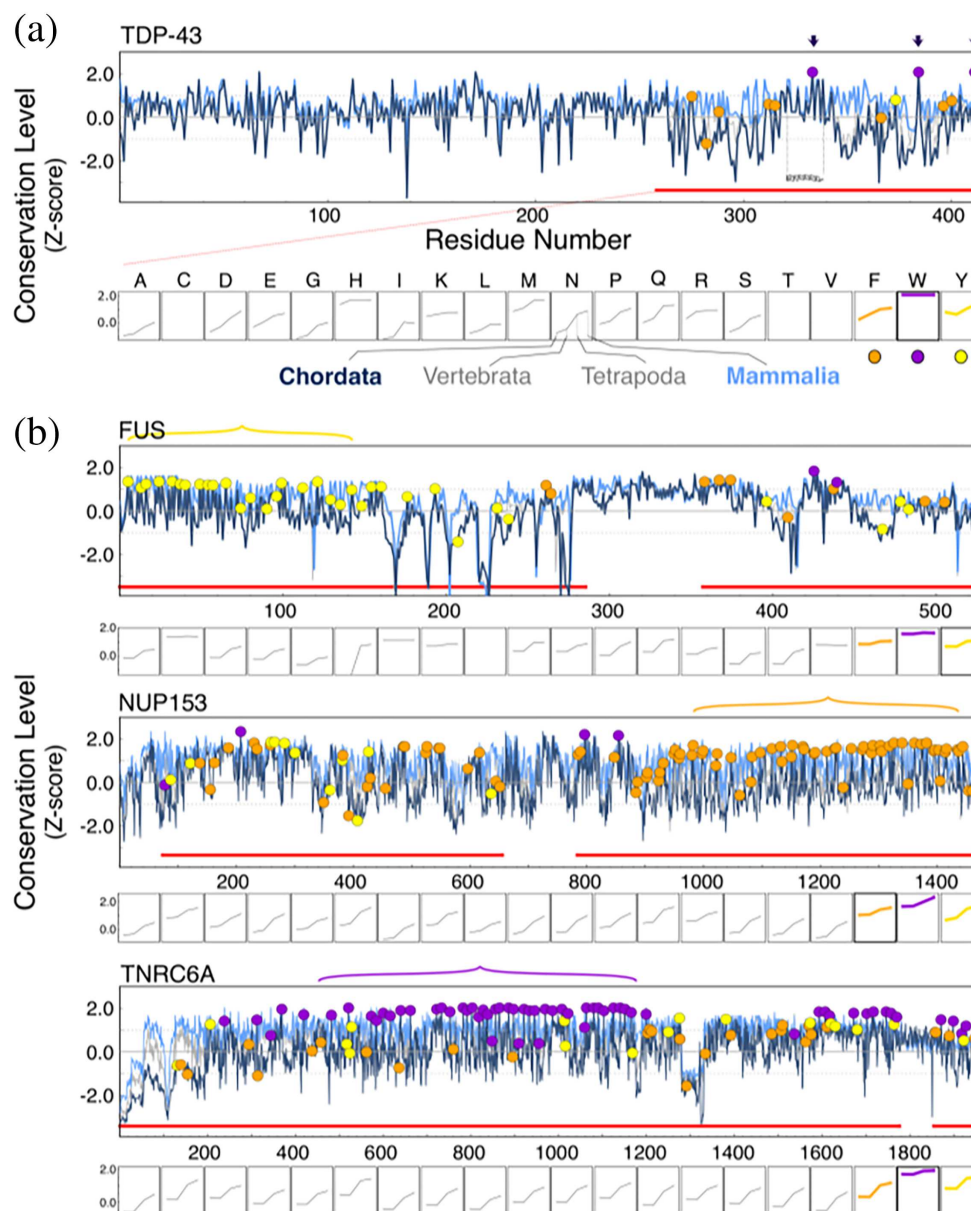


FIGURE 2 Sequence conservation in example proteins with highly conserved aromatic residues. (a) TDP-43, (b) FUS, NUP153, and TNRC6A. (*Upper panels*) Levels of sequence conservation are quantified by the Jensen–Shannon divergence score³¹ and normalized using the Z-score function (the mean of all values toward chordates is 0; the value in the y-axes is the standard deviation, positive values mean more conserved.). Levels of conservation in chordates (dark blue), vertebrates (gray), tetrapods (gray), and mammals (light blue), plotted versus the corresponding residue number in the human sequence. Predicted disordered regions longer than 40 residues are indicated with red bars. Aromatic residues are labeled on the chordate line: Phe (orange), Trp (purple), Tyr (yellow). (a) The three arrows indicate tryptophans experimentally identified as being crucial to liquid–liquid phase separation; the transient α -helical region that also contributes to self-assembly is also labeled. (b) The colored horizontal curly brackets indicate highly conserved aromatic-residue-rich regions. (*Lower panels*) Average levels of conservation as a function of decreasing taxonomic rank for amino-acid types in regions predicted to be disordered (indicated by red bars in the upper panel)

Many other RBPs have highly conserved aromatic residues whose functional importance have been reported (Figure 2b): Tyrosines are known to be involved in the LLPS of the N-terminal domain of the RBP FUS,^{5,7,38} and a recent analysis of mammalian FUS proteins found the same trend as observed here.³⁹ Phe-Gly repeats are a

common feature of nucleoporins, which regulate nucleocytoplasmic transport in the nuclear pore complex.^{40–42} The nucleoporin NUP153 is also categorized as an RBP because its Phe-Gly repeats mediate mRNA trafficking.⁴³ Phenylalanines are conserved in NUP153 (Figure 2b) and it is reasonable to suppose that

this is also the case in other nucleoporins with many Phe-Gly repeats. The tryptophan-rich region in TNRC6 interacts with Ago2 to promote the miRNA-induced silencing complex's phase separation⁴⁴; most of these tryptophans are highly conserved in chordates (Figure 2b). Other examples include the LLPS of CPEB⁴⁵ and HNRNP. Although the role of aromatic residues has not been explicitly studied in these proteins, their conservation in chordates hints at a possible role in LLPS related functions (Figure S5). Many RBPs that have not been reported to undergo LLPS also have conserved aromatic patterns, including RBM19, DDX18, KHDRBS1, ABT1, RSL24D1, SMAD5, and TAF9 (Figures S4b and S5). We suggest that these conserved aromatic residues in IDRs may also be involved in LLPS-related functions. However, multiple sequence alignment is limited in identifying the conserved residues in many other IDRs via current algorithms. The analysis above thus underestimates the numbers of RBPs having functionally important aromatic residues. These "unalignable" aromatic residues, however, may also be functionally important. For example, recent studies have shown that aromatic residues' amount and patterning are the keys for hnRNP-A1's LLPS regardless of its orthologs' IDRs cannot be aligned.⁴⁷ Moreover, the "spacers" between the stickers are also essential factors in driving LLPS,^{47,48} and current sequence alignment algorithms for IDRs cannot easily pick the importance of spacers as well. Although a recent work using machine learning approaches to collect the "features" of IDRs might be an alternative approach to find their traits,⁴⁹ it is still limited in capturing the distal interaction, such as the prevailing aromatic residues.

In summary, IDRs appeared late in proteins, and RBPs evolved under selective pressure to assemble precisely in specific cellular locations. This spatiotemporal control can be achieved by mimicking prion properties⁵⁰ or by having short α -helical motifs,³² blocked charged-pattern,⁵¹ or coiled-coil domains,⁵² features that provide multivalency and promote the formation of higher-order oligomers. Aromatic residues, with weak π - π or cation- π interactions, appear to have been selected in this role, affording the weak, reversible interactions required for biomolecular condensate formation. After contributing a first time in the evolution of folded proteins, the subsequent return of the (aromatic) rings in RBPs appears to have been crucial to the emergence of LLPS.

ACKNOWLEDGMENTS

This research was funded by the Ministry of Science and Technology of Taiwan, grant numbers 109-2113-M-010-003 and 110-2113-M-A49A-504-MY3.

AUTHOR CONTRIBUTIONS

Wen-Lin Ho: Data curation (lead); formal analysis (lead); investigation (equal); methodology (equal); software (lead); visualization (equal); writing – review and editing (supporting). **Jie-rong Huang:** Conceptualization (lead); formal analysis (equal); funding acquisition (lead); investigation (equal); methodology (equal); project administration (lead); resources (lead); software (supporting); supervision (lead); validation (equal); visualization (equal); writing – original draft (lead); writing – review and editing (lead).

DATA AVAILABILITY STATEMENT

All scripts and data generated or analyzed in this study are available in the repository: http://github.com/allmwh/the_return_of_the_rings.

ORCID

Jie-rong Huang  <https://orcid.org/0000-0003-3674-2228>

REFERENCES

- Burley SK, Petsko GA. Aromatic-aromatic interaction: A mechanism of protein structure stabilization. *Science*. 1985;229:23–28.
- Dunker AK, Lawson JD, Brown CJ, et al. Intrinsically disordered protein. *J Mol Graph Model*. 2001;19:26–59.
- Yan J, Cheng J, Kurgan L, Uversky VN. Structural and functional analysis of "non-smelly" proteins. *Cell Mol Life Sci*. 2020;77:2423–2440.
- Li HR, Chiang WC, Chou PC, Wang WJ, Huang JR. TAR DNA-binding protein 43 (TDP-43) liquid-liquid phase separation is mediated by just a few aromatic residues. *J Biol Chem*. 2018;293:6090–6098.
- Lin Y, Currie SL, Rosen MK. Intrinsically disordered sequences enable modulation of protein phase separation through distributed tyrosine motifs. *J Biol Chem*. 2017;292:19110–19120.
- Kwon I, Kato M, Xiang S, et al. Phosphorylation-regulated binding of RNA polymerase II to fibrous polymers of low-complexity domains. *Cell*. 2013;155:1049–1060.
- Wang J, Choi JM, Holehouse AS, et al. A molecular grammar governing the driving forces for phase separation of prion-like RNA binding proteins. *Cell*. 2018;174:688–699.e16.
- Eck RV, Dayhoff MO. Evolution of the structure of ferredoxin based on living relics of primitive amino acid sequences. *Science*. 1966;152:363–366.
- Jeffery CJ. Moonlighting proteins. *Trends Biochem Sci*. 1999;24:8–11.
- Oldfield CJ, Dunker AK. Intrinsically disordered proteins and intrinsically disordered protein regions. *Annu Rev Biochem*. 2014;83:553–584.
- Smock RG, Gierasch LM. Sending signals dynamically. *Science*. 2009;324:198–203.
- Alberti S, Hyman AA. Biomolecular condensates at the nexus of cellular stress, protein aggregation disease and ageing. *Nat Rev Mol Cell Biol*. 2021;22:196–213.
- Trifonov EN. The triplet code from first principles. *J Biomol Struct Dyn*. 2004;22:1–11.

14. Katsnelson A. Did disordered proteins help launch life on earth? *ACS Cent Sci.* 2020;6:1854–1857.
15. Kulkarni P, Uversky VN. Intrinsically disordered proteins: The dark horse of the dark proteome. *Proteomics.* 2018;18:e1800061.
16. Zhu H, Sepulveda E, Hartmann MD, et al. Origin of a folded repeat protein from an intrinsically disordered ancestor. *Elife.* 2016;5:e16761.
17. Uversky VN. A decade and a half of protein intrinsic disorder: Biology still waits for physics. *Protein Sci.* 2013;22:693–724.
18. Mohan A, Oldfield CJ, Radivojac P, et al. Analysis of molecular recognition features (MoRFs). *J Mol Biol.* 2006;362:1043–1059.
19. Vacic V, Oldfield CJ, Mohan A, et al. Characterization of molecular recognition features, MoRFs, and their binding partners. *J Proteome Res.* 2007;6:2351–2366.
20. Wiedner HJ, Giudice J. It's not just a phase: Function and characteristics of RNA-binding proteins in phase separation. *Nat Struct Mol Biol.* 2021;28:465–473.
21. Kloc M, Zearfoss NR, Etkin LD. Mechanisms of subcellular mRNA localization. *Cell.* 2002;108:533–544.
22. Martin KC, Ephrussi A. mRNA localization: Gene expression in the spatial dimension. *Cell.* 2009;136:719–730.
23. Gerstberger S, Hafner M, Tuschl T. A census of human RNA-binding proteins. *Nat Rev Genet.* 2014;15:829–845.
24. UniProt C. UniProt: A worldwide hub of protein knowledge. *Nucleic Acids Res.* 2019;47:D506–D515.
25. Romero O, Dunker K. Sequence data analysis for long disordered regions prediction in the calcineurin family. *Genome Inform Ser Workshop Genome Inform.* 1997;8:110–124.
26. Radivojac P, Obradovic Z, Brown CJ, Dunker AK. Prediction of boundaries between intrinsically ordered and disordered protein regions. *Pac Symp Biocomput.* 2003;216–227.
27. Dunker AK, Obradovic Z, Romero P, Garner EC, Brown CJ. Intrinsic protein disorder in complete genomes. *Genome Inform Ser Workshop Genome Inform.* 2000;11:161–171.
28. Zagrovic B, Bartonek L, Polyansky AA. RNA-protein interactions in an unstructured context. *FEBS Lett.* 2018;592:2901–2916.
29. Varadi M, Zsolyomi F, Guharoy M, Tompa P. Functional advantages of conserved intrinsic disorder in RNA-binding proteins. *PLoS One.* 2015;10:e0139731.
30. Zhao B, Katuwawala A, Oldfield CJ, et al. Intrinsic disorder in human RNA-binding proteins. *J Mol Biol.* 2021;433:167229.
31. Capra JA, Singh M. Predicting functionally important residues from sequence conservation. *Bioinformatics.* 2007;23:1875–1882.
32. Conicella AE, Zerze GH, Mittal J, Fawzi NL. ALS mutations disrupt phase separation mediated by alpha-helical structure in the TDP-43 low-complexity C-terminal domain. *Structure.* 2016;24:1537–1549.
33. McGurk L, Gomes E, Guo L, et al. Poly(ADP-ribose) prevents pathological phase separation of TDP-43 by promoting liquid demixing and stress granule localization. *Mol Cell.* 2018;71:703–717.e9.
34. Babinchak WM, Haider R, Dumm BK, et al. The role of liquid-liquid phase separation in aggregation of the TDP-43 low-complexity domain. *J Biol Chem.* 2019;294:6306–6317.
35. Sun Y, Medina Cruz A, Hadley KC, et al. Physiologically important electrolytes as regulators of TDP-43 aggregation and droplet-phase behavior. *Biochemistry.* 2019;58:590–607.
36. Li HR, Chen TC, Hsiao CL, Shi L, Chou CY, Huang JR. The physical forces mediating self-association and phase-separation in the C-terminal domain of TDP-43. *Biochim Biophys Acta.* 2018;1866:214–223.
37. Chen TC, Hsiao CL, Huang SJ, Huang JR. The nearest-neighbor effect on random-coil NMR chemical shifts demonstrated using a low-complexity amino-acid sequence. *Protein Pept Lett.* 2016;23:967–975.
38. Qamar S, Wang GZ, Randle SJ, et al. FUS phase separation is modulated by a molecular chaperone and methylation of arginine cation- π interactions. *Cell.* 2018;173:720–734 e15.
39. Dasmeh P, Wagner A. Natural selection on the phase-separation properties of FUS during 160 my of mammalian evolution. *Mol Biol Evol.* 2021;38:940–951.
40. Milles S, Mercadante D, Aramburu IV, et al. Plasticity of an ultrafast interaction between nucleoporins and nuclear transport receptors. *Cell.* 2015;163:734–745.
41. Onischenko E, Tang JH, Andersen KR, et al. Natively unfolded FG repeats stabilize the structure of the nuclear pore complex. *Cell.* 2017;171:904–917.e19.
42. Hayama R, Sparks S, Hecht LM, et al. Thermodynamic characterization of the multivalent interactions underlying rapid and selective translocation through the nuclear pore complex. *J Biol Chem.* 2018;293:4555–4563.
43. Bastos R, Lin A, Enarson M, Burke B. Targeting and function in mRNA export of nuclear pore complex protein Nup153. *J Cell Biol.* 1996;134:1141–1156.
44. Sheu-Gruttadauria J, MacRae IJ. Phase transitions in the assembly and function of human miRISC. *Cell.* 2018;173:946–957.e16.
45. Ford L, Ling E, Kandel ER, Fioriti L. CPEB3 inhibits translation of mRNA targets by localizing them to P bodies. *Proc Natl Acad Sci USA.* 2019;116:18078–18087.
46. Battle C, Yang P, Coughlin M, et al. hnRNPD phase separation is regulated by alternative splicing and disease-causing mutations accelerate its aggregation. *Cell Rep.* 2020;30:1117–1128.e5.
47. Martin EW, Holehouse AS, Peran I, et al. Valence and patterning of aromatic residues determine the phase behavior of prion-like domains. *Science.* 2020;367:694–699.
48. Dasmeh P, Doronin R, Wagner A. The length scale of multivalent interactions is evolutionarily conserved in fungal and vertebrate phase-separating proteins. *Genetics.* 2022;220, iyab184. <https://academic.oup.com/genetics/article/220/1/iyab184/6410649>
49. Lu AX, Lu AX, Pritisanac P, Zarin T, Forman-Kay JD, Moses AM. Discovering molecular features of intrinsically disordered regions by using evolution for contrastive learning. *bioRxiv.* 2021. <https://doi.org/10.1101/2021.07.29.454330>
50. Patel A, Lee HO, Jawerth L, et al. A liquid-to-solid phase transition of the ALS protein FUS accelerated by disease mutation. *Cell.* 2015;162:1066–1077.
51. Greig JA, Nguyen TA, Lee M, et al. Arginine-enriched mixed-charge domains provide cohesion for nuclear speckle condensation. *Mol Cell.* 2020;77:1237–1250.e4.

52. Fang X, Wang L, Ishikawa R, et al. Arabidopsis FLL2 promotes liquid-liquid phase separation of polyadenylation complexes. *Nature*. 2019;569:265–269.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Ho W-L, Huang J. The return of the rings: Evolutionary convergence of aromatic residues in the intrinsically disordered regions of RNA-binding proteins for liquid–liquid phase separation. *Protein Science*. 2022;31(5): e4317. <https://doi.org/10.1002/pro.4317>

Supplementary Materials

**The return of the rings: evolutionary convergence of
aromatic residues in the intrinsically disordered regions of
RNA-binding proteins for liquid-liquid phase separation**

Wen-Lin Ho and Jie-rong Huang

Methods

Disorder prevalence and disorder odds ratios relative to the human proteome

All human protein sequences were retrieved from UniProt (UniProtKB_2021_01, download date: 2021/02/06, 20396 sequences in total). Disorder predictions were performed using the VLXT, VL3, and VSL2 algorithms on the PONDR webserver^{1,2} and fIDPnn.³ According to these algorithms, each residue in a sequence was annotated as ordered or disordered. The criteria of consecutive disordered regions (longer than 30, 40, or 50 residues) or the type of amino acids within these regions were analyzed using in-house scripts (deposited in GitHub). RNA-binding proteins were based on the definition of the consensus studies.⁴ Sub-groups of RNA-binding proteins, targeting to mRNA, irRNA, tRNA, rRNA, ncRNA, snRNA, are defined based on the same study (the nomenclature follows the recent analysis⁵). Names of these RBPs and sequences used in this analysis are in an Excel file in the extended datasets. Nine gene names in the census study are not inconsistent with the UniProt and thus were not analyzed (also indicated in the Excel file). Missing these sequences does not affect our analysis.

Sequence conservation analysis

Human RBP orthologues were obtained from the Orthologous Matrix (OMA) database⁶ using the human protein's UniProt ID as input to grep its OMA group members. The orthologues were filtered by taxonomy. The sequences of the orthologues were aligned using the Clustal Omega (v.1.2.4)⁷ module in Biopython⁸. The aligned sequence with a gap longer than 20 consecutive residues will be removed unless more than three orthologs in the alignment have the same gap. The aligned sequences were mapped to the human sequence's residue number, and the level of conservation of each residue was quantified using the Jensen-Shannon divergence score (D^{JS})⁹:

$$D^{JS} = \frac{1}{2} \sum_{\alpha \in \text{a.a.}} p_C(\alpha) \log_2 \frac{p_C(\alpha)}{r(\alpha)} + \frac{1}{2} \sum_{\alpha \in \text{a.a.}} q(\alpha) \log_2 \frac{q(\alpha)}{r(\alpha)}$$

where α is one of the twenty amino acids (a.a.), $p_C(\alpha)$ is the column population of amino-acid α , $q(\alpha)$ is the background distribution in the BLOSUM62 alignment, $r(\alpha) = 0.5 p_C(\alpha) + 0.5 q(\alpha)$. The orthologs used for the analysis in Figure 2 and Figure S5 are listed in Table S8 or deposited the Github server. The Jensen-Shannon divergence scores were converted using the Z-score function. The mean and standard deviation derived from the chordates were used for normalization.

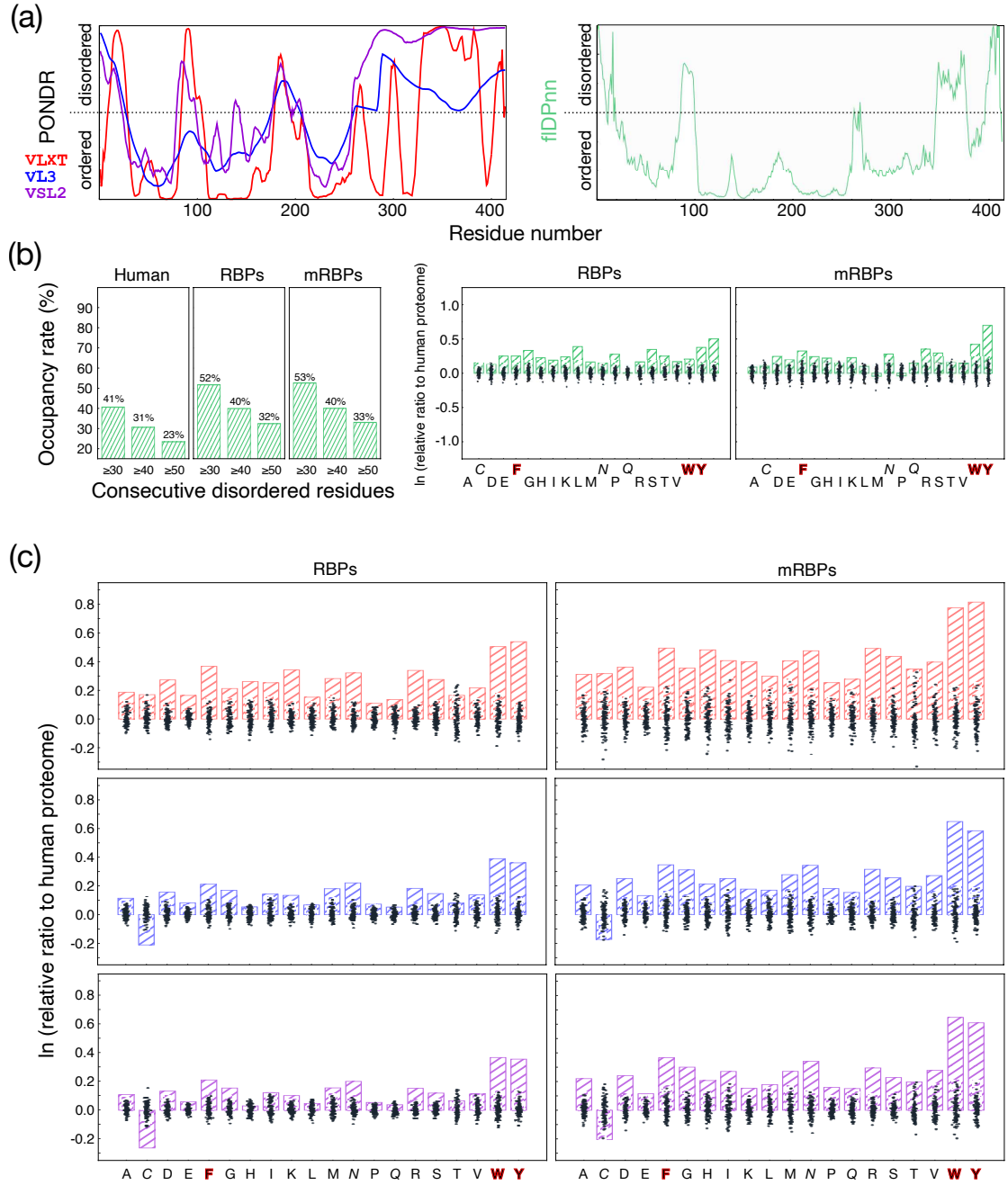


Figure S1. The analysis is independent of the algorithms chosen or the length criteria. One of the latest IDP predictors, fIDPnn³, which shows good performance¹⁰ is included for comparison. (a) Examples of using different algorithms: three from the POND server (left) and fIDPnn (right) on a single protein (TDP-43). (b) Compared to Figure 1, using other algorithms such as the latest fIDPnn or (c) analyzed with criteria of consecutive lengths (in this example, ≥ 40 residues). The overall trends are similar.

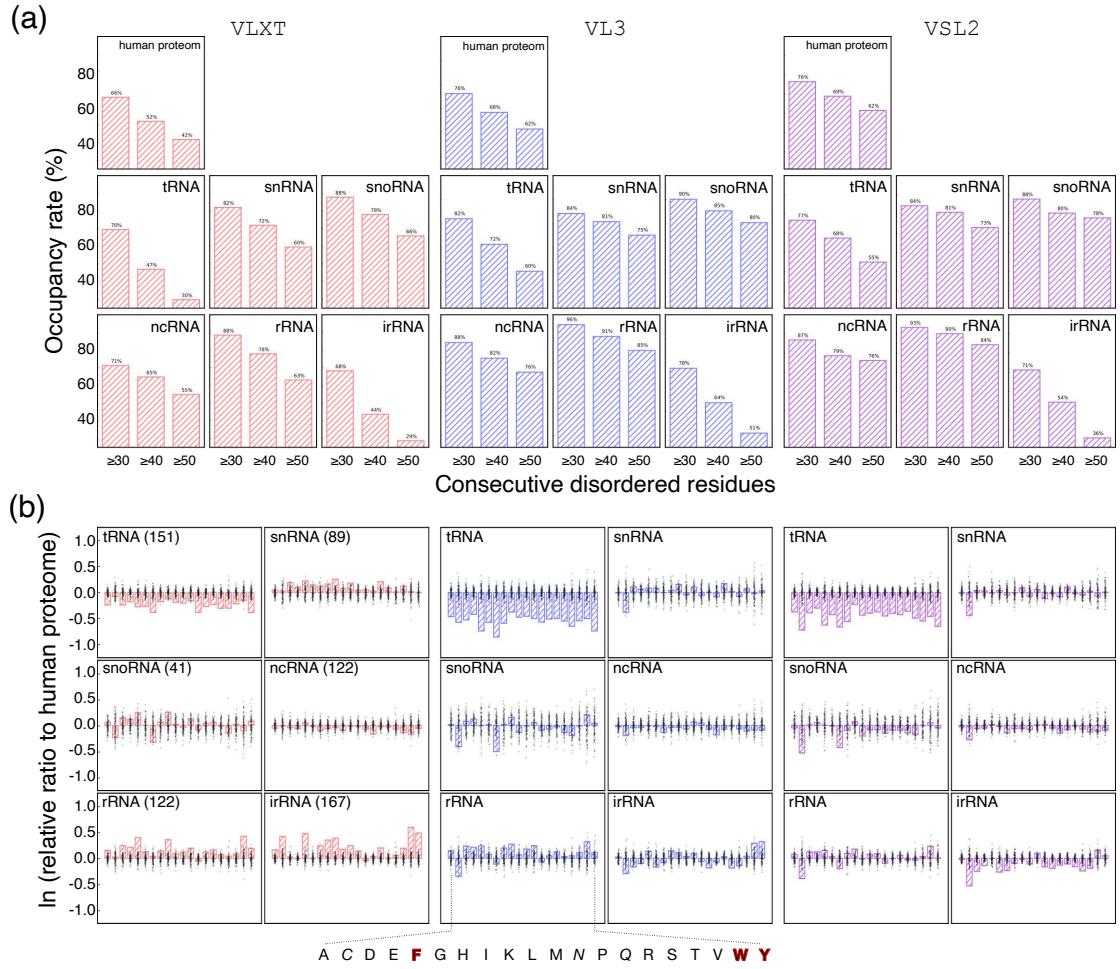


Figure S2. Prevalence of disorder and disorder odds ratios relative to the human proteome by amino acid type for different RNA-binding proteins. (a) The proportion of proteins with disordered regions longer than 30, 40, or 50 consecutive residues, as predicted using different algorithms, in the human proteome (upper panel) and other six groups of protein targeting to different types of RNAs based on the census study.⁵ (b) Log-odds ratios relative to the human proteome of being in an IDR for amino acids in different types of RNA targeting proteins. The dots represent the values obtained for randomly selected (negative control) subsamples of the human proteome. The number of random sample sizes depends on the RBPs' number in the census data,⁴ as indicated.

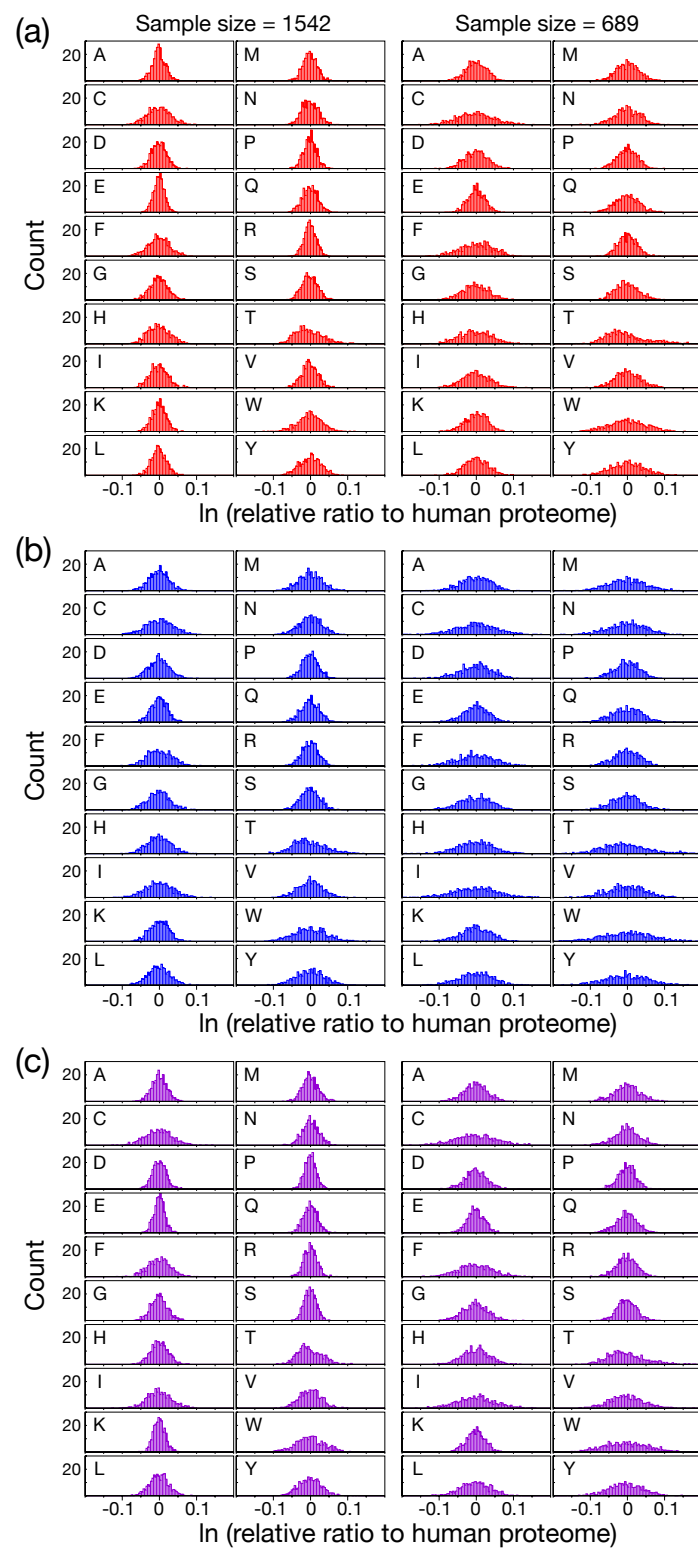


Figure S3. Distribution of the log-odds ratio relative to the human proteome of being in an intrinsically disordered region as predicted by the (a) VLXT, (b) VL3, and (c) VSL2 algorithm for randomly selected subgroups of the human proteome (left, N =1542; right, N = 689)

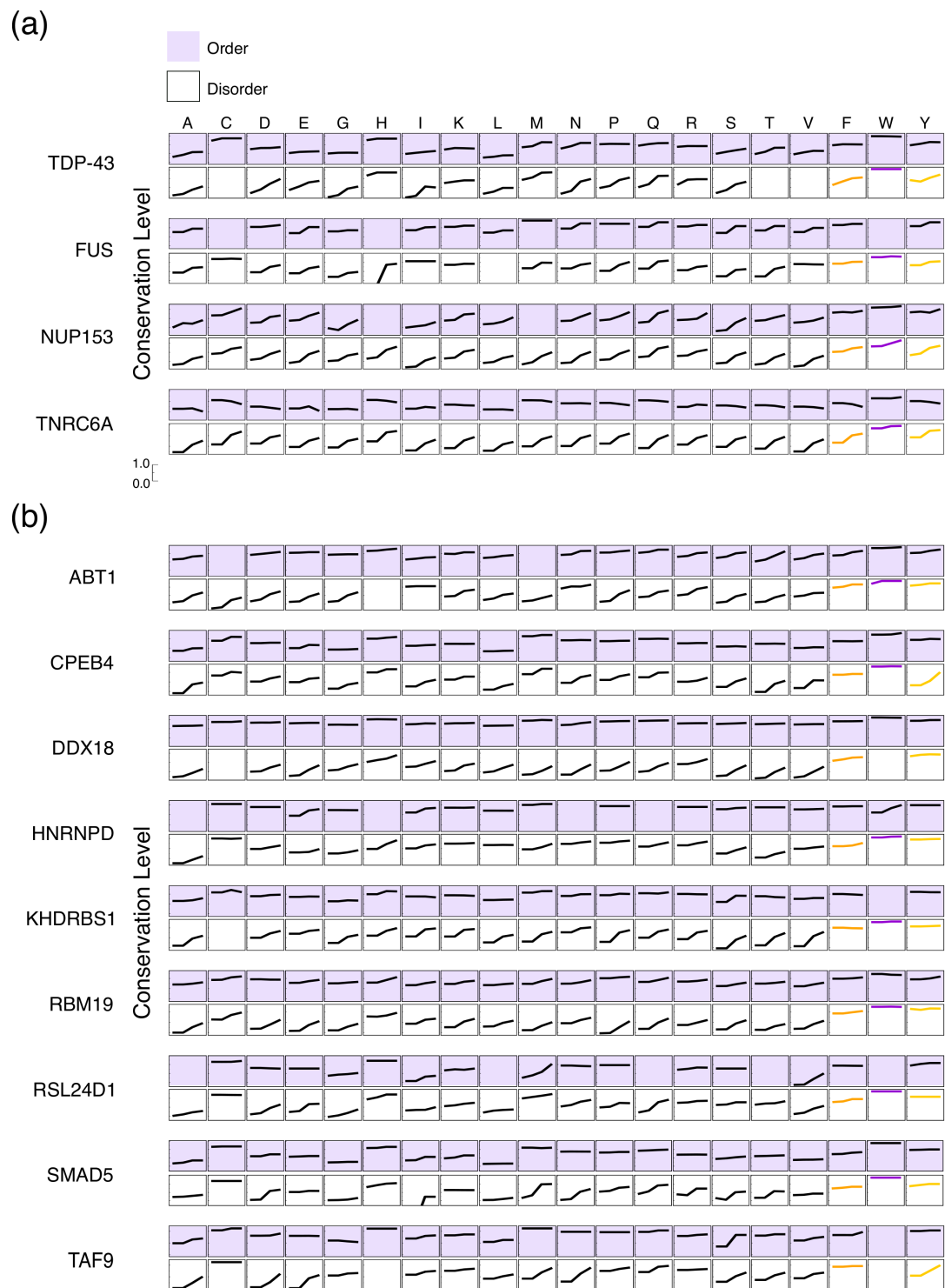


Figure S4. Average level of sequence conservation by amino acid type for residues in ordered (purple) or disordered regions (white) as a function of decreasing taxonomic ranks (Chordata, Vertebrata, Tetrapoda, Mammalia), for (a) the example proteins presented in fig. 2 of the main text and (b) the example proteins presented in figure S3 (below). Only residues in disordered (ordered) regions longer than 40 (10) consecutive amino acids were considered.

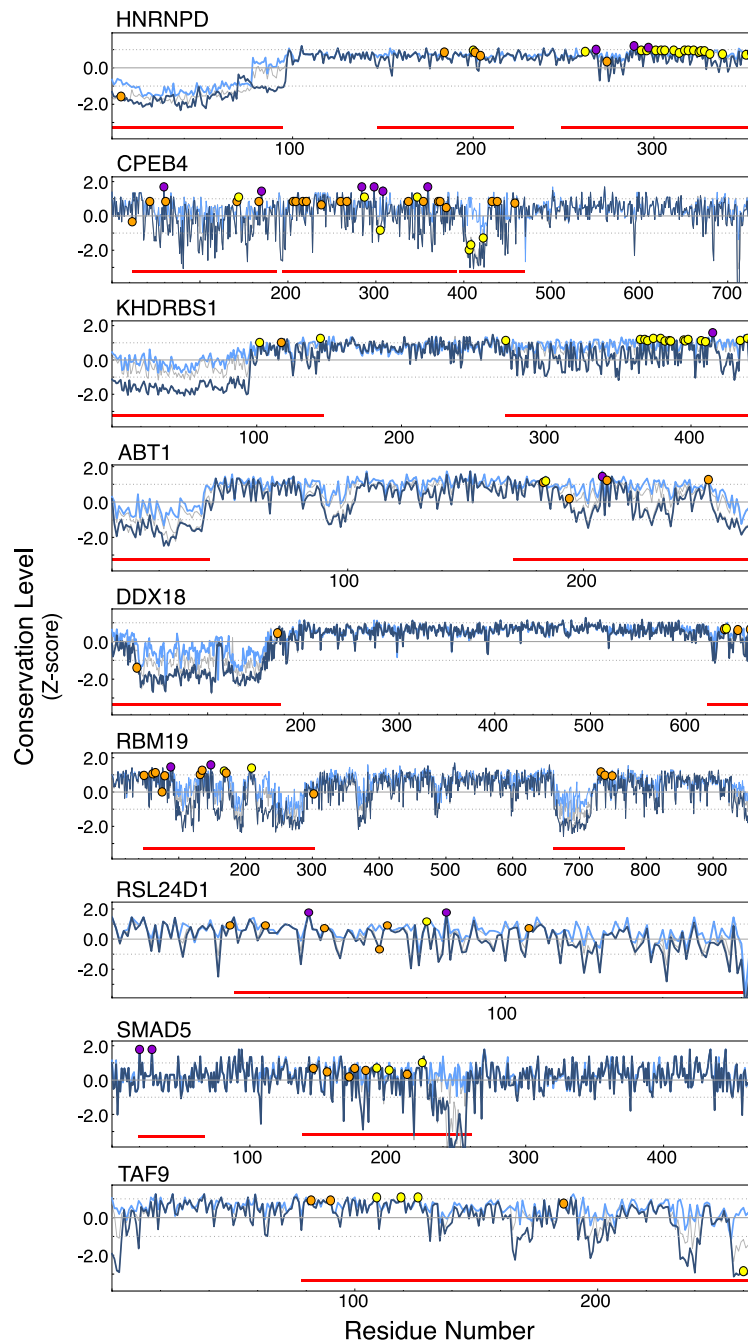


Figure S5. Sequence conservation in example proteins with highly conserved aromatic residues. Levels of sequence conservation are quantified by the Jensen-Shannon divergence score and normalized using the Z-score function (the mean of all values toward chordates is 0; the value in the y-axis is the standard deviation; positive values mean more conserved.) Levels of conservation in chordates (dark blue), vertebrates (gray), tetrapods (gray), and mammals (light blue) are plotted versus the corresponding residue number in the human sequence. Predicted disordered regions longer than 40 residues are indicated with red bars. Aromatic residues are labeled on the chordate line: Phe (orange), Trp (purple), Tyr (yellow).

Table S1. Proportions of proteins with predicted IDRs longer than certain lengths.

Algorithm	Length longer than	Human proteins (%)	RNA binding proteins (%)	(RBP – Human) (%)	mRNA binding proteins (%)	(mRBP – Human) (%)
VLXT	20	81.8	92.6	10.9	93.6	11.8
	30	65.6	79.5	13.9	84.5	18.9
	40	52.0	66.2	14.2	73.0	21.0
	50	41.7	54.7	12.9	62.6	20.8
	60	33.7	45.9	12.2	55.4	21.7
VL3	20	83.3	93.4	10.1	93.9	10.6
	30	76.3	88.0	11.7	90.1	13.8
	40	68.5	80.9	12.4	84.3	15.8
	50	61.6	74.4	12.8	79.7	18.1
	60	55.7	66.6	10.9	74.2	18.5
VSL2	20	84.1	91.8	7.7	93.9	9.8
	30	76.5	85.3	8.8	88.4	11.9
	40	68.9	78.1	9.2	82.6	13.7
	50	61.5	71.2	9.7	78.7	17.1
	60	55.6	64.3	8.7	73.4	17.7

Table S2. Number of residues predicted to be in ordered/disordered regions by the VLXT algorithm

	RBPs		mRBPs		Human proteins	
	Ordered	Disordered	Ordered	Disordered	Ordered	Disordered
A	33767	27760	15945	14705	474415	321897
C	12353	2662	5358	1244	220009	41333
D	27224	18481	13262	9798	349268	188768
E	31856	36215	15343	18489	409580	396968
F	25340	4883	12221	2586	362796	51625
G	32066	26379	15778	15089	452912	293467
H	15438	6544	6995	3577	220858	76954
I	26308	11070	12352	5838	369522	123116
K	34806	26359	16136	12385	417435	233546
L	54425	25927	24430	12995	802230	329184
M	10887	8169	5332	4419	151790	90374
N	21112	11254	10520	6336	287931	119746
P	25046	31437	11796	18099	341273	375933
Q	25311	18377	12474	10261	330112	211467
R	26230	30610	12211	16772	347223	292901
S	34805	36396	16745	21500	529363	416555
T	26013	17415	12199	9710	388457	219237
V	34359	17768	15600	9331	474983	202342
W	6853	2006	2992	1062	114749	23332
Y	18116	5406	8934	3333	251740	50805

Table S3. Number of residues predicted to be in ordered/disordered regions by the VL3 algorithm

	RBPs		mRBPs		Human proteins	
	Ordered	Disordered	Ordered	Disordered	Ordered	Disordered
A	34295	27232	16011	14639	479203	317109
C	10891	4124	4765	1837	175643	85699
D	25476	20229	11988	11072	330080	207956
E	31199	36872	14727	19105	399940	406608
F	22220	8003	10423	4384	324013	90408
G	29712	28733	13694	17173	431350	315029
H	13748	8234	6048	4524	191475	106337
I	27452	9926	12931	5259	378838	113800
K	30335	30830	13731	14790	360043	290938
L	53010	27342	23653	13772	771505	359909
M	11504	7552	5595	4156	159982	82182
N	19550	12816	9461	7395	274701	132976
P	23244	33239	10572	19323	320476	396730
Q	22870	20818	10930	11805	295531	246048
R	27151	29689	12085	16898	355342	284782
S	31099	40102	14545	23700	477217	468701
T	25201	18227	11733	10176	370569	237125
V	34809	17318	15676	9255	478562	198763
W	6619	2240	2768	1286	113612	24469
Y	16578	6944	7858	4409	238289	64256

Table S4. Number of residues predicted to be in ordered/disordered regions by the VLS2 algorithm

	RBPs		mRBPs		Human proteins	
	Ordered	Disordered	Ordered	Disordered	Ordered	Disordered
A	30642	30885	13975	16675	429548	366764
C	11246	3769	4884	1718	178491	82851
D	22064	23641	10146	12914	281602	256434
E	26089	41982	12175	21657	326165	480383
F	21578	8645	10046	4761	313891	100530
G	25897	32548	11723	19144	374088	372291
H	12557	9425	5427	5145	170785	127027
I	27085	10293	12597	5593	370211	122427
K	25112	36053	11275	17246	290910	360071
L	52104	28248	22962	14463	750188	381226
M	10076	8980	4834	4917	138373	103791
N	16953	15413	7988	8868	237124	170553
P	19526	36957	8703	21192	263731	453475
Q	19952	23736	9358	13377	255138	286441
R	24648	32192	10737	18246	314028	326096
S	24465	46736	11073	27172	373370	572548
T	22428	21000	10245	11664	328426	279268
V	34945	17182	15635	9296	473431	203894
W	6355	2504	2622	1432	108987	29094
Y	15972	7550	7451	4816	228179	74366

Table S5. Summary statistics for the log-odds ratios relative to the human proteome of being in a disordered region as predicted by the VXLT algorithm for randomly sampled subgroups of the human proteome

	Random sample size = 1542				Random sample size =689			
	mean	std	max	min	mean	std	max	min
A	0.00	0.02	0.05	-0.05	0.00	0.03	0.08	-0.09
C	0.00	0.03	0.10	-0.09	0.00	0.05	0.15	-0.15
D	0.00	0.02	0.08	-0.06	0.00	0.03	0.09	-0.11
E	0.00	0.01	0.04	-0.05	0.00	0.02	0.07	-0.08
F	0.00	0.03	0.08	-0.08	0.00	0.04	0.12	-0.13
G	0.00	0.02	0.08	-0.07	0.00	0.04	0.13	-0.12
H	0.00	0.03	0.08	-0.08	0.00	0.04	0.18	-0.13
I	0.00	0.02	0.08	-0.06	0.00	0.04	0.13	-0.10
K	0.00	0.02	0.06	-0.06	0.00	0.03	0.09	-0.10
L	0.00	0.02	0.06	-0.07	0.00	0.03	0.10	-0.11
M	0.00	0.02	0.08	-0.06	0.00	0.03	0.11	-0.08
N	0.00	0.02	0.07	-0.07	0.00	0.03	0.11	-0.13
P	0.00	0.02	0.06	-0.05	0.00	0.03	0.10	-0.09
Q	0.00	0.02	0.07	-0.06	0.00	0.03	0.09	-0.12
R	0.00	0.02	0.05	-0.06	0.00	0.02	0.09	-0.08
S	0.00	0.02	0.07	-0.06	0.00	0.03	0.11	-0.11
T	0.00	0.04	0.16	-0.09	0.00	0.06	0.22	-0.14
V	0.00	0.02	0.06	-0.07	0.00	0.03	0.09	-0.11
W	0.00	0.03	0.12	-0.13	0.00	0.05	0.15	-0.13
Y	0.00	0.03	0.09	-0.08	0.00	0.04	0.13	-0.12

The values are derived from 1000 randomly selected subgroups of the human proteome with sample sizes of 1542 or 689 (the same as those of the samples of RBPs and mRBPs considered) using eq. (1). The mean, standard deviation (std), maximum (max), and minimum (min) values are listed.

Table S6. Summary statistics for the log-odds ratios relative to the human proteome of being in a disordered region as predicted by the VL3 algorithm for randomly sampled subgroups of the human proteome

	Random sample size = 1542				Random sample size =689			
	mean	std	max	min	mean	std	max	min
A	0.00	0.03	0.08	-0.08	0.00	0.04	0.11	-0.13
C	0.00	0.04	0.11	-0.10	0.00	0.06	0.17	-0.19
D	0.00	0.03	0.10	-0.08	0.00	0.04	0.11	-0.17
E	0.00	0.02	0.06	-0.07	0.00	0.03	0.10	-0.11
F	0.00	0.04	0.14	-0.12	0.00	0.05	0.14	-0.17
G	0.00	0.03	0.12	-0.11	0.00	0.04	0.10	-0.16
H	0.00	0.03	0.10	-0.10	0.00	0.05	0.13	-0.19
I	0.00	0.04	0.14	-0.12	0.00	0.06	0.17	-0.19
K	0.00	0.03	0.08	-0.07	0.00	0.04	0.11	-0.14
L	0.00	0.03	0.10	-0.08	0.00	0.05	0.14	-0.14
M	0.00	0.03	0.09	-0.09	0.00	0.05	0.14	-0.17
N	0.00	0.03	0.09	-0.09	0.00	0.05	0.14	-0.18
P	0.00	0.02	0.06	-0.06	0.00	0.03	0.09	-0.11
Q	0.00	0.02	0.08	-0.07	0.00	0.04	0.13	-0.13
R	0.00	0.02	0.07	-0.09	0.00	0.04	0.10	-0.13
S	0.00	0.03	0.08	-0.07	0.00	0.04	0.12	-0.13
T	0.00	0.04	0.14	-0.11	-0.01	0.06	0.21	-0.19
V	0.00	0.03	0.12	-0.10	0.00	0.05	0.13	-0.17
W	0.00	0.04	0.16	-0.12	0.00	0.07	0.20	-0.23
Y	0.00	0.04	0.12	-0.12	0.00	0.06	0.16	-0.21

The values are derived from 1000 randomly selected subgroups of the human proteome with sample sizes of 1542 or 689 (the same as those of the samples of RBPs and mRBPs considered) using eq. (1). The mean, standard deviation (std), maximum (max), and minimum (min) values are listed.

Table S7. Summary statistics for the log-odds ratios relative to the human proteome of being in a disordered region as predicted by the VSL2 algorithm for randomly sampled subgroups of the human proteome

	Random sample size = 1542				Random sample size =689			
	mean	std	max	min	mean	std	max	min
A	0.00	0.02	0.06	-0.07	0.00	0.03	0.11	-0.09
C	0.00	0.04	0.14	-0.13	0.00	0.06	0.14	-0.21
D	0.00	0.02	0.07	-0.05	0.00	0.03	0.12	-0.10
E	0.00	0.01	0.05	-0.05	0.00	0.02	0.07	-0.08
F	0.00	0.03	0.10	-0.11	0.00	0.05	0.15	-0.14
G	0.00	0.02	0.07	-0.07	0.00	0.03	0.12	-0.10
H	0.00	0.02	0.08	-0.07	0.00	0.04	0.12	-0.11
I	0.00	0.03	0.10	-0.13	0.00	0.05	0.16	-0.16
K	0.00	0.02	0.05	-0.05	0.00	0.03	0.10	-0.07
L	0.00	0.03	0.09	-0.08	0.00	0.04	0.15	-0.12
M	0.00	0.02	0.07	-0.07	0.00	0.03	0.11	-0.10
N	0.00	0.02	0.06	-0.08	0.00	0.03	0.10	-0.11
P	0.00	0.02	0.05	-0.05	0.00	0.02	0.09	-0.08
Q	0.00	0.02	0.07	-0.07	0.00	0.03	0.10	-0.09
R	0.00	0.02	0.07	-0.05	0.00	0.03	0.08	-0.10
S	0.00	0.02	0.05	-0.05	0.00	0.03	0.09	-0.08
T	0.00	0.03	0.11	-0.09	0.00	0.05	0.18	-0.14
V	0.00	0.03	0.10	-0.10	0.00	0.05	0.18	-0.13
W	0.00	0.04	0.12	-0.12	0.00	0.06	0.17	-0.16
Y	0.00	0.03	0.12	-0.10	0.00	0.05	0.15	-0.16

The values are derived from 1000 randomly selected subgroups of the human proteome with sample sizes of 1542 or 689 (the same as those of the samples of RBPs and mRBPs considered) using eq. (1). The mean, standard deviation (std), maximum (max), and minimum (min) values are listed.

Table 8. Orthologs used in the analysis of Figure 2.

OMA Protein ID	Species	OMA cross reference	Index ¹
TDP-43			
BRAFL24270	<i>Branchiostoma floridae</i>	C3ZIE9	1
CIOA12468	<i>Ciona savignyi</i>	H2ZBU8	1
LATCH14939	<i>Latimeria chalumnae</i>	H3BC22	2
LEPOC09981	<i>Lepisosteus oculatus</i>	W5MHW9	2
ANGAN38603	<i>Anguilla anguilla</i>	XP_035237286	2
DANRE36783	<i>Danio rerio</i>	A0A2R8Q8T5	2
ESOLU39540	<i>Esox lucius</i>	ENSELUG00000009090.1	2
SALSA63007	<i>Salmo salar</i>	C0H962	2
ORYLA13842	<i>Oryzias latipes</i>	ENSORLG00000003593	2
ORENI03731	<i>Oreochromis niloticus</i>	ENSONIG00000004027	2
TAKRU15372	<i>Takifugu rubripes</i>	ENSTRUG000000012063	2
NEOBR03357	<i>Neolamprologus brichardi</i>	A0A3Q4HV10	2
AMPCI13505	<i>Amphilophus citrinellus</i>	A0A3Q0RHE0	2
MOLML02764	<i>Mola mola</i>	A0A3Q4B8R2	2
NOTFU16587	<i>Nothobranchius furzeri</i>	XP_015814443	2
HIPCM06019	<i>Hippocampus comes</i>	A0A3Q2YIA0	2
AMPPE26829	<i>Amphiprion percula</i>	A0A3P8RUU9	2
XENLA32018	<i>Xenopus laevis</i>	A0A1L8FFB1	3
XENTR12916	<i>Xenopus tropicalis</i>	TADBP_XENTR	3
CHRP132751	<i>Chrysemys picta bellii</i>	ENSCPBG000000022012.1	3
SPHPU16313	<i>Sphenodon punctatus</i>	ENSPUG00000006688.1	3
ANAPL05491	<i>Anas platyrhynchos</i>	ENSAPLG00000006001	3
ANAPP02225	<i>Anas platyrhynchos platyrhynchos</i>	U3IED4	3
CHICK10318	<i>Gallus gallus</i>	F1NBY1	3
MELGA07585	<i>Meleagris gallopavo</i>	G1MZJ1	3
SERCA17445	<i>Serinus canaria</i>	ENSSCAG00000016954.1	3
PARMJ11107	<i>Parus major</i>	ENSPMJG00000014837.1	3
MELUD13787	<i>Melopsittacus undulatus</i>	ENSMUNG00000016204.1	3
PELSI14462	<i>Pelodiscus sinensis</i>	K7FJ67	3
JUNHY18871	<i>Junco hyemalis</i>	ENSJHYG00000010677.1	3
TAEGU06701	<i>Taeniopygia guttata</i>	H0YYI1	3
FICAL11802	<i>Ficedula albicollis</i>	U3JXL3	3
CHEAB01735	<i>Chelonoidis abingdonii</i>	ENSCABG00000017429.1	3
ORNAN13991	<i>Ornithorhynchus anatinus</i>	F7EDX1	4
SARHA02491	<i>Sarcophilus harrisii</i>	G3WM39	4
DASNO07425	<i>Dasyus novemcinctus</i>	ENSDNOG00000008531	4
ERIEU01545	<i>Erinaceus europaeus</i>	ENSEEUG00000003755	4
CALJA18973	<i>Callithrix jacchus</i>	F7CL81	4
CERAT28967	<i>Cercopithecus atys</i>	A0A2K5LQ78	4
MACFA23554	<i>Macaca fascicularis</i>	A0A2K5WEW9	4
MACMU00269	<i>Macaca mulatta</i>	ENSMUG000000007456	4
MACNE36802	<i>Macaca nemestrina</i>	A0A2K6BK18	4
PAPAN00174	<i>Papio anubis</i>	A0A096NBQ0	4
MANLE13078	<i>Mandrillus leucophaeus</i>	A0A2K6A5S4	4
PANPA00278	<i>Pan paniscus</i>	A0A2R9C1F8	4
HUMAN47411	<i>Homo sapiens</i>	TADBP_HUMAN	4
CANLF07184	<i>Canis lupus familiaris</i>	ENSCAFG000000016759	4
VULVU24847	<i>Vulpes vulpes</i>	ENSVVUG000000027110.1	4
URSAM28564	<i>Ursus americanus</i>	A0A452R9Z0	4
AILME09206	<i>Ailuropoda melanoleuca</i>	D2HKA5	4
FELCA08644	<i>Felis catus</i>	A0A2I2U8C8	4
TURTR12812	<i>Tursiops truncatus</i>	ENSTTRG000000001312	4
LOXAF11400	<i>Loxodonta africana</i>	G3TD75	4
HORSE08842	<i>Equus caballus</i>	F6WAU6	4
PROCA06978	<i>Procavia capensis</i>	ENSPCAG000000010442	4
PIGXX29002	<i>Sus scrofa</i>	I3LNA4	4
BOVIN08303	<i>Bos taurus</i>	G3MX91	4
CAPHI06804	<i>Capra hircus</i>	A0A452G5Y7	4
MANJA47321	<i>Manis javanica</i>	XP_036855904	4
OCHPR06808	<i>Ochotona princeps</i>	ENSOPRG000000006352	4
DIPOR02872	<i>Dipodomys ordii</i>	ENSDORG000000007465	4
CRIGR20744	<i>Cricetulus griseus</i>	ENSCGRG000000007923.1	4
MOUSE42872	<i>Mus musculus</i>	TADBP_MOUSE	4
CAVPO10357	<i>Cavia porcellus</i>	A0A286XX33	4
HETGA00594	<i>Heterocephalus glaber</i>	G5C522	4
VICPA09903	<i>Vicugna pacos</i>	ENSVPAG000000009543	4
OTOGA09169	<i>Otolemur garnettii</i>	H0XFZ2	4
CHILA18976	<i>Chinchilla lanigera</i>	ENSCLAG000000014547.1	4
AOTNA20136	<i>Aotus nancymaae</i>	A0A2K5EQY0	4
TUPBE02937	<i>Tupaia belangeri</i>	ENSTBEG000000000263	4
CAVAP09087	<i>Cavia aperea</i>	ENSCAPG000000015189.1	4
PHACI11189	<i>Phascogalea cinereus</i>	A0A6P5JEZ4	4
SAIBB26155	<i>Saimiri boliviensis boliviensis</i>	A0A2K6TZ91	4
ICTTR08691	<i>Ictidomys tridecemlineatus</i>	I3MBK5	4
JACJA09000	<i>Jaculus jaculus</i>	ENSJJAG000000014196.1	4
MYOLU14708	<i>Myotis lucifugus</i>	G1P208	4
CHLSB09400	<i>Chlorocebus sabaeus</i>	A0A0D9S8N2	4

RHIBE13964	<i>Rhinopithecus bieti</i>	A0A2K6LE20	4
RHIRO33581	<i>Rhinopithecus roxellana</i>	A0A2K6QV39	4
EQUAS11333	<i>Equus asinus</i>	ENSEASG00005019017.1	4
COLAP32291	<i>Colobus angolensis palliatus</i>	A0A2K5IA07	4
FUKDA15705	<i>Fukomys damarensis</i>	A0A091DKG3	4
NANGA15865	<i>Nannospalax galili</i>	ENSNGAG00000023416.1	4
CARSF02449	<i>Carlito syrichta</i>	A0A1U7SU45	4
FUS			
LEPOC18149	<i>Lepisosteus oculatus</i>	W5LVA3	2
ANGAN53300	<i>Anguilla anguilla</i>	XP_035254071	2
ANATE28040	<i>Anabas testudineus</i>	A0A3Q1JBK0	2
SERDU24279	<i>Seriola dumerili</i>	A0A3B4UPE4	2
SCOMX14580	<i>Scophthalmus maximus</i>	ENSSMAG00000011685.1	2
TETNG10220	<i>Tetraodon nigroviridis</i>	H3D156	2
NOTFU30467	<i>Nothobranchius furzeri</i>	A0A1A7ZC85	2
KRYMA15889	<i>Kryptolebias marmoratus</i>	A0A3Q2ZW58	2
ASTCA29489	<i>Astatotilapia calliptera</i>	A0A3P8QDK7	2
HAPBU09804	<i>Haplochromis burtoni</i>	A0A3Q2W7N5	2
NEOBR22012	<i>Neolamprologus brichardi</i>	A0A3Q4H4B7	2
AMPOC31093	<i>Amphiprion ocellaris</i>	A0A3Q1B6C3	2
AMPPE11049	<i>Amphiprion percula</i>	A0A3P8SA41	2
ESOLU17742	<i>Esox lucius</i>	A0A3P8Z6T5	2
SALSA21612	<i>Salmo salar</i>	A0A1S3R6J7	2
ASTMX07328	<i>Astyanax mexicanus</i>	W5KQF2	2
PYGNA09213	<i>Pygocentrus nattereri</i>	A0A3B4E8M4	2
ICTPU24291	<i>Ictalurus punctatus</i>	ENSIPUG00000007342.1	2
DANRE29660	<i>Danio rerio</i>	F1R0M4	2
LATCH06576	<i>Latimeria chalumnae</i>	H3A671	2
ECHE00861	<i>Echinops telfairi</i>	ENSETEG00000015358	2
PONAB04910	<i>Pongo abelii</i>	H2NQS4	2
CHRP129771	<i>Chrysemys picta bellii</i>	ENSCPBG00000008979.1	3
CHEAB27054	<i>Chelonoidis abingdonii</i>	ENSCABG00000010374.1	3
SPHPU24372	<i>Sphenodon punctatus</i>	ENSSPUG00000002998.1	3
ANOCA17128	<i>Anolis carolinensis</i>	ENSACAG00000006776	3
XENTR14904	<i>Xenopus tropicalis</i>	Q28EL3	3
LOXAF05575	<i>Loxodonta africana</i>	G3SNY8	4
RABIT15423	<i>Oryctolagus cuniculus</i>	ENSOCUG00000021526.2	4
DIPOR01258	<i>Dipodomys ordii</i>	ENSODRG00000014218	4
FUKDA20682	<i>Fukomys damarensis</i>	ENSFDAG00000007552.1	4
HETGA21202	<i>Heterocephalus glaber</i>	ENSHGLG00100006254.1	4
CHILA19792	<i>Chinchilla lanigera</i>	ENSCLAG00000010457.1	4
OCTDE00228	<i>Octodon degus</i>	A0A6P3EP53	4
JACJA23117	<i>Jaculus jaculus</i>	ENSJJAG00000011802.1	4
CRIGR21309	<i>Cricetulus griseus</i>	ENSCGRG00000012106.1	4
MOUSE56790	<i>Mus musculus</i>	FUS_MOUSE	4
RATNO02010	<i>Rattus norvegicus</i>	Q5PQK2	4
NANGA16163	<i>Nannospalax galili</i>	ENSNGAG00000023361.1	4
ICTTR12793	<i>Ictidomys tridecemlineatus</i>	ENSSTOG00000005943	4
CERAT36652	<i>Cercopithecus atys</i>	A0A2K5M862	4
CHLSB14622	<i>Chlorocebus sabaeus</i>	A0A0D9R082	4
MACFA24759	<i>Macaca fascicularis</i>	A0A2K5UY14	4
MACMU14458	<i>Macaca mulatta</i>	ENSMMUG00000019637	4
MACNE37188	<i>Macaca nemestrina</i>	A0A2K6B648	4
MANLE13941	<i>Mandrillus leucophaeus</i>	A0A2K5ZIM2	4
PAPAN11413	<i>Papio anubis</i>	A0A096NHV1	4
COLAP30700	<i>Colobus angolensis palliatus</i>	A0A2K5K1V9	4
RHIBE34040	<i>Rhinopithecus bieti</i>	A0A2K6MN44	4
RHIRO31968	<i>Rhinopithecus roxellana</i>	A0A2K6PBV8	4
GORGO07778	<i>Gorilla gorilla gorilla</i>	ENSGGOG000000003707	4
HUMAN27992	<i>Homo sapiens</i>	FUS_HUMAN	4
PANPA15069	<i>Pan paniscus</i>	A0A2R9A818	4
PANTR12223	<i>Pan troglodytes</i>	H2RA80	4
NOMLE17198	<i>Nomascus leucogenys</i>	ENSNLEG00000016218	4
AOTNA37715	<i>Aotus nancymae</i>	A0A2K5CEH5	4
CALJA04844	<i>Callithrix jacchus</i>	ENSCJAG00000003169	4
SAIBB16827	<i>Saimiri boliviensis boliviensis</i>	A0A2K6UY94	4
CARSF03981	<i>Carlito syrichta</i>	ENSTSYG00000011757	4
PROCO05597	<i>Propithecus coquereli</i>	A0A2K6G321	4
BOVIN20650	<i>Bos taurus</i>	A0A140T861	4
CAPHI17070	<i>Capra hircus</i>	A0A452DQG8	4
PIGXX20573	<i>Sus scrofa</i>	ENSSSCG000000030798.2	4
VICPA09706	<i>Vicugna pacos</i>	ENSPVAG00000004211	4
CANLF16361	<i>Canis lupus familiaris</i>	ENSCAFG00000016862	4
VULVU24268	<i>Vulpes vulpes</i>	ENSVVUG000000027797.1	4
MUSPF04187	<i>Mustela putorius furo</i>	M3YJ27	4
AILME07018	<i>Ailuropoda melanoleuca</i>	G1LKT4	4
URSAM07884	<i>Ursus americanus</i>	A0A452QX30	4
FELCA17000	<i>Felis catus</i>	ENSFCAG000000027928	4
PTEVA04324	<i>Pteropus vampyrus</i>	ENSPVAG00000003097	4
MYOLU16715	<i>Myotis lucifugus</i>	G1P397	4
ERIEU03865	<i>Erinaceus europaeus</i>	ENSEEUG00000004934	4

EQUAS04518	<i>Equus asinus</i>	ENSEASG00005011180.1	4
MANJA23313	<i>Manis javanica</i>	XP_017521732	4
DASNO08858	<i>Dasyus novemcinctus</i>	ENSDNOG00000017091	4
CHOHO04231	<i>Choloepus hoffmanni</i>	ENSCHOG00000010780	4
SARHA12777	<i>Sarcophilus harrisii</i>	G3WEM9	4
MONDO11342	<i>Monodelphis domestica</i>	ENSMODG00000001889	4
MACEU03227	<i>Macropus eugenii</i>	ENSMEUG00000000968	4
PHACI08601	<i>Phascolarctos cinereus</i>	ENSPCIG00000015897.1	4
Nup153			
XENTR02157	<i>Xenopus tropicalis</i>	ENSXETG00000014197	1
CIOIN15442	<i>Ciona intestinalis</i>	F6ZT40	1
LEPOC02490	<i>Lepisosteus oculatus</i>	W5MNS7	2
ANGAN03227	<i>Anguilla anguilla</i>	XP_035291407	2
ANATE05308	<i>Anabas testudineus</i>	A0A3Q1JXD7	2
SERDU08610	<i>Seriola dumerili</i>	A0A3B4U9G7	2
CYNSE14136	<i>Cynoglossus semilaevis</i>	A0A3P8WRB1	2
SCOMX19706	<i>Scophthalmus maximus</i>	ENSSMAG00000008418.1	2
GASAC05602	<i>Gasterosteus aculeatus</i>	G3NP66	2
MOLML09099	<i>Mola mola</i>	A0A3Q3X2J2	2
TAKRU10012	<i>Takifugu rubripes</i>	ENSTRUG00000010163	2
TETNG09217	<i>Tetraodon nigroviridis</i>	H3CUV8	2
ORYLA02168	<i>Oryzias latipes</i>	ENSORLG00000012674	2
ORYME27144	<i>Oryzias melastigma</i>	A0A3B3BZH3	2
NOTFU28633	<i>Nothobranchius furzeri</i>	XP_015829300	2
KRYMA01537	<i>Kryptolebias marmoratus</i>	A0A3Q3F4A6	2
CYPVA16762	<i>Cyprinodon variegatus</i>	A0A3Q2EGJ3	2
POEFO03854	<i>Poecilia formosa</i>	A0A087Y5I5	2
POERE03600	<i>Poecilia reticulata</i>	A0A3P9NMY4	2
XIPMA17891	<i>Xiphophorus maculatus</i>	ENSXMAG00000003437	2
ASTCA22790	<i>Astatotilapia calliptera</i>	A0A3P8Q5S6	2
HAPBU04455	<i>Haplochromis burtoni</i>	A0A3Q2V3R9	2
NEOBR07702	<i>Neolamprologus brichardi</i>	A0A3Q4H186	2
ORENI21884	<i>Oreochromis niloticus</i>	I3J2S0	2
AMPCI22627	<i>Amphilophus citrinellus</i>	A0A3Q0RT68	2
AMPOC02144	<i>Amphiprion ocellaris</i>	A0A3Q1ASF8	2
AMPPE25635	<i>Amphiprion percula</i>	A0A3P8SWR4	2
HIPCM26771	<i>Hippocampus comes</i>	A0A3Q2YQJ3	2
GADMO09197	<i>Gadus morhua</i>	ENSGMOG00000010422	2
ESOLU15511	<i>Esox lucius</i>	A0A3P9AD61	2
ASTMX04317	<i>Astyanax mexicanus</i>	ENSAMXG00000021025	2
PYGNA00871	<i>Pygocentrus nattereri</i>	A0A3B4D0Y3	2
ICTPU18872	<i>Ictalurus punctatus</i>	A0A2D0PV24	2
DANRE15074	<i>Danio rerio</i>	A0A0R4IVZ0	2
LATCH05943	<i>Latimeria chalumnae</i>	H3ACD6	2
OCHPR04515	<i>Ochotona princeps</i>	ENSOPRG00000006221	3
COLAP01419	<i>Colobus angolensis palliatus</i>	A0A2K5IW93	3
CARSF08065	<i>Carlito syrichta</i>	ENSTSYG00000001251	3
URSMA09101	<i>Ursus maritimus</i>	A0A452USF3	3
ERIEU07856	<i>Erinaceus europaeus</i>	ENSEEUG00000010183	3
MELGA08705	<i>Meleagris gallopavo</i>	G1MW17	3
CHICK14841	<i>Gallus gallus</i>	A0A1D5PXQ1	3
FICAL02256	<i>Ficedula albicollis</i>	U3KG17	3
PARMJ14580	<i>Parus major</i>	ENSPMJG00000018241.1	3
JUNHY01031	<i>Junco hyemalis</i>	ENSJHYG00000015545.1	3
TAEGU05586	<i>Taeniopygia guttata</i>	ENSTGUG00000005917	3
SERCA00052	<i>Serinus canaria</i>	ENSSCAG00000000585.1	3
MELUD05600	<i>Melopsittacus undulatus</i>	ENSMUNG00000009250.1	3
CHRP105922	<i>Chrysemys picta bellii</i>	ENSCPBG00000008333.1	3
CHEAB03264	<i>Chelonoidis abingdonii</i>	ENSCABG000000005639.1	3
PELSI16509	<i>Pelodiscus sinensis</i>	K7GGR9	3
SPHPU04687	<i>Sphenodon punctatus</i>	ENSSPUG00000016687.1	3
ANOCA05697	<i>Anolis carolinensis</i>	G1KCK7	3
ORNAN12721	<i>Ornithorhynchus anatinus</i>	ENSOANG00000011858	4
LOXAF08097	<i>Loxodonta africana</i>	G3TFY7	4
RABIT14444	<i>Oryctolagus cuniculus</i>	G1SIZ4	4
DIPOR15101	<i>Dipodomys ordii</i>	ENSDORG00000001736	4
HETGA17013	<i>Heterocephalus glaber</i>	ENSHGLG00100001529.1	4
CAVPO07965	<i>Cavia porcellus</i>	ENSCPOG00000004007	4
CHILA18633	<i>Chinchilla lanigera</i>	ENSCLAG00000001915.1	4
OCTDE06300	<i>Octodon degus</i>	ENSODEG00000005620.1	4
JACJA19740	<i>Jaculus jaculus</i>	ENSJJAG00000016760.1	4
CRIGR01418	<i>Cricetulus griseus</i>	ENSCGRG00000009244.1	4
MOUSE10788	<i>Mus musculus</i>	E9Q3G8	4
RATNO08849	<i>Rattus norvegicus</i>	G3V662	4
NANGA14919	<i>Nannospalax galili</i>	ENSNGAG00000013817.1	4
ICTTR03195	<i>Ictidomys tridecemlineatus</i>	ENSTTOG000000005807	4
CERAT09565	<i>Cercocebus atys</i>	A0A2K5NJ22	4
CHLSB06650	<i>Chlorocebus sabaues</i>	A0A0D9R5K0	4
MACFA33008	<i>Macaca fascicularis</i>	A0A2K5TL19	4
MACMU16280	<i>Macaca mulatta</i>	ENSMMUG00000001993	4
MACNE10264	<i>Macaca nemestrina</i>	A0A2K6CZZ2	4

MANLE36126	<i>Mandrillus leucophaeus</i>	A0A2K5YCI0	4
RHIBE28287	<i>Rhinopithecus bieti</i>	A0A2K6MPT9	4
RHIRO11751	<i>Rhinopithecus roxellana</i>	A0A2K6P3I1	4
GORG017159	<i>Gorilla gorilla gorilla</i>	ENSGGOG00000014261	4
HUMAN83246	<i>Homo sapiens</i>	NU153 HUMAN	4
PANPA33599	<i>Pan paniscus</i>	A0A2R9A8E2	4
PANTR37299	<i>Pan troglodytes</i>	A0A2I3SL61	4
PONAB13047	<i>Pongo abelii</i>	H2PI06	4
NOMLE06341	<i>Nomascus leucogenys</i>	G1QKG7	4
AOTNA16236	<i>Aotus nancymae</i>	A0A2K5BV75	4
CALJA14506	<i>Callithrix jacchus</i>	F7EHG8	4
SAIBB12360	<i>Saimiri boliviensis boliviensis</i>	A0A2K6T874	4
MICMU04339	<i>Microcebus murinus</i>	ENSMICG00000006089	4
PROCO19645	<i>Propithecus coquereli</i>	A0A2K6EY30	4
OTOGA13338	<i>Otolemur garnettii</i>	H0XC65	4
BOVIN19169	<i>Bos taurus</i>	A0A3Q1MAS6	4
CAPHI15081	<i>Capra hircus</i>	A0A452FV25	4
SHEEP10664	<i>Ovis aries</i>	W5PPG7	4
PIGXX30834	<i>Sus scrofa</i>	A0A480MLT8	4
VICPA01837	<i>Vicugna pacos</i>	ENSVFAG00000002072	4
CANLF13754	<i>Canis lupus familiaris</i>	ENSCAFG00000010146	4
VULVU13037	<i>Vulpes vulpes</i>	A0A3Q7S734	4
MUSPF09744	<i>Mustela putorius furo</i>	ENSMFUG00000005159	4
AILME04788	<i>Ailuropoda melanoleuca</i>	G1LIN5	4
FELCA05138	<i>Felis catus</i>	ENSFCAG00000011583	4
PTEVA10137	<i>Pteropus vampyrus</i>	ENSPVAG00000012632	4
MYOLU06869	<i>Myotis lucifugus</i>	G1P376	4
SORAR06050	<i>Sorex araneus</i>	ENSSARG00000013269	4
EQUAS08047	<i>Equus asinus</i>	ENSEASG00005009945.1	4
HORSE09415	<i>Equus caballus</i>	ENSECAG00000016010	4
MANJA39534	<i>Manis javanica</i>	XP_036850301	4
DASNO13918	<i>Dasyurus novemcinctus</i>	ENSDNOG00000001153	4
MONDO04745	<i>Monodelphis domestica</i>	ENSMODG00000010951	4
PHACI10774	<i>Phascogale carolinensis</i>	ENSPICG00000029680.1	4

TNRC6A

ANATE29102	<i>Anabas testudineus</i>	A0A3Q1HCA4	2
SERDU24303	<i>Seriola dumerili</i>	A0A3B4UQH3	2
CYNSE12687	<i>Cynoglossus semilaevis</i>	A0A3P8WUF2	2
SCOMX13694	<i>Scophthalmus maximus</i>	ENSSMAG00000008293.1	2
GASAC08536	<i>Gasterosteus aculeatus</i>	G3PMS9	2
MOLML21320	<i>Mola mola</i>	A0A3Q3WUT0	2
TAKRU11363	<i>Takifugu rubripes</i>	ENSTRUG00000009839	2
ORYLA16918	<i>Oryzias latipes</i>	ENSORLG00000012798	2
ORYME17187	<i>Oryzias melastigma</i>	A0A3B3CNE4	2
NOTFU05655	<i>Nothobranchius furzeri</i>	A0A1A8A0C2	2
KRYMA10889	<i>Kryptolebias marmoratus</i>	A0A3Q3B4A0	2
CYPVA18089	<i>Cyprinodon variegatus</i>	A0A3Q2DHY7	2
POEFO13649	<i>Poecilia formosa</i>	A0A087XTC6	2
POERE31547	<i>Poecilia reticulata</i>	A0A3P9Q3C9	2
XIPMA10115	<i>Xiphophorus maculatus</i>	ENSXMAG0000001697	2
ASTCA30898	<i>Astatotilapia calliptera</i>	A0A3P8PDD9	2
HAPBU25992	<i>Haplochromis burtoni</i>	A0A3Q2WY78	2
ORENI21771	<i>Oreochromis niloticus</i>	I3KF62	2
AMPCI16532	<i>Amphilophus citrinellus</i>	A0A3Q0SBA3	2
AMPOC21905	<i>Amphiprion ocellaris</i>	A0A3Q1DGT8	2
AMPPE09980	<i>Amphiprion percula</i>	A0A3P8T1Z7	2
HIPCM18813	<i>Hippocampus comes</i>	A0A3Q2Z453	2
ESOLU15775	<i>Esox lucius</i>	ENSELUG00000013855.1	2
ASTMX21398	<i>Astyanax mexicanus</i>	ENSAMXG00000019919	2
PYGNA21286	<i>Pygocentrus nattereri</i>	A0A3B4DXJ0	2
ICTPU25584	<i>Ictalurus punctatus</i>	ENSIPUG0000001602.1	2
DANRE29842	<i>Danio rerio</i>	A0A0R4IJ26	2
LATCH11830	<i>Latimeria chalumnae</i>	H3ABQ4	2
PROCA10536	<i>Procavia capensis</i>	ENSPCAG00000008260	2
OCHPR12848	<i>Ochotona princeps</i>	ENSOPRG00000005819	2
DIPOR07487	<i>Dipodomys ordii</i>	ENDORG00000014734	2
ICTTR12692	<i>Ictidomys tridecemlineatus</i>	ENSTTSG00000023394	2
CARSF10784	<i>Carlito syrichta</i>	ENSTSYG00000012711	2
MICMU06378	<i>Microcebus murinus</i>	ENSMICG00000001883	2
CHOHO09049	<i>Choloepus hoffmanni</i>	ENSCHOG00000009374	2
MELGA04507	<i>Meleagris gallopavo</i>	G1N3Z6	2
PELSI06952	<i>Pelodiscus sinensis</i>	ENSPSIG00000003965	2
PAPAN11340	<i>Papio anubis</i>	ENSPANG00000018523	3
MUSPF02299	<i>Mustela putorius furo</i>	M3YLM1	3
SARHA15034	<i>Sarcophilus harrisii</i>	G3W445	3
MONDO11658	<i>Monodelphis domestica</i>	ENSMODG00000016348	3
ANAPL06809	<i>Anas platyrhynchos</i>	ENSAPLG00000009978	3
ANAPP26723	<i>Anas platyrhynchos platyrhynchos</i>	A0A493TP73	3
CHICK02910	<i>Gallus gallus</i>	F1NAU7	3
FICAL04956	<i>Ficedula albicollis</i>	U3JMB2	3
PARMJ04275	<i>Parus major</i>	ENSPMJG00000019495.1	3

JUNHY15710	<i>Junco hyemalis</i>	ENSIHYG00000002230.1	3
TAEGU02638	<i>Taeniopygia guttata</i>	ENSTGUG00000006142	3
SERCA11590	<i>Serinus canaria</i>	ENSSCAG00000003490.1	3
MELUD20429	<i>Melopsittacus undulatus</i>	ENSMUNG00000002613.1	3
CHRP101613	<i>Chrysemys picta bellii</i>	ENSCPBG00000001881.1	3
CHEAB20009	<i>Chelonoidis abingdonii</i>	ENSCABG000000016518.1	3
SPHPU07887	<i>Sphenodon punctatus</i>	ENSSPUG000000010525.1	3
ANOCA11291	<i>Anolis carolinensis</i>	ENSACAG000000011857	3
XENTR01823	<i>Xenopus tropicalis</i>	ENSXETG000000030437	3
ORNAN00398	<i>Ornithorhynchus anatinus</i>	ENSOANG000000007672	4
LOXAF05468	<i>Loxodonta africana</i>	G3STU5	4
ECHE12195	<i>Echinops telfairi</i>	ENSETEG000000014658	4
RABIT11241	<i>Oryctolagus cuniculus</i>	ENSOCUG000000004325.3	4
HETGA19908	<i>Heterocephalus glaber</i>	ENSHGLG00100012496.1	4
CHILA19632	<i>Chinchilla lanigera</i>	ENSLAG000000010217.1	4
OCTDE00070	<i>Octodon degus</i>	ENSODEG000000012802.1	4
CRIGR22441	<i>Cricetulus griseus</i>	ENSCGRG000000013351.1	4
MOUSE56368	<i>Mus musculus</i>	TNR6A_MOUSE	4
RATNO1902	<i>Rattus norvegicus</i>	A0A0G2JZJ3	4
NANGA18433	<i>Nannospalax galili</i>	ENSNGAG000000020702.1	4
CERAT31849	<i>Cercopithecus atys</i>	A0A2K5L032	4
CHLSB14521	<i>Chlorocebus sabaeus</i>	A0A0D9R2K7	4
MACFA24538	<i>Macaca fascicularis</i>	A0A2K5WC69	4
MACMU14332	<i>Macaca mulatta</i>	ENSMUG0000000020428	4
MACNE37716	<i>Macaca nemestrina</i>	A0A2K6CFH5	4
MANLE09220	<i>Mandrillus leucophaeus</i>	A0A2K6AAZ0	4
COLAP38355	<i>Colobus angolensis palliatus</i>	A0A2K5K3P5	4
RHIBE00537	<i>Rhinopithecus bieti</i>	A0A2K6JV14	4
GORG007659	<i>Gorilla gorilla gorilla</i>	ENSGGOG000000005137	4
HUMAN27314	<i>Homo sapiens</i>	TNR6A_HUMAN	4
PANPA14856	<i>Pan paniscus</i>	A0A2R9A4V3	4
PANTR11987	<i>Pan troglodytes</i>	H2R4F2	4
PONAB04808	<i>Pongo abelii</i>	H2NQF8	4
NOMLE05264	<i>Nomascus leucogenys</i>	ENSNLEG000000011851	4
AOTNA13844	<i>Aotus nancymae</i>	A0A2K5D1C1	4
CALJA04720	<i>Callithrix jacchus</i>	ENSCJAG000000020677	4
SAIBB17092	<i>Saimiri boliviensis boliviensis</i>	A0A2K6V8D8	4
PROCO17866	<i>Propithecus coquereli</i>	A0A2K6ENQ1	4
OTOGA06942	<i>Otolemur garnettii</i>	H0WUN8	4
TUPBE12837	<i>Tupaia belangeri</i>	ENSTBEG000000014965	4
BOVIN20475	<i>Bos taurus</i>	G3N258	4
CAPHI16912	<i>Capra hircus</i>	A0A452F9V6	4
SHEEP12355	<i>Ovis aries</i>	W5Q977	4
PIGXX38979	<i>Sus scrofa</i>	A0A287BNE0	4
CANLF16466	<i>Canis lupus familiaris</i>	ENSCAFG000000017531	4
VULVU34014	<i>Vulpes vulpes</i>	ENSVVUG000000005380.1	4
AILME02472	<i>Ailuropoda melanoleuca</i>	G1LRX8	4
URSAM01166	<i>Ursus americanus</i>	A0A452QQI8	4
URSM28208	<i>Ursus maritimus</i>	A0A452TSR0	4
PTEVA04038	<i>Pteropus vampyrus</i>	ENSPVAG000000012066	4
MYOLU03029	<i>Myotis lucifugus</i>	G1NYH2	4
ERIEU09113	<i>Erinaceus europaeus</i>	ENSEEUG000000004530	4
EQUAS29758	<i>Equus asinus</i>	ENSEASG00005017693.1	4
HORSE04775	<i>Equus caballus</i>	ENSECAG000000025056	4
MANJA36779	<i>Manis javanica</i>	XP_036847877	4
DASNO11072	<i>Dasyurus novemcinctus</i>	ENSDNOG000000013989	4
PHACI20675	<i>Phascolarctos cinereus</i>	ENSPCIG000000012106.1	4

¹The taxonomic rank. 1: (Cordata; 2: (Vertebrata; 3: (Tetrapoda; 4: (Mammalia))).

References

1. Romero, Obradovic & Dunker, K. Sequence Data Analysis for Long Disordered Regions Prediction in the Calcineurin Family. *Genome Inform Ser Workshop Genome Inform* **8**, 110-124 (1997).
2. Radivojac, P., Obradovic, Z., Brown, C.J. & Dunker, A.K. Prediction of boundaries between intrinsically ordered and disordered protein regions. *Pac Symp Biocomput*, 216-27 (2003).
3. Hu, G. et al. fIDPnn: Accurate intrinsic disorder prediction with putative propensities of disorder functions. *Nat Commun* **12**, 4438 (2021).
4. Gerstberger, S., Hafner, M. & Tuschl, T. A census of human RNA-binding proteins. *Nat Rev Genet* **15**, 829-45 (2014).
5. Zhao, B. et al. Intrinsic Disorder in Human RNA-Binding Proteins. *J Mol Biol* **433**, 167229 (2021).
6. Altenhoff, A.M. et al. OMA orthology in 2021: website overhaul, conserved isoforms, ancestral gene order and more. *Nucleic Acids Res* **49**, D373-D379 (2021).
7. Sievers, F. et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* **7**, 539 (2011).
8. Cock, P.J. et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics* **25**, 1422-3 (2009).
9. Capra, J.A. & Singh, M. Predicting functionally important residues from sequence conservation. *Bioinformatics* **23**, 1875-82 (2007).
10. Necci, M., Piovesan, D., Predictors, C., DisProt, C. & Tosatto, S.C.E. Critical assessment of protein intrinsic disorder prediction. *Nat Methods* **18**, 472-481 (2021).

Phase separation driven by interchangeable properties in the intrinsically disordered regions of protein paralogs

Shih-Hui Chiu¹, Wen-Lin Ho¹, Yung-Chen Sun¹, Jean-Cheng Kuo¹ & Jie-rong Huang^{1,2,3}  

Paralogs, arising from gene duplications, increase the functional diversity of proteins. Protein functions in paralog families have been extensively studied, but little is known about the roles that intrinsically disordered regions (IDRs) play in their paralogs. Without a folded structure to restrain them, IDRs mutate more diversely along with evolution. However, how the diversity of IDRs in a paralog family affects their functions is unexplored. Using the RNA-binding protein Musashi family as an example, we applied multiple structural techniques and phylogenetic analysis to show how members in a paralog family have evolved their IDRs to different physicochemical properties but converge to the same function. In this example, the lower prion-like tendency of Musashi-1's IDRs, rather than Musashi-2's, is compensated by its higher α -helical propensity to assist their assembly. Our work suggests that, no matter how diverse they become, IDRs could evolve different traits to a converged function, such as liquid-liquid phase separation.

¹Institute of Biochemistry and Molecular Biology, National Yang Ming Chiao Tung University, No. 155 Section 2, Li-nong Street, Taipei, Taiwan. ²Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, No. 155 Section 2, Li-nong Street, Taipei, Taiwan. ³Department of Life Sciences and Institute of Genome Sciences, National Yang Ming Chiao Tung University, No. 155 Section 2, Li-nong Street, Taipei, Taiwan. ✉email: jierongh@nycu.edu.tw

Recurrent gene duplications over many generations create paralogs in a genome, giving rise to a myriad of new protein functions¹. The redundant copy, if not silenced as a pseudogene (the non-functional copy from its ancestor), enhances survival, and many duplicated genes mutate toward new functions². The hemoglobin, composing paralog α - and β -globin chains, is a textbook example³. The existence of two copies of the α -globin gene (*HBA1* and *HBA2*) reduces the effect of α -thalassemia, but not the β -thalassemia, which results from the loss of the only one β -globin encoding gene (*HBB*); These paralogs, which have evolved from an ancestral oxygen-binding globin, collaborate in their new function of oxygen transport³. Although protein paralogs have been studied for decades, the paralogs of intrinsically disordered proteins (IDPs) or proteins with intrinsically disordered regions (IDRs), account for more than half of the eukaryotic proteome⁴, remain largely ignored.

Our particular interest is in RNA-binding proteins (RBPs), which are at the core of gene regulation and RNA metabolism and whose dysfunction is implicated in many diseases⁵. In addition to the common feature of having RNA interacting motifs⁶, RBPs often contain IDRs^{7,8}. The role of these IDRs had been unclear until it was recently shown that they can organize cellular structures without a lipid membrane⁹. These membraneless organelles, which assemble via liquid-liquid phase separation (LLPS), have since been extensively studied and have become a working model of spatiotemporal control for many cellular functions¹⁰. Although some RBPs' IDRs have been reported involved in forming membraneless organelles (such as RNA or stress granules), a large proportion of them are still of unknown function. The RBPs themselves are often paralogs, many of which share highly conserved RNA-binding domains but with different IDRs. In the manually curated list of RBPs (1542 RBPs in total)¹¹, more than 22% (341/1542) are annotated in one of the paralog families in the OrthoMCL database¹², and more than 48% are paralogs (749/1542) as defined in the original census study based on sequence similarity¹¹ (see Supplementary Data 1 and Supplementary Data 2 for the lists). These paralogs have similar folded domains, hinting at similar RNA recognition mechanisms, whereas many disordered regions have greater sequence diversity.

To understand the different properties of IDRs in paralogs, we focus on those families of proteins that have two members and found that the Musashi protein family is a suitable example for this purpose. The Musashi gene was originally identified in the fruit fly (*Drosophila melanogaster*), responsible for sensory organ development¹³ and is highly conserved in animals^{14–18}. In mammals, the two Musashi paralogs are translational regulators of cell fate and are involved in maintaining the stem-cell state, differentiation, and tumorigenesis¹⁹. In addition to their role in neural stem-cell development^{17,18,20}, they also regulate several types of cancer^{21,22}. The C-terminal domain (an IDR) is critical for forming chemoresistant stress granules in glioblastoma²³ and colorectal cancer cells²⁴. Musashi proteins cannot join stress granules without it^{23,24}. Furthermore, the toxic oligomers formed by Musashi proteins have been implicated in Alzheimer's disease^{25,26}. We noticed through sequence analysis that the IDRs of Musashi-1 and -2 have different physical properties, which are conserved among vertebrate orthologs. Here, our data suggest that the decreased prion-likeness (the level of amino acid composition resembling that of prion proteins) of one paralog is compensated in assembly formation by an increase in α -helical propensity. We also compare the IDRs' properties of other well-studied RBPs related to Musashi proteins, including TDP-43 and hnRNP A1. These results show how different properties may have evolved in IDR paralogs for the same biophysical mechanism.

Results

The IDRs of the Musashi family have different prion propensities. We used the primary sequence of the folded domain of fruit fly Musashi (residues 29–195, a predicted RNA recognition motif (RRM)) to identify orthologs in the UniProt database (i.e. ignoring its IDR). We selected the model organisms²⁷ (listed in Supplementary Table 1) with similar RRM sequences and constructed a phylogenetic tree using their full-length sequences (Fig. 1a). The nematode and fruit fly have only one Musashi gene, but vertebrates have two paralogs (Fig. 1a). The C-terminal half of all these proteins are intrinsically disordered (purple bars in Fig. 1a and Supplementary Fig. 1). Note that the IDR was not used to identify orthologs but is a common feature of all Musashi proteins. Results suggest the IDR is involved in Musashi-1 joining stress granules^{23,24}. The stress-granule-related proteins often possess a prion-like domain^{28,29}, which is similar to prion behavior to assist assembly but does not necessarily aggregate. We, therefore, used the PLAAC algorithm³⁰ to predict prion-likeness. Although all the IDRs of Musashi-2 in the vertebrates and lower animals are prion-like, the IDRs of Musashi-1 orthologs are less so (Fig. 1a). Figure 1b compares the human Musashi paralogs, where the IDRs cover residues 237–362 (Msi-1C) and 235–328 (Msi-2C) (red box). We separated all Musashi protein sequences into IDRs and RRM based on their alignment to the corresponding human ortholog (Fig. 1b and Supplementary Fig. 2). The RRM in Musashi-1 and Musashi-2 have very similar amino acid compositions (Fig. 1c, d and Supplementary Table 2). On the contrary, although the IDRs of the two paralogs have many glycines, prolines, and alanines (Fig. 1c and Supplementary Table 2), their amino acid sequences differ substantially (Fig. 1d, lower panel). Sorting the amino acids in terms of prion-forming propensity³⁰ reveals that the Musashi-1 orthologs have fewer prion-promoting amino acids, such as glutamine and asparagine than Musashi-2 orthologs do. This amino acid composition analysis reinforces the suggestion that Musashi-2 is more prion-like (Fig. 1d).

The disordered regions of Musashi proteins have a polyalanine region that forms an α -helix.

We purified the human Musashi proteins' IDRs to investigate their differences experimentally (Supplementary Fig. 3). Their circular dichroism (CD) spectra show that they are mostly unstructured (Fig. 2a). The CD patterns of Msi-1C and Msi-2C at pH 5.5, 283 K (the conditions at which the proteins were most stable and soluble, see below), are similar overall but differ slightly around 220 nm, hinting at a potential difference in α -helicity^{31,32}. We assigned the NMR chemical shifts of the two IDRs to obtain residue-specific structural propensities (BMRB accesses: 51207 and 51208). In the ¹⁵N-edited heteronuclear single-quantum coherence (HSQC) spectra (Fig. 2b), most of the amide proton signals are within 1 ppm, confirming the disordered nature of these domains³³. Nevertheless, both regions contain a stretch of residues in which the secondary chemical shifts (the differences between the measured chemical shifts and random-coil values) are positive for C α and C' atoms and negative for C β atoms, indicating a propensity of an α -helix (Fig. 2c). The deviations from random-coil values are smaller for Msi-2C than for Msi-1C (Fig. 2c). δ 2D predictions³⁴ based only on chemical shifts (with no missing assignments in the α -helical region) indicate that the α -helical propensity of Msi-1C is higher than Msi-2C's (residue-specific values up to ~50% vs up to ~20%; red bars in Fig. 2d). These α -helical forming regions correspond to the polyalanine stretches of the two proteins (Fig. 2d, e). Msi-1C contains an eight-alanine-repeat whereas Msi-2C's polyalanine stretch is interrupted (and its α -helical propensity reduced) by a valine^{35,36}. Importantly, the

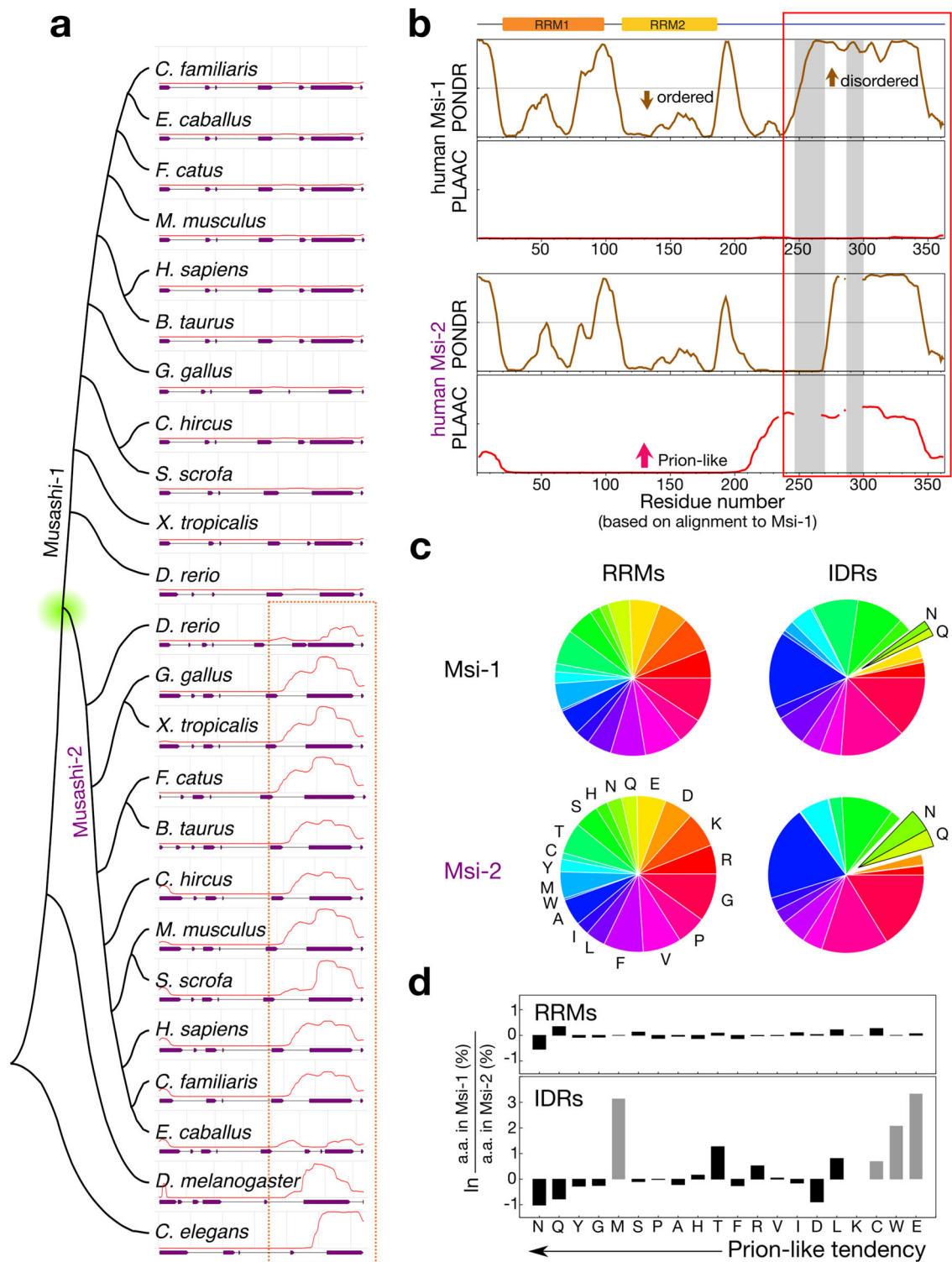


Fig. 1 Prion-like nature of Musashi protein homologs. **a** Phylogenetic tree (constructed using MEGA X⁶²) of Musashi proteins in model organisms. The red lines indicate the PLAAC prion propensity scores³⁰. The purple bars indicate regions predicted to be disordered by the VSL2 algorithm in PONDR⁶¹. The dashed box highlights the most prion-like regions. **b** Disorder and prion-likeness analysis of human Musashi proteins. Residues are numbered according to the alignment with Musashi-1 and the gray shading indicates the regions that are missing in Musashi-2. The red box contains the consensus intrinsically disordered regions (residues 237–362 of Musashi-1 and 235–328 of Musashi-2) considered in this study. **c** Amino acid pie charts of Musashi proteins in the model organisms listed in panel (a). The RRM and IDR were defined as shown in panel (b). **d** Single amino acid population differences in Musashi-1 and -2 with the amino acids sorted by prion-likeness³⁰. Gray bars indicate amino acids that appear fewer than ten times in the sequence.

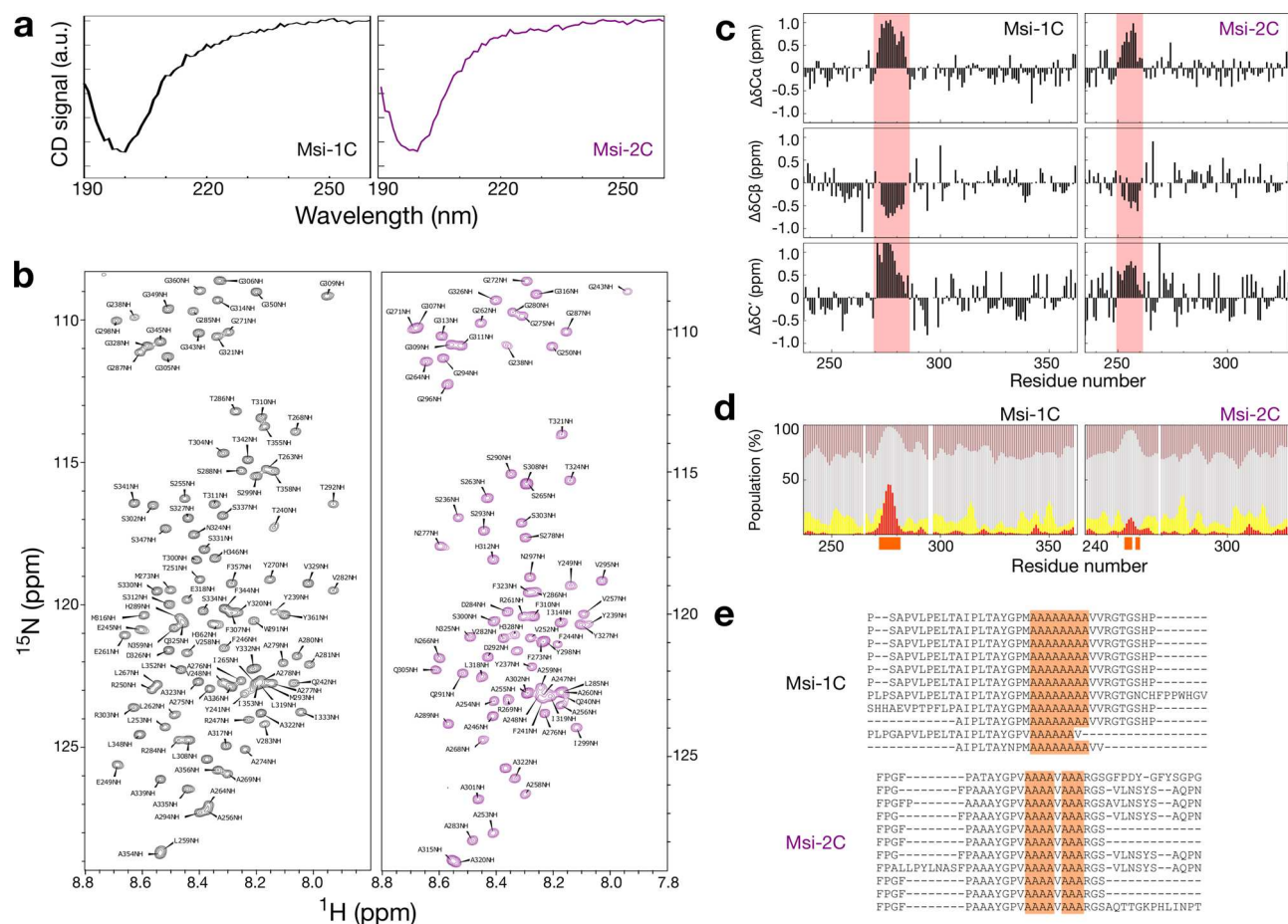


Fig. 2 The C-terminal domains of Musashi proteins are intrinsically disordered and have helix-forming polyaniline tracts. **a** Circular dichroism (CD) spectra of the C-terminal domains of Musashi-1 (black; Msi-1C) and Musashi-2 (purple; Msi-2C). **b** ^{15}N -edited NMR HSQC spectra of Msi-1C and Msi-2C with resonance assignments. **c** Secondary chemical shifts of $\text{C}\alpha$, $\text{C}\beta$, and carbonyl-carbon (C'). The most pronounced deviations are highlighted in red. **d** Stacked plots of the secondary structural populations derived from the chemical shifts using the $\delta 2\text{D}$ algorithm³⁴. Red: α -helix; yellow: β -sheet; gray: random coil; brown: polyproline II helix. The alanines in the polyaniline tracts are indicated by orange blocks on the x-axis. **e** Multiple sequence alignment of Musashi homologs. The sequences are in the same order as shown in Fig. 1a. The polyaniline stretch is highlighted in orange. The CD and HSQC spectra were repeated at least three times for both protein samples.

respectively continuous and disrupted polyaniline tracts are conserved in Musashi-1 and Musashi-2 orthologs (Fig. 2e and Supplementary Fig. 2).

The non-conserved regions flanking the polyaniline tracts tune their α -helical propensity. Musashi-1 has two additional regions not found in Musashi-2 around the polyaniline tract (denoted Seq1 and Seq2; Fig. 3a). We created three constructs without Seq1, Seq2, or both, to investigate their effect (Fig. 3b; ΔSeq1 , ΔSeq2 , and $\Delta\text{Seq1}\Delta\text{Seq2}$). The CD patterns of these variants are similar to those of Msi-1C (Fig. 3c) and in the ^{15}N - ^1H HSQC spectra, most peaks overlap with those of Msi-1C. However, there are pronounced changes between Seq1 and Seq2 (Fig. 3d). The adjacency of these regions to the mutation sites is not the only cause because residues further away from the polyaniline region show smaller chemical shift perturbations than those within the polyaniline region (as highlighted in Fig. 3d as an example and quantified in Fig. 3e). These changes show systematic patterns with downfield chemical shifts in the proton and nitrogen dimensions for ΔSeq1 and the opposite trend for ΔSeq2 (Fig. 3d, e), indicating a shifting equilibrium between different conformations in the fast exchange regime. To confirm this, we assigned the ^{13}C chemical shifts of the three variants (BMRB accesses: 51204, 51205, and 51206; Supplementary Fig. 4a), which

are mostly similar, except those close to the polyaniline region (Supplementary Fig. 4b). We calculated the secondary chemical shift difference of $\text{C}\alpha$ and $\text{C}\beta$ atoms ($\Delta\delta\text{C}\alpha$ - $\Delta\delta\text{C}\beta$; which minimizes the error from chemical shift referencing) and compared these values in the variants to those in the wildtype ($\Delta(\Delta\delta\text{C}\alpha$ - $\Delta\delta\text{C}\beta)$; Fig. 3f). Since larger ($\Delta\delta\text{C}\alpha$ - $\Delta\delta\text{C}\beta$) values indicate a stronger α -helical tendency, these results indicate that the wildtype's α -helical propensity is lower than ΔSeq1 's but higher than ΔSeq2 's. The estimated α -helical propensities (using the $\delta 2\text{D}$ program) are up to 6% higher than in the wildtype for ΔSeq1 but up to 5% lower than in the wildtype for ΔSeq2 (Fig. 3g). These results are in keeping with the indistinguishable CD data because, relative to the entire length of the sequences, the difference in α -helical propensity is negligible (<10% difference over just 10% of the sequence). The $\Delta\text{Seq1}\Delta\text{Seq2}$ variant has a similar predicted secondary structural population as the wildtype (Fig. 3g). These results indicate that α -helical propensity around the polyaniline stretch is altered by these non-conserved variations in primary sequence with, in order of α -helical propensity, $\Delta\text{Seq1} > \text{wildtype} \approx \Delta\text{Seq1}\Delta\text{Seq2} > \Delta\text{Seq2}$. Both these regions modify the α -helical propensity probably by altering the equilibrium between the monomeric and condensed states because of their physical properties, either many charged residues (Seq1) or a majority of hydrophobic residues (Seq2), would change the tendency to

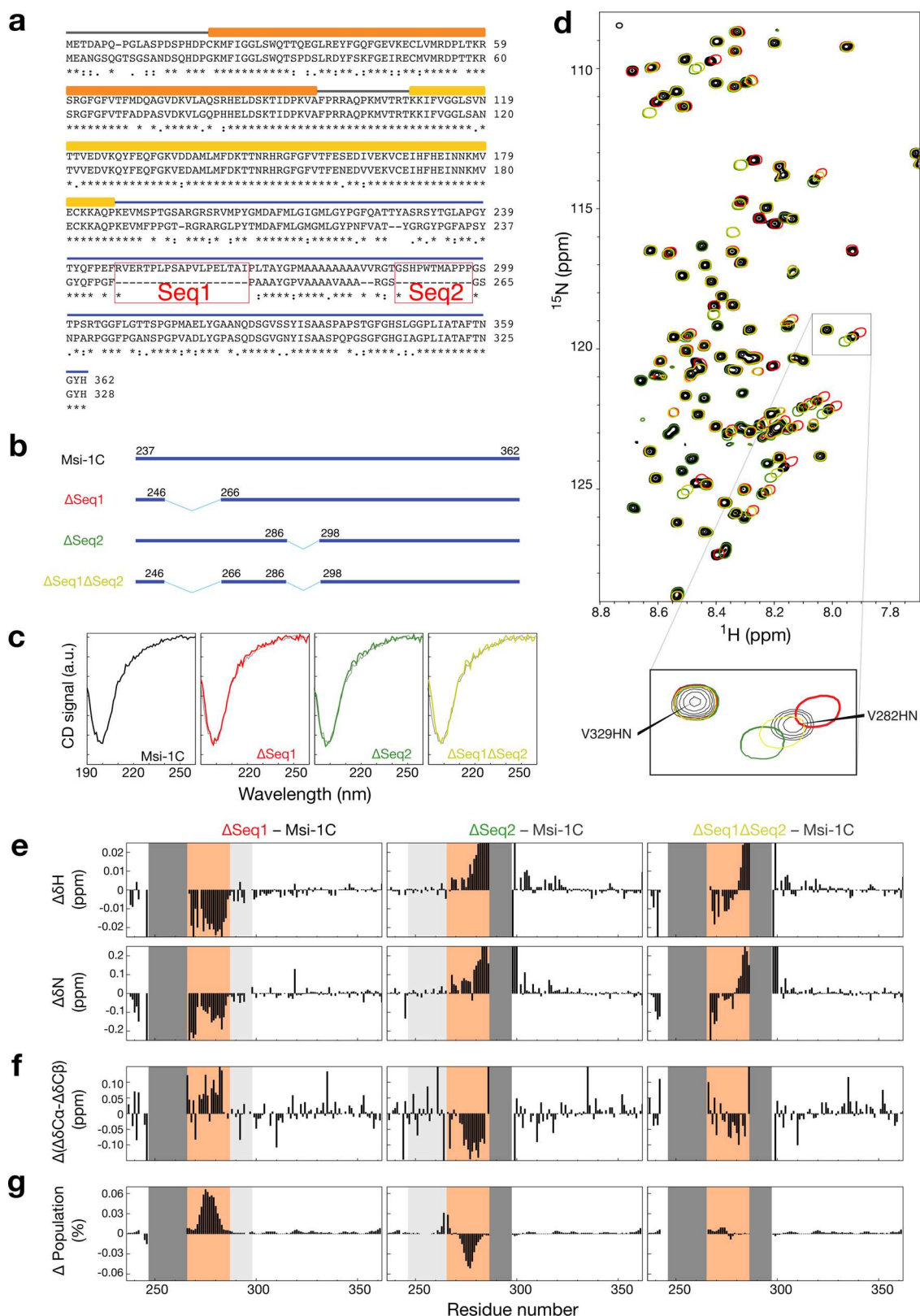


Fig. 3 The peptides flanking the polyaniline tract tune its α -helical propensity. **a** Sequence alignment of the human Musashi proteins. The main difference is the absence of the Seq1 and Seq2 regions in Musashi-2. **b** The designed constructs, with only the preserved residues indicated. **c** Circular dichroism spectra of the different constructs (in black, red, green, and yellow) overlaid on the wild types (in gray). **d** Overlaid NMR HSQC spectra (color-coded as in panel **b**), with an expanded view of overlapping (Val-329) and shifted (Val-282) cross-peaks. **e–g** Difference between the deletion constructs and the wildtype for **e** proton ($\Delta\delta\text{H}$) and nitrogen ($\Delta\delta\text{N}$) chemical shifts, **f** secondary chemical shift differences between $\text{C}\alpha$ and $\text{C}\beta$; **g** $\delta 2\text{D}^{34}$ α -helical propensity scores. The Seq1 and Seq2 regions are indicated in light gray if present, dark gray if missing; the polyaniline region is highlighted in orange. The CD and HSQC spectra were repeated at least three times for all protein samples.

assemble. Although the helicity correlates with self-assembly in some cases³⁷, the correlation is difficult to identify in our studies. Nevertheless, it is notable that there are no species in which Seq1 or Seq2 are missing alone, hinting that this α -helical propensity may have been fine-tuned.

The polyalanine region promotes Musashi-1 assembly. Msi-1C and Msi-2C differ in α -helical propensity and the regions in which they differ most change this propensity, and thus we investigated the effects of removing the polyalanine stretch between Seq1 and Seq2. The CD curve of this variant, denoted Δ SeqA, differs from that of Msi-1C around 220 nm (Fig. 4a), in a similar manner as Msi-2C's does (Fig. 2a), indicating a loss of α -helicity. Except for the truncation sites, the HSQC spectrum overlaps well with Msi-1C's, indicating no further structural change (Fig. 4b, c).

In all the NMR and CD studies, we noticed that the signal intensities did not correlate with protein concentrations. For example, the intensity ratio of HSQC spectra did not match their molar ratio (Supplementary Fig. 5). We thus speculate that higher-order oligomers (NMR-undetectable) are present. We used optical microscopy and indeed observed condensates in each case (Fig. 4d and Supplementary Fig. 6). At pH 5.5, the condensates were all spherical, and fluorescence recovery after photobleaching (FRAP) measurements on proteins labeled with a fluorescent probe (Cy3-NHS) showed that the condensates are dynamic (Fig. 4e). At pH 6.5, on the contrary, the condensates observed for Msi-1C were irregular and showed no fluorescence recovery (Fig. 4d, e), whereas, for Msi-2C and Δ Seq, the condensates remained spherical and dynamic (Fig. 4d and Supplementary Fig. 6). In order to avoid ambiguity in the following discussion, the term “condensates” is used for the dynamic states and “aggregates” for the less dynamic and irreversible state (irregular shapes under the microscope), as suggested^{38,39}. We also noticed that the level of fluorescence recovery varied between samples (gray lines in Fig. 4e), indicating that the condensates undergo very rapid sol-gel transitions (Fig. 4e), especially for Msi-1C at pH 5.5 (i.e., quantifying the level of recovery would not be informative). Rapid aggregation has also been reported for other proteins^{40–42}. Therefore, to compare the aggregation tendency, we prepared 20 μ M samples at pH 5.5 and 6.5 and incubated them at room temperature for different periods. We then centrifuged the samples (at $\sim 12,000 \times g$, to remove large aggregates but leave small condensates in the supernatant, as confirmed by microscopy) and measured the concentration of the soluble fraction. At pH 5.5, the proteins remained mostly in the supernatant for all incubation times (Fig. 4f, as determined by the absorbance at 280 nm). Condensates in the soluble fraction may affect the absorbance accuracy, thus we have also confirmed the amount of supernatant protein using SDS-PAGE gel (Fig. 4f; uncropped images in Supplementary Fig. 6). At pH 6.5, nearly all Msi-1C molecules aggregated within 1 h of incubation but Msi-2C, and to a lesser extent, the Δ SeqA construct remained partially in the supernatant for longer (at least 24 h; Fig. 4f). These results suggest that the polyalanine region in Msi-1C promotes aggregation. Figure 4g shows an energy landscape representation of the metastable nature of LLPS in these proteins⁴⁰. Their three main states are the monomeric form, the LLPS state (dynamic condensate), and the aggregate form. At pH 5.5, the energy barrier between LLPS and aggregation is high, leaving Msi-1C trapped in the dynamic condensate. At pH 6.5, however, the energy barrier is lower, and Msi-1C aggregates quickly. However, at the same pH, Δ SeqA remains in the dynamic soluble state for longer, indicating that removing the α -helical region restores the energy barrier between

the LLPS and aggregate states. These results suggest that although Msi-1C is less prion-like, its stronger α -helical propensity promotes assembly, regardless of the “price paid” in terms of aggregation^{42,43}.

Discussion

Various properties contribute to functional assembly in IDRs.

Without the constraints of a fixed shape, the IDRs in a paralog family can evolve more freely than structured regions, either gaining new functions or compensating for lost ones. The results obtained here for the Musashi protein family suggest IDRs may have evolved different means of functional assembly. Musashi-1 is found in stress granules, but its IDR has fewer prion-promoting amino acids than many others with this property (e.g., FUS, TDP-43, and hnRNP A1). However, our results suggest that the stronger α -helical propensity of Musashi-1's polyalanine stretch may assist its assembly, as polyalanine tracts are known to contribute to protein self-assembly⁴⁴. On the other hand, Musashi-2's IDR is sufficiently prion-like for assembly despite a lower α -helical propensity (Fig. 4).

Our results also explain a number of biological observations. Musashi proteins are overexpressed for cell renewal and stemness maintenance²². Although certain cell types express one or other Musashi paralogs (e.g., Musashi-2 in hematopoietic stem cells⁴⁵), they are functionally redundant when they appear together. For example, Musashi-2 compensates for the proliferation of neural progenitor cells in Musashi-1 double-knockout mice²⁰; both Musashi proteins promote colorectal cancer cell growth through the same signaling pathway⁴⁶; complete loss of visual function in photoreceptor cells is only observed in Musashi-1,2 double-knockout mice⁴⁷. The IDRs' different physical properties compensate for the loss of the other, but these differences (e.g., α -helix dominant vs more prion-like) may nevertheless lead to specific interaction mechanisms, for instance, tau protein only interacts with Musashi-1 for transportation into nuclei, whereas tau's interaction with Musashi-1 or -2 leads to different pathological stages in tauopathies²⁶.

Polyalanine is commonly involved in promoting self-assembly.

Polyalanine sequences may be a common evolved trait through which RBPs assemble or join membraneless organelles. For example, a sufficiently long polyalanine tract in its IDR enhances the subnuclear targeting properties of RBM4 (RNA-binding motif 4)⁴⁸. Indeed, with glutamine and asparagine, alanine is one of the most frequently repeated amino acids in proteins, and polyalanine stretches promote self-assembly⁴⁴. Several other amino acids are also helix-promoting, such as methionine, leucine, and glutamine³⁵. In the IDR of the extensively studied TDP-43, for example, the “AMMAAAQAALQ” amino acid motif has a strong tendency to form an α -helix, promoting condensation^{49–51} regardless of its short polyalanine tract. In order to estimate the frequency of polyalanine appearance, we rapidly searched for this trait in the proteome using a simple scoring function based on amino acid α -helical propensity values derived from our experimental observations and consensus studies (Fig. 5a)^{35,36}. Using these tentative scores and aggregating for repeats (examples shown in Fig. 5b), we calculated the portion of all IDRs in the human proteome and in RBPs (1542 in total) in which the aggregated polyalanine score reaches 1.5, 2.0, or 2.5 (Fig. 5c). We also analyzed a group of 692 mRNA-binding proteins (mRBPs) because many reported RBPs^{42,52,53} with LLPS-related functions reside in this category. Our analysis shows that polyalanine stretches are more common in RBPs, especially mRBPs, than in the general human proteome (Fig. 5c). We also used RaptorX⁵⁴ to predict the residues with α -helical propensity among the IDRs of

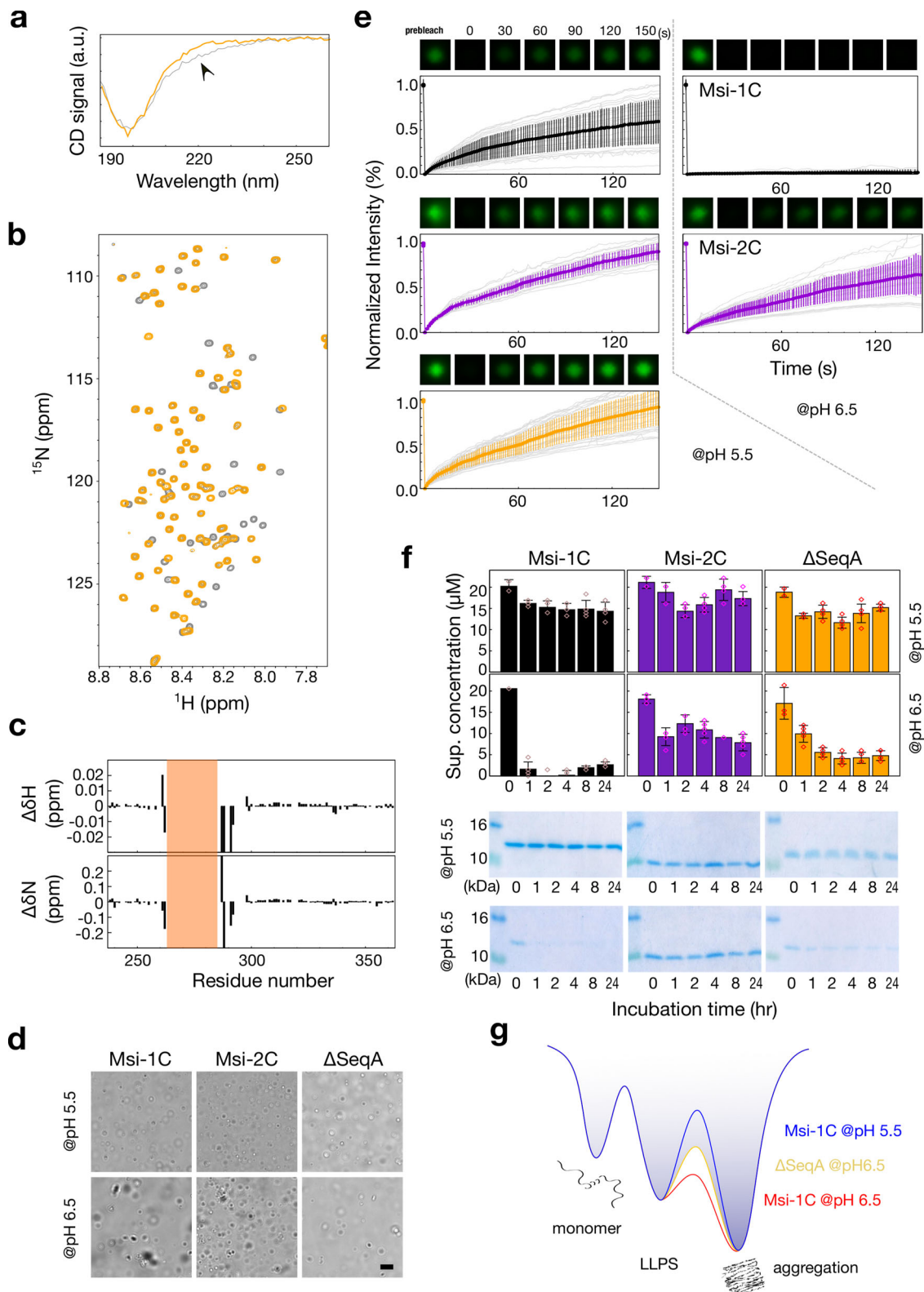


Fig. 4 The α -helical region promotes self-assembly in Musashi-1's C-terminal domain (Msi-1C). Overlaid **a** circular dichroism and **b** NMR HSQC spectra of the Δ SeqA construct (orange) and Msi-1C (gray). **c** Chemical shift differences in the proton ($\Delta\delta\text{H}$) and nitrogen ($\Delta\delta\text{N}$) dimensions between Δ SeqA and Msi-1C. The polyaniline region is highlighted in orange. **d** Light micrographs of Msi-1C, Msi-2C, and Δ SeqA at pH 5.5 or 6.5. Scale bar: 10 μm . Experiments were performed at least three times for each protein sample. **e** Fluorescence recovery after photobleaching (FRAP) results in pH 5.5 and 6.5. The colored lines represent the mean $\pm$ SD and the gray lines are individual recovery profiles. At least 20 condensates were recorded for each sample. **f** Precipitation assays of Msi-1C, Msi-2C, and Δ SeqA at pH 5.5 and 6.5 after incubation at room temperature for different time periods. Supernatant concentrations were determined after centrifugation (triplicate; mean $\pm$ SD) and confirmed by SDS-PAGE. **g** Energy landscape representation of the results.

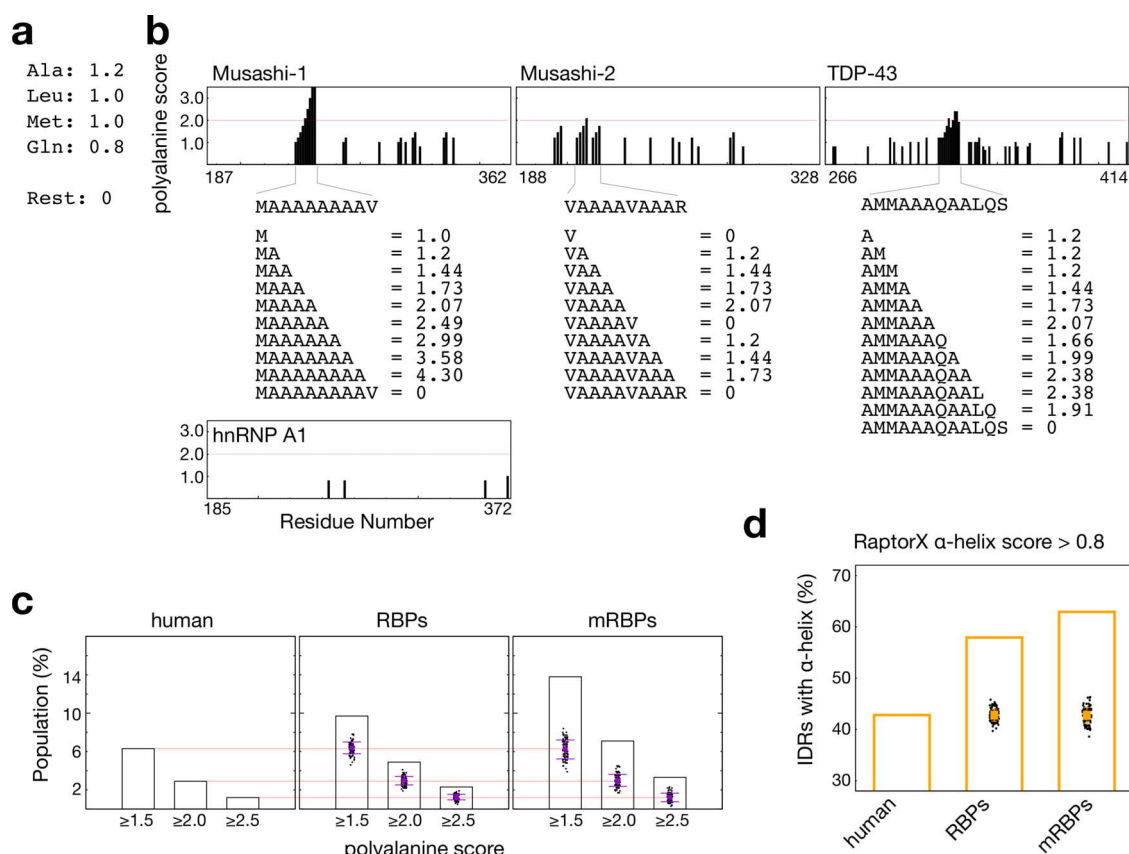


Fig. 5 Estimated prevalence of polyaniline and α -helix-forming residue stretches in the intrinsically disordered regions. **a** Tentative scores are assigned to the amino acids according to the experimental observation to estimate the frequency of polyaniline appearance. **b** Examples of how polyaniline propensity scores were calculated for continuous residue stretches of increasing length in proteins with α -helix-forming regions (Musashi-1, Musashi-2, and TDP-43) or without (hnRNP A1). **c** Prevalence of IDRs with polyaniline scores greater than 2.5, 2.0, and 1.5, in the human proteome, RNA-binding proteins (RBPs), and mRNA-binding proteins (mRBPs). **d** Population of IDRs with α -helical elements longer than five consecutive residues in the human proteome, RBPs, and mRBPs. The point clouds, square points, and error bars in the RBP and mRBP panels are the distribution, mean, and standard deviation of the polyaniline (panel **c**) and α -helix scores (panel **d**) of 100 random selections of sets of 1542 or 689 proteins (the RBP and mRBP sample sizes) from the human proteome (as negative controls).

the human proteome, RBPs, and mRBPs. We set arbitrary criteria (all other criteria show the same trend) to count those IDRs with at least five consecutive residues predicted higher than 0.8 (1.0 is the full scale) α -helical propensity in the program. Figure 5d shows the same tendency as the polyaniline score: mRBPs' or RBPs' IDRs are more likely to have α -helical elements within them. In these analyses, the difference is higher than the deviation of 100 random selections of 1542 or 689 proteins from the human proteome (the sample sizes considered for the RBPs and mRBPs, respectively; Fig. 5c, d). We attribute the results to that RBPs often join biomolecular condensates for their functions⁵⁵, and polyaniline/ α -helix is one feature of IDRs that contributes to self-assembly.

When did Musashi IDRs diverge? The IDRs of Musashi proteins may have respectively gained α -helicity or become less prion-like. Which came first? The answer is perhaps hidden in the primary sequence. Using the RRM from *D. melanogaster* Musashi once again, we searched for orthologs in the phylum Chordata, a higher taxonomic rank than the subphylum of vertebrates in which the Musashi paralogs arise (Fig. 1a). *Amphioxus* (the lancelet), a model organism for primitive chordates, has a Musashi homolog which does not have a long polyaniline tract but is prion-like, similar to that of the nematode and fruit fly (Fig. 6a, b and Supplementary Fig. 7). This suggests that the Musashi

paralogs may have arisen after the appearance of chordates. In ghost sharks indeed, primitive vertebrates with the two Musashi paralogs, Musashi-1 has a polyaniline tract that is not prion-like (Fig. 6c and Supplementary Fig. 7), whereas interestingly, Musashi-2 has a polyaniline tract and is prion-like. Collectively, these results suggest that the polyaniline stretch may have appeared in the paralogs alongside primitive prion propensity before Musashi-1 lost its prion-promoting amino acids, whereas a disruption in the α -helical region reduced Musashi-2's tendency to self-assemble.

“Distant relatives” in the Musashi family. According to the RBP census study¹¹, many other RBPs were grouped in the paralog family of Musashi proteins based on sequence similarity. We confirmed this by searching for sequences similar to Musashi-1's in the human proteome. Other than Musashi-2 (71.3% identity), the top hits are DAZ-associated protein 1 (DAZAP-1, 42.6%), several heterogeneous nuclear ribonucleoproteins (hnRNP A0, A1, A2...; 39.0–42.5%) and TDP-43 (34.9%) (Fig. 6d), agreeing with the previous study¹¹. Accordingly, we compared the IDRs' properties of these Musashi protein's “distant relatives”: Studies of the low complexity IDRs of hnRNP A1 and A2 are pioneering examples in the emerging field of protein LLPS^{41,52,56}. They are also a good model for studying LLPS theory, such as the effects of the number and distribution of aromatic residues⁵⁷. TDP-43's

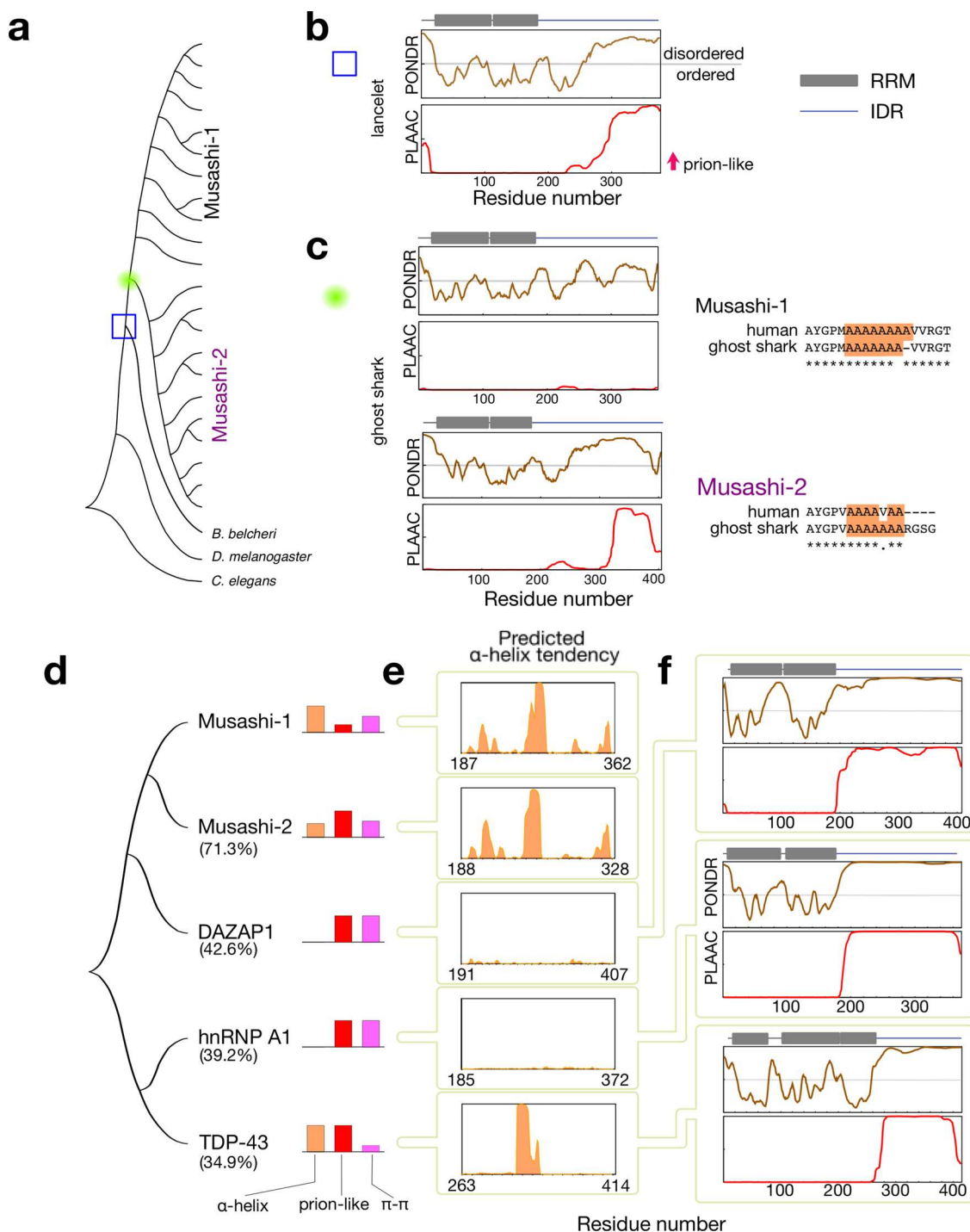


Fig. 6 Sequence analysis of the Musashi family in primitive model organisms and distantly related paralogs. **a** Phylogenetic tree of the Musashi family as shown in Fig. 1a with the additional lineages analyzed: a primitive chordate (the lancelet, blue box, panel **b**) and a primitive vertebrate (the ghost shark, green circle, panel **c**). **b, c** PONDR sequence disorder and PLAAC prion-likeness scores of **b** the lancelet (*Buccinum belcheri*) and **c** the ghost shark (*Callorhynchus milii*). The RNA recognition motifs (based on PROSITE analysis⁷³) are indicated by gray squares. The alignment of the polyaniline regions (highlighted in orange) of human and ghost shark Musashi proteins are also shown. **d** Phylogenetic tree of Musashi paralogs with high sequence similarity found in the human proteome. The percentages shown are the levels of sequence similarity. The importance of different drivers of assembly in their IDRs is indicated by colored bars representing their α -helical propensity (orange), prion-likeness (red), and the number of aromatic residues (purple). **e** α -helical propensity predicted using RaptorX⁵⁴. **f** Sequence disorder and prion-likeness for DAZAP1, hnRNP A1, and TDP-43 (c.f. Fig. 1b for Musashi-1 and -2).

IDR is another extensively studied example. TDP-43's IDR differs from hnRNP A1 or A2's in that, as well as being prion-like, it encompasses a short α -helical region that contributes to self-association and LLPS^{49,51}. Furthermore, TDP-43's LLPS is mediated by just a few aromatic residues⁵⁰, a feature we have

attributed to the presence of the α -helix, which promotes inter-molecular contacts⁵⁰ (Fig. 6d). The present study adds the biophysical properties of the two Musashi proteins to existing information on this distantly related protein family. Although DAZPA1 has been far less studied than other members, its

C-terminal domain is critical to its function in interacting with eIF4G⁵⁸ and potentially interacts with many RBPs, including hnRNP A1⁵⁹. Moreover, bioinformatic analysis shows that this protein's C-terminal domain is disordered and prion-like, with many aromatic residues but no predicted α -helix, similar to that of hnRNP A1 (Fig. 6e, f and Supplementary Fig. 8). Although the IDRs of these RBPs have different properties, they undergo LLPS. As a conclusion, we suggest that IDRs evolve whatever traits are beneficial to function, which accords with François Jacob's statement that "evolution does not produce novelties from scratch"⁶⁰. He was referring to the diversity of all lifeforms but his words also apply to IDRs, which acquire new functions through evolutionary "tinkering"⁶⁰ between prion-likeness, α -helicity, aromatic residues, etc. Our work could be used as a template to investigate IDRs in other paralogs and how their functions have diversified or been preserved during evolution.

Methods

Bioinformatics analysis. The primary sequences were obtained from UniProt with the associated entries listed in Supplementary Table 1. Levels of structural disorder and prion-likeness were respectively analyzed with the PONDR⁶¹ and PLAAC³⁰ web servers. The phylogenetic trees were constructed using the neighbor-joining method in MEGA X⁶².

In analyzing the polyaniline and α -helical propensities, all human protein sequences were retrieved from UniProt (UniProtKB_2021_01, download date: 2021/02/06, 20396 sequences in total) and separated into disordered and ordered regions based on PONDR predictions (VSL2)⁶¹. The predicted disordered regions longer than 40 consecutive residues were analyzed. The tentative polyaniline scoring (Fig. 5a) of the sequences was done using in-house scripts to estimate the frequency of polyaniline appearance. The α -helical propensities were predicted using RaptorX⁵⁴. An IDR in a protein having at least five residues in a raw predicted as α -helix (with a value higher than 0.8 out of 1.0) is counted in our analysis.

DNA constructs. The Msi-1C construct in this study (residues number 237 to 362) was cloned from a previous longer construct (194–362)⁶³. We removed a strongly hydrophobic region reported to contain binding sites for PABP and GLD2²², which is irrelevant to the present study and whose removal improves the protein's solubility. The Δ Seq1, Δ Seq2, Δ Seq1 Δ Seq2, and Δ SeqA constructs were prepared with designed primers (Supplementary Table 3). Msi-2 cDNA (residues 234–328, according to the alignment with Msi-1C) was purchased from OriGene. All these variants were constructed in a pET21 vector backbone with a hexahistidine tag on the C-terminus of the expressed protein. All constructs were fully sequenced.

Protein expression and purification. All constructs were purified using the same protocol, which has been described in detail elsewhere⁶³. In short, transformed *E. coli* BL21(DE3) cells were grown at 37 °C until the OD reached 0.6 and were induced with a final concentration of 1 mM isopropyl- β -D-1-thiogalactopyranoside at 25 °C overnight. The cells were harvested and lysed by sonication. After centrifugation, the inclusion bodies were dissolved using 20 mM Tris buffer at pH 8 with 8 M urea (buffer A). After a second centrifugation, the supernatant was filtered with a 0.45 μ m filter and loaded onto a nickel-charged immobilized metal affinity chromatography column (Qiagen). After washing with ten column volumes of buffer A, the samples were eluted with 5 column volumes of buffer B (buffer A with 500 mM imidazole). The eluted samples were acidified with trifluoroacetic acid (down to pH ~3), loaded onto a C4 reverse-phase column (Thermo Scientific Inc.), and eluted with a gradient of acetonitrile (from 0 to 100%, mixed with triple-distilled water) by HPLC and then lyophilized. The lyophilized samples were stored in a dry cabinet until use. For all experiments, powder samples were dissolved in 20 mM MES-NaOH at pH 5.5 or 6.5. The protein concentration was determined from the absorbance at 280 nm measured with a Nanodrop spectrometer (Thermo Scientific Inc.).

Circular dichroism spectroscopy. Circular dichroism spectra were recorded using an AVIV model 410 spectropolarimeter with a 0.1 mm cuvette. Data were collected between 190 and 260 nm with an interval of 1 nm. Ten measurements were co-added for each data point. All spectra were recorded at 283 K and the samples were kept in a water bath at 283 K between measurements. All experiments were performed in triplicate.

NMR spectroscopy, chemical shift assignment, and data analysis. ¹⁵N-edited HSQC spectra were recorded using the standard pulse sequence with WATERGATE solvent suppression^{64,65}. Chemical shifts were assigned using standard HNCA, HN(CO)CA, HNCO, HN(CA)CO, CBCA(CO)NH, and HNCACB experiments acquired with non-uniform sampling (25%)^{66,67}. All data were recorded using a Bruker AVIII 600 MHz spectrometer with a cryogenic probe.

The data were processed using NMRPipe⁶⁸. Chemical shifts were assigned using the automated assignment scheme⁶⁹ implemented in NMRFAM-Sparky⁷⁰, and then confirmed manually. Secondary chemical shift analysis was performed using Kjaergaard et al.'s database of random-coil shifts⁷¹. Secondary structure populations were estimated using δ 2D³⁴. No chemical shifts were missing around the critical α -helical region, such that the secondary structure estimates for the constructs were made using the same number of chemical shifts.

Microscopy. Protein samples were loaded onto mPEG-passivated slides⁷². Micrographs were collected using an Olympus BX51 microscope with a 40 $\times$ long-working-distance objective lens.

Microscopy and fluorescence recovery after photobleaching (FRAP) experiments. For the fluorescent dye labeling, 2.5 mg lyophilized protein was dissolved in 0.1 M sodium phosphate with 6 M guanidine hydrochloride at pH 8.3 and mixed with 15.6 mM (~1 mg in total) of Cy3-NHS (Lumiprobe) overnight at room temperature. The excess Cy3-NHS was removed and the buffer was exchanged with a 20 mM MES buffer at pH 5.5 using a PD-10 column (GE Healthcare). The typical labeling efficiency, determined from the ratio of extinction coefficients measured for the protein and the fluorescence dye (Cy3: 150,000 M⁻¹ cm⁻¹ at 550 nm) was ~20%. The Cy3-labeled sample was aliquoted, flash-frozen, and stored at -80 °C before usage.

The lyophilized samples were dissolved in 20 mM MES buffer at pH 5.5 or pH 6.5 and mixed with a Cy3-labeled sample with a final concentration of 20 μ M and ~1% Cy3-labeled protein. The samples were then loaded onto ultraclean coverslips and observed with an *i*LAS multi-modal of total internal reflection fluorescence (Roper Scientific, Inc.)/spinning disk (CSUX1, Yokogawa) confocal microscope (Ti-E, Nikon) equipped with 100 $\times$ 1.49NA plan objective lens (Nikon). The condensates were bleached with a 561 nm laser. Images were acquired at one-second intervals using an Evolve EMCCD camera (Photometrics) and were analyzed using the Metamorph software (Molecular Devices, LLC).

Aggregation assays. Lyophilized samples were dissolved in 20 mM MES buffer at pH 5.5 or pH 6.5 and stored at room temperature for different times. The samples were then centrifuged at 12,000 $\times$ g for 5 min. Supernatant concentrations were determined using a NanoDrop spectrometer. Supernatant samples were also analyzed using SDS-PAGE gels to confirm the amount of protein present. All experiments were performed in triplicate.

Statistics and reproducibility. All NMR HSQC spectra, CD experiments, and microscope observations were repeated at least three times. The reproducibility of these types of biophysical experiments is high. The FRAP experiments were repeated from three independently prepared samples. At least ten condensates were recorded to obtain the mean $\pm$ standard deviation recovery curves. Precipitation assays were performed at least three times from independently prepared samples. The data were given as the mean $\pm$ standard deviation. The individual data points were also shown to present data distribution.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The chemical shift assignments for generating Figs. 2b, 2c, 2d, 3d, 3e, 3f, 3g are deposited in the Biological Magnetic Resonance Bank (BMRB): 51204, 51205, 51206, 51207, 51208. Scripts for reproducing Fig. 5 are deposited in http://github.com/allmwh/helix_score.

Received: 24 January 2022; Accepted: 10 April 2022;

Published online: 29 April 2022

References

- Ohno, S. *Evolution by Gene Duplication* (Springer, 1970).
- Wilson, E. B. The structure of protoplasm. *Science* **10**, 33–45 (1899).
- Lodish, H. et al. *Molecular Cell Biology* (W. H. Freeman and Company, 2016).
- Uversky, V. N. & Dunker, A. K. Understanding protein non-folding. *Biochim. Biophys. Acta* **1804**, 1231–1264 (2010).
- Gebauer, F., Schwarzl, T., Valcarcel, J. & Hentze, M. W. RNA-binding proteins in human genetic disease. *Nat. Rev. Genet.* **22**, 185–198 (2020).
- Dominguez, D. et al. Sequence, structure, and context preferences of human RNA binding proteins. *Mol. Cell* **70**, 854–867.e9 (2018).
- Varadi, M., Zsoltyomi, F., Guharoy, M. & Tompa, P. Functional advantages of conserved intrinsic disorder in RNA-binding proteins. *PLoS ONE* **10**, e0139731 (2015).
- Zagrovic, B., Bartonek, L. & Polyansky, A. A. RNA-protein interactions in an unstructured context. *FEBS Lett.* **592**, 2901–2916 (2018).

9. Kwon, I. et al. Phosphorylation-regulated binding of RNA polymerase II to fibrous polymers of low-complexity domains. *Cell* **155**, 1049–1060 (2013).
10. Alberti, S. & Hyman, A. A. Biomolecular condensates at the nexus of cellular stress, protein aggregation disease and ageing. *Nat. Rev. Mol. Cell Biol.* **22**, 196–213 (2021).
11. Gerstberger, S., Hafner, M. & Tuschl, T. A census of human RNA-binding proteins. *Nat. Rev. Genet.* **15**, 829–845 (2014).
12. Li, L., Stoeckert, C. J. Jr. & Roos, D. S. OrthoMCL: identification of ortholog groups for eukaryotic genomes. *Genome Res.* **13**, 2178–2189 (2003).
13. Nakamura, M., Okano, H., Blendy, J. A. & Montell, C. Musashi, a neural RNA-binding protein required for Drosophila adult external sensory organ development. *Neuron* **13**, 67–81 (1994).
14. Yoda, A., Sawa, H. & Okano, H. MSI-1, a neural RNA-binding protein, is involved in male mating behaviour in *Caenorhabditis elegans*. *Genes Cells* **5**, 885–895 (2000).
15. Hirota, Y. et al. Musashi and seven in absentia downregulate Tramtrack through distinct mechanisms in Drosophila eye development. *Mech. Dev.* **87**, 93–101 (1999).
16. Kawashima, T. et al. Expression patterns of musashi homologs of the ascidians, *Halocynthia roretzi* and *Ciona intestinalis*. *Dev. Genes Evol.* **210**, 162–165 (2000).
17. Shibata, S. et al. Characterization of the RNA-binding protein Musashi1 in zebrafish. *Brain Res.* **1462**, 162–173 (2012).
18. Sakakibara, S. et al. Mouse-Musashi-1, a neural RNA-binding protein highly enriched in the mammalian CNS stem cell. *Dev. Biol.* **176**, 230–242 (1996).
19. Okano, H., Imai, T. & Okabe, M. Musashi: a translational regulator of cell fate. *J. Cell Sci.* **115**, 1355–1359 (2002).
20. Sakakibara, S. et al. RNA-binding protein Musashi family: roles for CNS stem cells and a subpopulation of ependymal cells revealed by targeted disruption and antisense ablation. *Proc. Natl Acad. Sci. USA* **99**, 15194–15199 (2002).
21. Lagade, C. et al. The RNA-binding protein Musashi-1 regulates proteasome subunit expression in breast cancer- and glioma-initiating cells. *Stem Cells* **32**, 135–144 (2014).
22. Fox, R. G., Park, F. D., Koehlein, C. S., Kritzik, M. & Reya, T. Musashi signaling in stem cells and cancer. *Annu. Rev. Cell Dev. Biol.* **31**, 249–267 (2015).
23. Chiou, G. Y. et al. Musashi-1 promotes a cancer stem cell lineage and chemoresistance in colorectal cancer cells. *Sci. Rep.* **7**, 2172 (2017).
24. Chen, H. Y. et al. Musashi-1 promotes chemoresistant granule formation by PKR/eIF2 α signalling cascade in refractory glioblastoma. *Biochim. Biophys. Acta Mol. Basis Dis.* **1864**, 1850–1861 (2018).
25. Sengupta, U. et al. Formation of toxic oligomeric assemblies of RNA-binding protein: Musashi in Alzheimer's disease. *Acta Neuropathol. Commun.* **6**, 113 (2018).
26. Montalbano, M. et al. RNA-binding proteins Musashi and tau soluble aggregates initiate nuclear dysfunction. *Nat. Commun.* **11**, 4305 (2020).
27. Hedges, S. B. The origin and evolution of model organisms. *Nat. Rev. Genet.* **3**, 838–849 (2002).
28. King, O. D., Gitler, A. D. & Shorter, J. The tip of the iceberg: RNA-binding proteins with prion-like domains in neurodegenerative disease. *Brain Res.* **1462**, 61–80 (2012).
29. Harrison, A. F. & Shorter, J. RNA-binding proteins with prion-like domains in health and disease. *Biochem. J.* **474**, 1417–1438 (2017).
30. Lancaster, A. K., Nutter-Upham, A., Lindquist, S. & King, O. D. PLAAC: a web and command-line application to identify proteins with prion-like amino acid composition. *Bioinformatics* **30**, 2501–2502 (2014).
31. Greenland, K. N. et al. Order, disorder, and temperature-driven compaction in a designed elastin protein. *J. Phys. Chem. B* **122**, 2725–2736 (2018).
32. Uversky, V. N. Natively unfolded proteins: a point where biology waits for physics. *Protein Sci.* **11**, 739–756 (2002).
33. Prestel, A., Bugge, K., Staby, L., Hendus-Altenburger, R. & Kragelund, B. B. Characterization of dynamic IDP complexes by NMR spectroscopy. *Methods Enzymol.* **611**, 193–226 (2018).
34. Camilloni, C., De Simone, A., Vranken, W. F. & Vendruscolo, M. Determination of secondary structure populations in disordered states of proteins using nuclear magnetic resonance chemical shifts. *Biochemistry* **51**, 2224–2231 (2012).
35. Pace, C. N. & Scholtz, J. M. A helix propensity scale based on experimental studies of peptides and proteins. *Biophys. J.* **75**, 422–427 (1998).
36. Levitt, M. Conformational preferences of amino acids in globular proteins. *Biochemistry* **17**, 4277–4285 (1978).
37. Conicella, A. E. et al. TDP-43 alpha-helical structure tunes liquid-liquid phase separation and function. *Proc. Natl Acad. Sci. USA* **117**, 5883–5894 (2020).
38. Banani, S. F., Lee, H. O., Hyman, A. A. & Rosen, M. K. Biomolecular condensates: organizers of cellular biochemistry. *Nat. Rev. Mol. Cell Biol.* **18**, 285–298 (2017).
39. Shin, Y. & Brangwynne, C. P. Liquid phase condensation in cell physiology and disease. *Science* **357**, eaaf4382 (2017).
40. Hardenberg, M., Horvath, A., Ambrus, V., Fuxreiter, M. & Vendruscolo, M. Widespread occurrence of the droplet state of proteins in the human proteome. *Proc. Natl Acad. Sci. USA* **117**, 33254–33262 (2020).
41. Molliex, A. et al. Phase separation by low complexity domains promotes stress granule assembly and drives pathological fibrillization. *Cell* **163**, 123–133 (2015).
42. Patel, A. et al. A liquid-to-solid phase transition of the ALS protein FUS accelerated by disease mutation. *Cell* **162**, 1066–1077 (2015).
43. Itakura, A. K., Futia, R. A. & Jarosz, D. F. It pays to be in phase. *Biochemistry* **57**, 2520–2529 (2018).
44. Polling, S. et al. Polyalanine expansions drive a shift into alpha-helical clusters without amyloid-fibril formation. *Nat. Struct. Mol. Biol.* **22**, 1008–1015 (2015).
45. Kharas, M. G. et al. Musashi-2 regulates normal hematopoiesis and promotes aggressive myeloid leukemia. *Nat. Med.* **16**, 903–908 (2010).
46. Li, N. et al. The Msi family of RNA-binding proteins function redundantly as intestinal oncoproteins. *Cell Rep.* **13**, 2440–2455 (2015).
47. Sundar, J., Matalakah, F., Jeong, B., Stoilov, P. & Ramamurthy, V. The Musashi proteins MSI1 and MSI2 are required for photoreceptor morphogenesis and vision in mice. *J. Biol. Chem.* **296**, 100048 (2020).
48. Chang, S. H., Chang, W. L., Lu, C. C. & Tarn, W. Y. Alanine repeats influence protein localization in splicing speckles and paraspeckles. *Nucleic Acids Res.* **42**, 13788–13798 (2014).
49. Li, H. R. et al. The physical forces mediating self-association and phase-separation in the C-terminal domain of TDP-43. *Biochim. Biophys. Acta* **1866**, 214–223 (2018).
50. Li, H. R., Chiang, W. C., Chou, P. C., Wang, W. J. & Huang, J. R. TAR DNA-binding protein 43 (TDP-43) liquid-liquid phase separation is mediated by just a few aromatic residues. *J. Biol. Chem.* **293**, 6090–6098 (2018).
51. Conicella, A. E., Zerze, G. H., Mittal, J. & Fawzi, N. L. ALS mutations disrupt phase separation mediated by alpha-helical structure in the TDP-43 low-complexity C-terminal domain. *Structure* **24**, 1537–1549 (2016).
52. Xiang, S. et al. The LC domain of hnRNP A2 adopts similar conformations in hydrogel polymers, liquid-like droplets, and nuclei. *Cell* **163**, 829–839 (2015).
53. Sun, Y. & Chakrabarty, A. Phase to phase with TDP-43. *Biochemistry* **56**, 809–823 (2017).
54. Yang, Y. et al. Sixty-five years of the long march in protein secondary structure prediction: the final stretch? *Brief. Bioinform.* **19**, 482–494 (2018).
55. Wiedner, H. J. & Giudice, J. It's not just a phase: function and characteristics of RNA-binding proteins in phase separation. *Nat. Struct. Mol. Biol.* **28**, 465–473 (2021).
56. Kato, M. et al. Cell-free formation of RNA granules: low complexity sequence domains form dynamic fibers within hydrogels. *Cell* **149**, 753–767 (2012).
57. Martin, E. W. et al. Valence and patterning of aromatic residues determine the phase behavior of prion-like domains. *Science* **367**, 694–699 (2020).
58. Smith, R. W. et al. DAZAP1, an RNA-binding protein required for development and spermatogenesis, can regulate mRNA translation. *RNA* **17**, 1282–1295 (2011).
59. Yang, H. T., Pegg, M., Cohen, P. & Rousseau, S. DAZAP1 interacts via its RNA-recognition motifs with the C-termini of other RNA-binding proteins. *Biochem. Biophys. Res. Commun.* **380**, 705–709 (2009).
60. Jacob, F. Evolution and tinkering. *Science* **196**, 1161–1166 (1977).
61. Obradovic, Z., Peng, K., Vucetic, S., Radivojac, P. & Dunker, A. K. Exploiting heterogeneous sequence properties improves prediction of protein disorder. *Proteins* **61**, 176–182 (2005).
62. Kumar, S., Stecher, G., Li, M., Nkay, C. & Tamura, K. MEGA X: molecular evolutionary genetics analysis across computing platforms. *Mol. Biol. Evol.* **35**, 1547–1549 (2018).
63. Chen, T. C. & Huang, J. R. Musashi-1: an example of how polyalanine tracts contribute to self-association in the intrinsically disordered regions of RNA-binding proteins. *Int. J. Mol. Sci.* **21**, 2289 (2020).
64. Piotto, M., Saudek, V. & Sklenar, V. Gradient-tailored excitation for single-quantum NMR spectroscopy of aqueous solutions. *J. Biomol. NMR* **2**, 661–665 (1992).
65. Bodenhausen, G. & Ruben, D. J. Natural abundance N-15 NMR by enhanced heteronuclear spectroscopy. *Chem. Phys. Lett.* **69**, 185–189 (1980).
66. Hyberts, S. G., Milbradt, A. G., Wagner, A. B., Arthanari, H. & Wagner, G. Application of iterative soft thresholding for fast reconstruction of NMR data non-uniformly sampled with multidimensional Poisson Gap scheduling. *J. Biomol. NMR* **52**, 315–327 (2012).
67. Hyberts, S. G., Frueh, D. P., Arthanari, H. & Wagner, G. FM reconstruction of non-uniformly sampled protein NMR data at higher dimensions and optimization by distillation. *J. Biomol. NMR* **45**, 283–294 (2009).
68. Delaglio, F. et al. NMRPipe: a multidimensional spectral processing system based on UNIX pipes. *J. Biomol. NMR* **6**, 277–293 (1995).
69. Lee, W., Westler, W. M., Bahrami, A., Eghbalnia, H. R. & Markley, J. L. PINE-SPARKY: graphical interface for evaluating automated probabilistic peak assignments in protein NMR spectroscopy. *Bioinformatics* **25**, 2085–2087 (2009).

70. Lee, W., Tonelli, M. & Markley, J. L. NMRFAM-SPARKY: enhanced software for biomolecular NMR spectroscopy. *Bioinformatics* **31**, 1325–1327 (2015).
71. Kjaergaard, M., Brander, S. & Poulsen, F. M. Random coil chemical shift for intrinsically disordered proteins: effects of temperature and pH. *J. Biomol. NMR* **49**, 139–149 (2011).
72. Alberti, S. et al. A user's guide for phase separation assays with purified proteins. *J. Mol. Biol.* **430**, 4806–4820 (2018).
73. Sigrist, C. J. et al. PROSITE, a protein domain database for functional characterization and annotation. *Nucleic Acids Res.* **38**, D161–D166 (2010).

Acknowledgements

The authors thank Prof. Won-Jing Wang (NYCU) for access to the microscope, Academia Sinica High-Field NMR Center for technical support (HFNMRC is funded by Academia Sinica Core Facility and Innovative Instrument Project (AS-CFII-108-112)), Chen-Yu Hung for microscope setting, and Dr. Tsai-Chen Chen for initial work on this project. This work was supported by the Ministry of Science and Technology of Taiwan (109-2113-M-010-003 and 110-2113-M-A49A-504-MY3 to J.-r.H. and 109-2326-B-010-002 -MY3 to J.-C.K.).

Author contributions

J.-r.H. conceived the project and wrote the manuscript. S.-H.C., W.-L.H., Y.-C.S., J.-C.K., and J.-r.H. collected and analyzed the data.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s42003-022-03354-4>.

Correspondence and requests for materials should be addressed to Jie-rong Huang.

Peer review information *Communications Biology* thanks Sourav Chowdhury and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Krishnananda Chattopadhyay and Manuel Breuer. Peer reviewer reports are available.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2022